GATES LEARJET CORPORATION

maintenance manual

CHAPTER

24

ELECTRICAL POWER

Record of Temporary Revisions

Rev. No.	Date Inserted	Date Removed	Page Number	Rev. No.	Date Inserted	Date Removed	Page Number
24-1	May 12/78	Apr 30/82 Rev 14	24-32-02 Page 201				
24-2	Sep 25/78	Apr 30/82 Rev 14	24-31-02 Page 201				
24-3	Jul 25/80	Apr 30/82 Rev 14	24-50-00 Page 1				
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GENERAL - DESCRIPTION AND OPERATION

1. DESCRIPTION

A. Primary aircraft DC electrical power is provided by two independent DC starter generator systems. During normal operation, both generator systems operate in parallel. Secondary aircraft electrical power is provided by either two nickel-cadmium batteries or two optional lead-acid batteries. For ground operation, an external power source provides electrical power requirements. AC power requirements are provided by two static inverters installed in the tailcone. In addition, the aircraft is equipped with an emergency power supply.

B. Two engine-driven DC starter-generators, one on each engine, provide the normal source of aircraft 28 vdc. Each starter-generator is an air cooled type, rated at 30 volts, 400 amperes.

C. Two nickel-cadmium or two optional lead-acid batteries, located in the tailcone provide the secondary source of 28 vdc. Battery power is utilized for engine starting whenever external power is not used. The nickel-cadmium batteries are equipped with an overheat warning system and an optional temperature indicating system.

D. AC electrical power is provided by two solid state inverters installed in the tailcone. Each inverter is rated at 115 volts, 400 Hz and 1000 VA. During normal operation, the inverters operate in parallel. An optional auxiliary inverter may also be installed.

E. External DC power can be connected to the aircraft through an external power receptacle located on the left side of the aircraft adjacent to the tailcone access door. On aircraft equipped with a single battery switch, external power can be connected directly to the power distribution buses. On aircraft equipped with dual battery switches, either or both battery switches must be on to connect external power to the power distribution buses.

F. Emergency power is provided by either single or dual emergency batteries installed in the tailcone or the nose compartment. The batteries provide both AC and DC power. AC power is 115 volts, 400 Hz at 100 VA. DC power is 28 vdc at either 1.9 amp hours or 3.8 amp hours depending upon the model installed.

G. Indicators

(1) A DC ammeter is installed for each generator. The ammeter indicates the load current supplied by its respecitve generator.

(2) A single DC voltmeter indicates DC voltage present on the battery charging bus.

H. Switches

(1) On aircraft equipped with a single battery switch, the switch controls a relay which applies power from the battery bus to the battery charging bus. The switch is a two position on-off switch. On aircraft equipped with dual battery switches, the switches, one RH and one LH, provide the aircraft with battery isolation capabilities. Each switch is a two position (on-off) switch and completes a ground circuit to its respective battery relay. The switches are labeled L ON-OFF and R ON-OFF.

(2) Two starter generator switches, one RH and one LH are installed. Each switch is a three position switch marked GEN-OFF-START. When in the START position, power is applied to close respective fuel motive flow valve and energize the standby fuel pump. After the motive flow valve closes, power is applied, energizing the start relay. When in the GEN position, power is removed from the start relay; the motive flow valve is opened and the standby pump is deenergized. Automatic setup of generator regulation, paralleling and protection takes place within the generator control box.

(3) Two generator reset switches, one RH and one LH, are installed. Each switch is a two position double-throw momentary type switch. When held in the RESET position, one set of contacts complete a ground circuit to energize and reset the generator over-voltage cutout relay. The remaining contacts complete a power circuit to the voltage regulator.

(4) Two inverter switches, one for the primary inverter and one for the secondary inverter, are installed. Each switch is a two position double-throw switch. When in the ON position, one set of contacts complete a ground circuit to energize the inverter control relay. The remaining set of contacts connect inverter remote switch and inverter switch return lines. This completes the inverter operation circuit and inverter paralleling circuits.

(5) A single AC Bus switch (PRI-SEC) is intalled. The switch, a two position type, is used to monitor either AC bus voltage depending upon switch setting.

I. Indicator Lights

(1) On aircraft with nickel-cadmium batteries, two warning lights (red) labeled BAT 140 and BAT 160 are installed in the glareshield. The warning lights are controlled by thermal switches within each battery.

(2) Two generator caution lights (amber) labeled LGEN and RGEN are installed in the glareshield. The lights will be illuminated if a generator is not functioning or has failed.

(3) Two inverter warning lights (red) labeled PRI INV and SEC INV are installed in the glareshield. If an inverter fails, its respective warning light will illuminate. The optional auxiliary inverter incorporates a caution light (amber) labeled AUX INV.

GENERAL - MAINTENANCE PRACTICES

1. INSPECTION/CHECK

NOTE: Inspection and/or checking of the electrical system and components should be conducted whenever a closed area is opened for maintenance purposes. In addition, electrical system and components shall be inspected in accordance with the current intervals specified in Chapter 5.

A. Check Electrical System

(1) Check electrical wiring for damage such as chafing, fraying, and cuts.
(2) Check wire clamps and supports for security.
(3) Check that all wiring is supported clear of sharp metal edges.
(4) Check wiring for liquid impregnation.
(5) Check that terminal connections are secure and that lugs are not cracked or touching adjacent terminal or structure.
(6) Check electrical bonding jumpers for security and for frayed or broken condition.
(7) Check electrical equipment for proper installation, security of mounting, physical damage, and evidence of overheating.
(8) Check security of safety wire on electrical equipment and electrical connectors, where applicable.

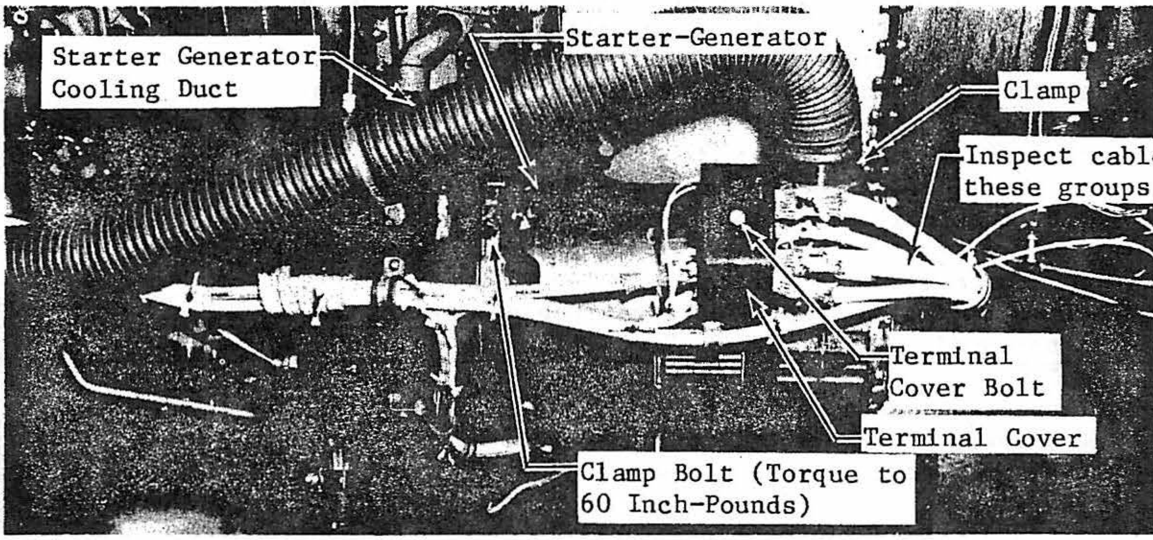
B. Inspection of No. 4 Gauge Electrical Cables (Aircraft 25-061, 25-070 thru 25-354 not modified per AMK 82-3, "Replacement of Number 4 Gage Electrical Cables.")

(1) Inspect No. 4 Gauge Electrical Cables (See figure 201.)
 (a) Lower or remove tailcone access door.
 (b) Lower engine nacelle cowls
 (c) Inspect for cracks in electrical cables from starter-generator to batteries. Inspect particularly the outside circumference at each of the cable bends.

 NOTE: ° Only the Number 4 electrical cables with extruded Teflon insulated wire need be inspected.

 ° Cracks are most likely to occur in the outside circumference of the cable bend where cables are routed under engine. Insulation cracks will appear as black hairlines running parallel with the cable.

 (d) If cracks are found, comply with AMK 82-3, "Replacement of Number 4 Gauge Electrical Cables" not later than the next 150 hour inspection. If cracks are not found, comply with this inspection until such time that cables are replaced.
 (e) Install engine nacelle cowls and secure tailcone access door.



Inspection of No. 4 Gauge Electrical Cables
Figure 201

AC GENERATION - DESCRIPTION AND OPERATION

1. DESCRIPTION
   A. An AC generation system is installed to provide 115 vac, 400 Hz to various using systems.
   B. Two autotransformers provide 26 vac for various using systems.
      (1) The autotransformers reduce 115 vac to 26 vac to power various electrical and electronic equipment. The transformers are installed on the RH and LH sides of the cockpit, aft of frame 8.
   C. The system consists of two static inverters, a paralleling control box, two overload sensors, two power relays, an AC voltmeter and switch, two control switches and two red warning lights.
      (1) The inverters are installed on the RH side of the tailcone between frames 29 and 30.
      (2) An optional auxiliary inverter is installed in the tailcone equipment section between frames 29 and 30 adjacent to the emergency batteries.
      (3) Each inverter is capable of delivering 115 vac, 400 Hz at 1000 VA. The inverters are powered directly from their respective generator bus through an overload sensor and a power relay. Each inverter incorporates a load equalizer and a frequency synchronizer sensing circuit which connect to the paralleling control box for inverter paralleling purposes.
      (4) The auxiliary inverter is rated at 115 vac, 400 Hz at 1000 VA. The auxiliary inverter is powered directly from the Battery Charging Bus through an overload sensor and power relay. The auxiliary inverter output is applied to the R AC BUS or the L AC BUS by using the AUX INVERTER L BUS - R BUS Switch.
   D. On aircraft equipped with an optional auxiliary inverter, an inverter, overload sensor, a power relay, a system switch and an amber caution light are installed in addition to the standard components.
   E. The paralleling control box is installed on the RH side of the tailcone between frames 27 and 28.
      (1) The paralleling control box serves as the central control unit for the AC generation system. The inverters output, load equalizer, and frequency synchronizer circuits are connected to the paralleling control box. The control box senses the inputs and through its load equalizer and frequency synchronizer circuits maintains the load and frequency balance between inverters.
   F. Overload sensors are located on a panel on the aft side of the aft engine beam.
      (1) The sensor consists basically of a 60-ampere thermal circuit breaker mechanically connected to a set of switch contacts. During normal operation, the primary inverter power relay is energized through the closed contacts of the overload control sensor and the PRI INV circuit breaker. Operation of the secondary and auxiliary inverters are the same as the primary. If an overload condition exists, the thermal breaker contained within the sensor positions the switch contacts to remove power from the power relay and grounds the applicable AC bus circuit breaker. The overload sensor will reset when the thermal

breaker has cooled and the switch contacts have been positioned to their original position. However, the 28 vdc required to energize the power relay will not be present due to the open AC bus circuit breaker. After the malfunction has been corrected and the applicable AC bus relay is energized, generator output voltage is connected to the main power buses.

G. The power relays are located on a panel on the aft side of the aft engine beam.

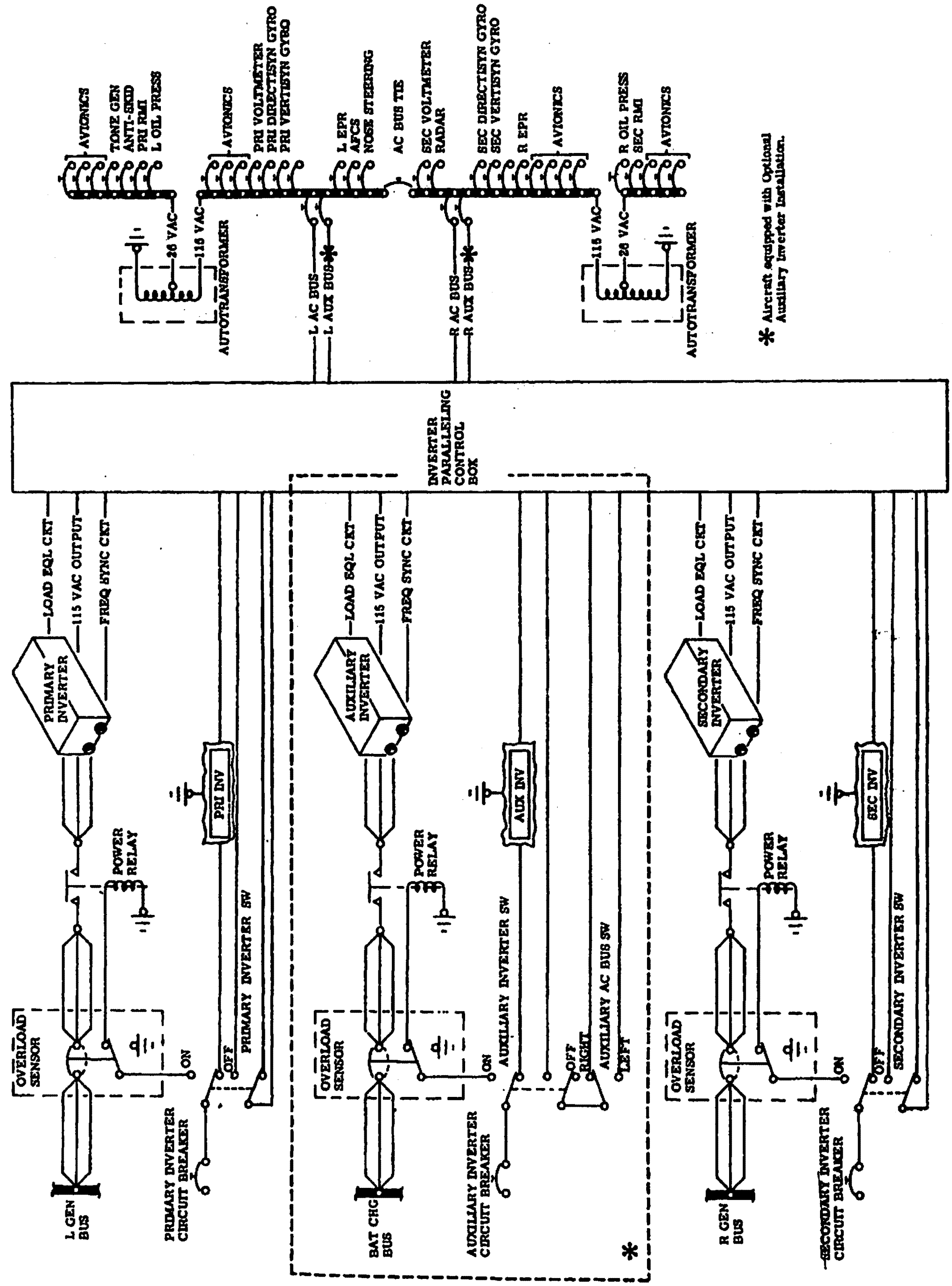
(1) The power relays are standard AN relays used to supply power to the inverters.

H. The voltmeter is installed on the instrument panel. The voltmeter indicates the output voltage supplied to the AC distribution buses. The range of the voltmeter is 0 to 150 volts. Refer to Chapter 39 for instrument markings.

2. OPERATION

A. When the Primary Inverter Switch is set to ON, a power circuit is completed through a closed switch within the overload sensor to energize the inverter power relay. With the inverter power relay energized, 28 vdc is applied to the inverter from its applicable generator bus. Inverter output is applied to its respective bus through the paralleling control box. Operation of the secondary inverter is the same. Both AC buses are connected through an AC bus tie circuit breaker.

B. Operation of the optional auxiliary inverter is the same as the primary and secondary inverter operation, except that the auxiliary inverters output is applied to either the R or L AC bus through the Auxiliary Inverter Bus Switch. In case of primary or secondary inverter failure, the Auxiliary Inverter ON-OFF Switch must be set to ON and the Auxiliary Inverter Bus Switch must be set to either R BUS or L BUS depending on which inverter has failed.



AC Power System Electrical Schematic
Figure 1

AC GENERATION - TROUBLE SHOOTING

1. TROUBLE SHOOTING

A. Tools and Equipment Required for Trouble Shooting

NOTE: Equivalent substitutes may be used in lieu of the following items.

NAME	PART NUMBER	MANUFACTURER	USE
Voltmeter	3430A	Hewlett Packard	To check voltages and continuity of circuits.
Voltmeter	260	Simpson	To check voltages and continuity of circuits.

B. See figure 101 for trouble shooting procedures.

(1) This figure is provided as an aid in isolating troubles in the AC power system.

TROUBLE:

EITHER INVERTER WARNING LIGHT ON WITH LOSS OF 115 VAC OUTPUT.

Note

The pilot squawk may also include AC BUS TIE circuit breaker open.

PROBABLE CAUSES

LOSS OF REMOTE INVERTER TURN-ON CAPABILITY (AIRCRAFT WIRING OR PARALLELING CONTROL BOX MALFUNCTION).

LOSS OF 28 VDC INPUT VOLTAGE (AIRCRAFT WIRING).

LOSS OF INVERTER OUTPUT POWER.

LOSS OF 28 VDC INPUT VOLTAGE (AIRCRAFT WIRING). Perform the following test:

1. Connect a voltmeter between terminal A2 of inverter power relay and aircraft ground.
2. Set Battery switch(es) to ON.
3. Check for 28 vdc between A2 and ground.

IF

NOT OK, check wiring from relay to generator bus. Replace or repair wiring as necessary. Check overload sensor for open circuit.

OK, check wiring inverter power relay to power source. Replace or repair wiring as necessary.

AC Generation System Trouble Shooting
Figure 101 (Sheet 1 of 3)

LOSS OF REMOTE INVERTER TURN-ON CAPABILITY (AIRCRAFT WIRING OR PARALLELING CONTROL BOX MALFUNCTION). Perform the following check:

1. Remove power input plug from affected inverter.
2. Connect a Simpson 260 voltmeter or equivalent across any of the power input pins (1, 2, or 3) and the ground pins (4, 5, or 6) of aircraft plug.
3. Set Battery and applicable Inverter switches to ON. Check for 28 vdc input.

IF

OK, perform the following check:

1. Disconnect power output plug (P41, P42, or P62) from applicable inverter.
2. Set applicable Inverter switch to ON.
3. Place positive lead of ohmmeter on pin 5 (Remote ON/OFF HOT) and negative lead on pin 6 (Remote ON/OFF RET) of aircraft plug. Set ohmmeter on RX100 scale and check for continuity between pins 5 and 6.

IF

NOT OK, check inverter power circuit or control circuit. (Refer to "Loss of 28 Vdc Input Voltage (Aircraft Wiring).")

OK, perform a check of all inverter interconnect wiring, paying particular attention to all grounds, pin connections, burnt or broken wires.
IF

NOT OK, repair or replace wiring as necessary.

OK, proceed as follows:

1. Tag affected inverter. Remove and install affected inverter in place of inverter that functions properly. IF

NOT OK (inverter does not function properly), the malfunction has occurred within the inverter. Remove and replace inverter.

OK (inverter functions properly), the malfunction has occurred within the paralleling control box. Remove and replace paralleling control box.

NOT OK, continuity check shows an open circuit. Perform the following check:

1. Disconnect applicable paralleling control box output plug (P212 or P213).
2. With applicable inverter switch ON, check continuity of circuit between pins 1 and 2 of aircraft plug. Place positive lead on pin 2 and negative load on pin 1. IF

NOT OK, circuit shows an open circuit. Check wiring from paralleling control box to inverter switch.

OK, the circuit is open within the paralleling control box. Remove and replace paralleling control box.

AC Generation System Trouble Shooting
Figure 101 (Sheet 2 of 3)

LOSS OF INVERTER OUTPUT POWER. Proceed as follows:

1. Remove affected inverter and install in place of the inverter that functions properly. Do not install good inverter. IF

NOT OK (inverter does not function properly), remove and replace inverter.

OK (inverter functions properly), remove and replace paralleling control box.

TROUBLE:

EITHER INVERTER WARNING LIGHT ON 115 VAC OUTPUT NORMAL (10 VA MINIMUM LOAD APPLIED TO INVERTER OUTPUT). Perform the following check:

1. Disconnect applicable plug from warning light control box (P529 for primary inverter or P530 for secondary and auxiliary inverter).
2. Set affected inverter switch to ON.
3. Check for 28 vdc at connector. IF

NOT OK (28 vdc absent), check circuit from warning light control box to paralleling control box. IF

OK, remove and replace paralleling control box.

NOT OK, repair or replace wiring as required.

OK (28 vdc present), refer to applicable wiring diagram and check warning light control box circuitry.

AC Generation System Trouble Shooting
Figure 101 (Sheet 3 of 3)

AC GENERATION - MAINTENANCE PRACTICES

1. INSPECTION/CHECK

NOTE: It is recommended that the AC power system be operationally checked after maintenance of any type is performed in the tailcone area of the aircraft and in accordance with the current inspection intervals specified in Chapter 5.

A. Operational Check (Two Inverter System)

(1) Set Battery Switches ON and pull AC Bus Tie circuit breaker.
(2) Set Primary Inverter Switch to ON and AC Bus Switch to PRI. Check voltage reading within green arc.
(3) Reset AC BUS TIE circuit breaker and set AC Bus Switch to SEC. Check voltage reading within green arc.
(4) Set Primary Inverter Switch to OFF and pull AC BUS TIE circuit breaker.
(5) Set Secondary Inverter Switch to ON and check voltage reading within green arc.
(6) Reset AC BUS TIE circuit breaker and set AC Bus Switch to PRI. Check voltage reading within green arc.
(7) In the following steps check the inverters for proper paralleling. If an inverter will not parallel, its applicable warning light will illuminate.
  (a) Set Primary Inverter Switch to ON.
  (b) Set AC Bus Switch to PRI then to SEC; voltage reading within green arc.
(8) Set Primary, Secondary, and Battery Switches to OFF.

B. Operational Check (Three Inverter System)

(1) Set Battery Switches to ON and pull AC BUS TIE circuit breaker.
(2) Set Primary Inverter Switch to ON and AC Bus Switch to PRI. Check voltage reading within green arc.
(3) Reset AC BUS TIE circuit breaker and set AC Bus Switch to SEC. Check voltage reading within green arc.
(4) Set Primary Inverter Switch to OFF and pull AC BUS TIE circuit breaker.
(5) Set Secondary Inverter Switch to ON and check voltage reading within green arc.
(6) Reset AC BUS TIE circuit breaker and set Bus Switch to PRI. Check voltage reading within green arc.
(7) Set Secondary Inverter Switch to OFF and Auxiliary Inverter Switch to ON.
(8) Pull AC BUS TIE circuit breaker and set Auxiliary Inverter Bus Switch to L BUS. Check voltage reading within green arc.
(9) Reset AC BUS TIE circuit breaker and set AC Bus Switch to SEC. Check voltage reading within green arc.
(10) Pull AC BUS TIE circuit breaker. Set Auxiliary Inverter Bus Switch to R BUS. Check voltage reading within green arc.

(11) Reset AC BUS TIE circuit breaker and set AC Bus Switch to PRI. Check voltage reading within green arc.

(12) In the following steps, check the inverters for proper paralleling. If an inverter will not parallel, its applicable Warning light will illuminate.

 (a) Set Primary and Secondary Inverter Switches to ON. Check voltage reading within green arc and assure that inverter warning lights are not illuminated.

 (b) Pull AC BUS TIE circuit breaker and set Auxiliary Inverter Bus Switch to L BUS.

 (c) Set Auxiliary Inverter Switch to ON. Check voltage reading within green arc and assure inverter warning lights are not illuminated.

 (d) Set Auxiliary Inverter Bus Switch to R BUS and AC Bus Switch to SEC. Check voltage reading within green arc.

 (e) Reset AC BUS TIE circuit breaker and set Auxiliary Inverter Bus Switch to L BUS and the R BUS. In each position, check voltage reading within green arc and assure warning lights are not illuminated.

(13) Set Primary, Secondary, Auxiliary and Battery Switch(es) to OFF.

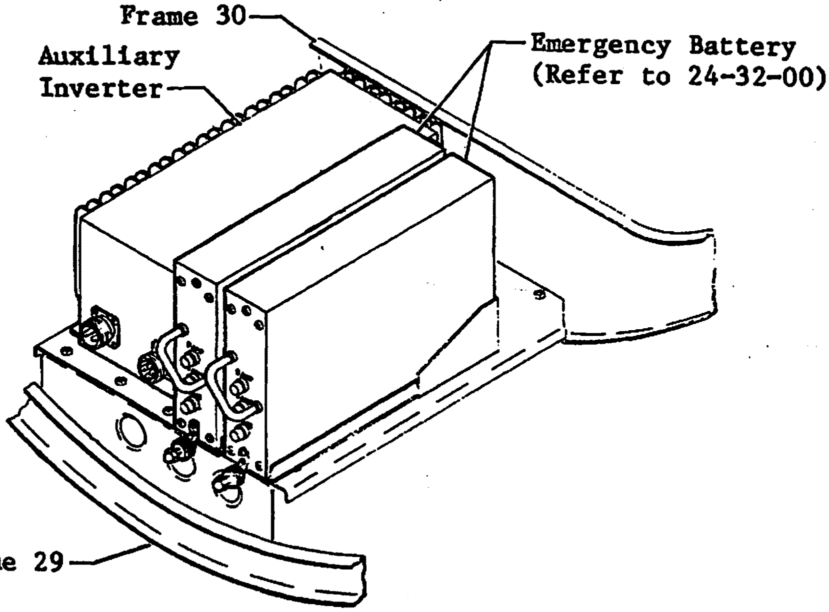
INVERTER - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

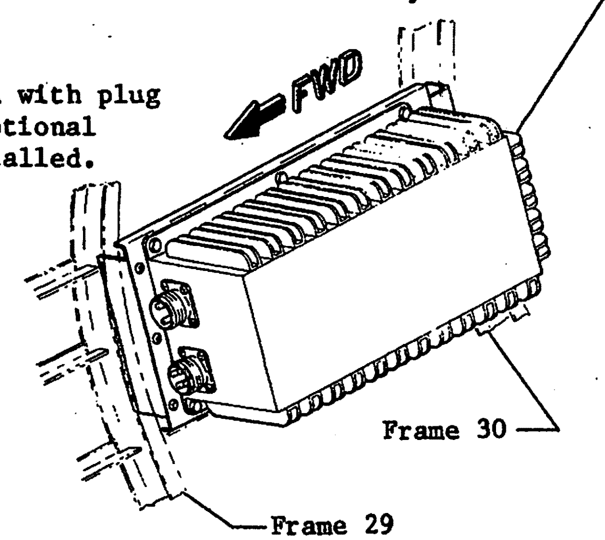
NOTE: Removal and installation procedures for both inverters are identical.

A. Remove Inverter (See figure 201.)
(1) Lower tailcone access door.
(2) Disconnect both aircraft batteries.
(3) Disconnect electrical plugs from inverters.
(4) Remove attaching parts and inverter from aircraft.

B. Install Inverter (See figure 201.)
(1) Install inverter and secure with attaching parts.
(2) Connect electrical plugs to inverters.
(3) Connect aircraft batteries.
(4) Perform operational check of AC power system if any system component was replaced.



AUXILIARY INVERTER INSTALLATION



* Inverter installed with plug on aft end when optional spare tire is installed.

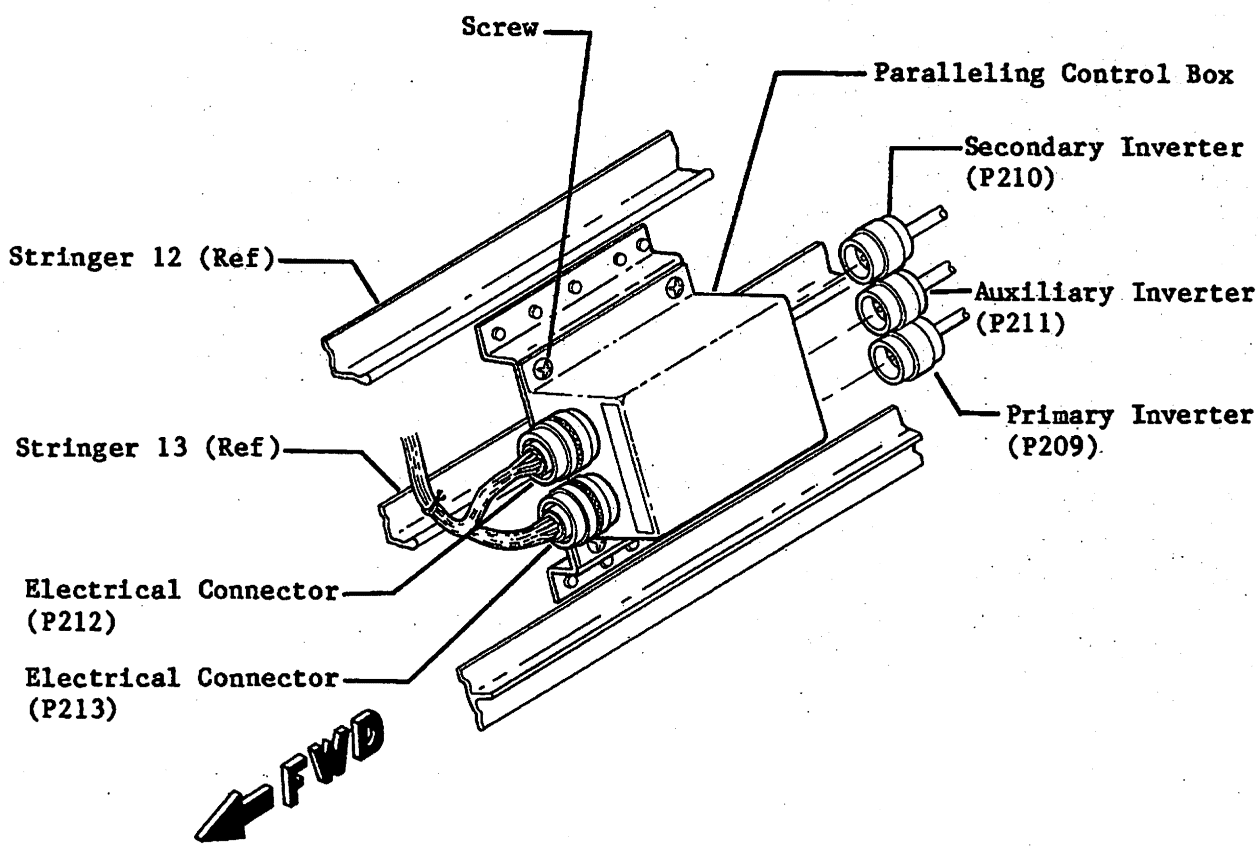
NOTE: Primary Inverter installed opposite Secondary Inverter.

PRIMARY AND SECONDARY INVERTER INSTALLATION

Primary, Secondary, and Auxiliary Inverter Installation
Figure 201

PARALLELING CONTROL BOX - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   A. **Remove Paralleling Control Box** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Disconnect electrical connectors from control box.
      (4) Remove attaching parts and control box from aircraft.
   B. **Install Paralleling Control Box** (See figure 201.)
      (1) Install control box and secure with attaching parts.
      (2) Connect electrical connectors to control box.
      (3) Connect aircraft batteries.
      (4) Perform operational check of AC power system if any system component was replaced.
      (5) Secure tailcone access door.



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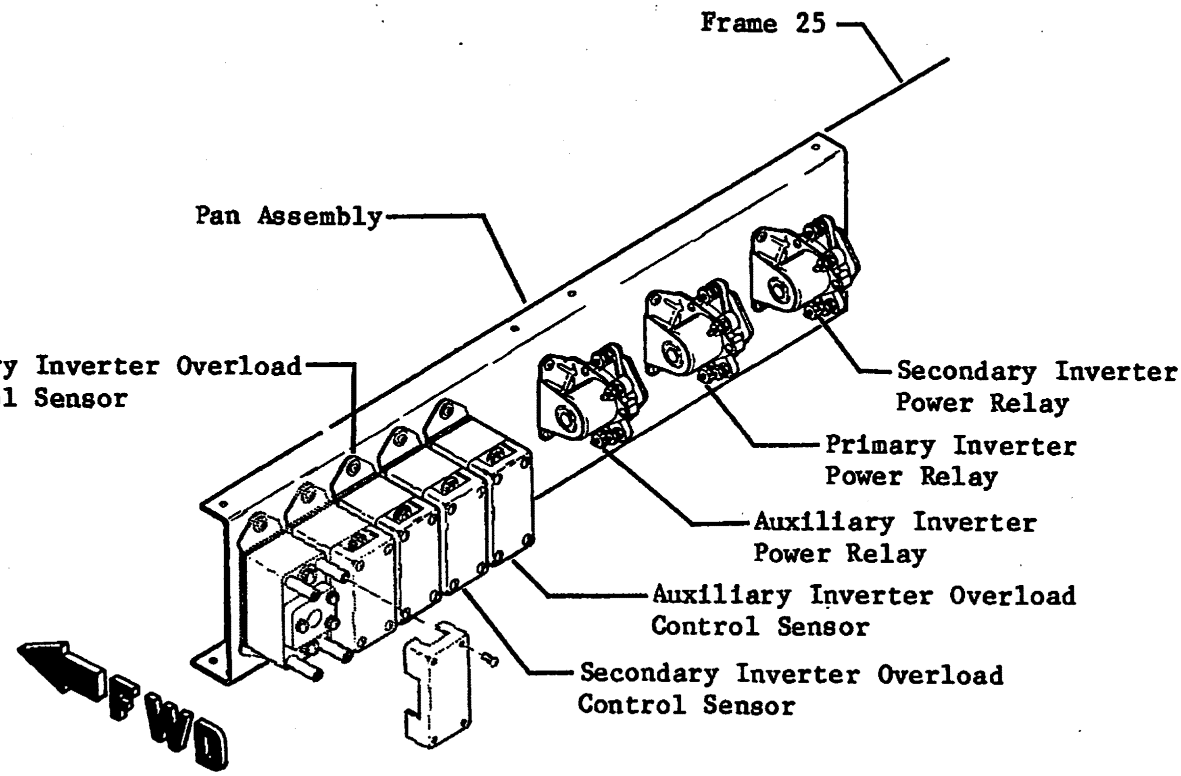
Paralleling Control Box Installation
Figure 201

AC OVERLOAD SENSOR - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   A. **Remove Overload Sensor** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Remove attaching parts and cover from sensor.
      (4) Disconnect electrical wiring.
      (5) Remove attaching parts and sensor from aircraft.
   B. **Install Overload Sensor** (See figure 201.)
      (1) Install sensor and secure with attaching parts.
      (2) Connect electrical wiring to sensor.
      (3) Install cover and secure with attaching parts.
      (4) Connect aircraft batteries.
      (5) Perform operational check of AC power system. (Refer to 24-20-00.)
      (6) Secure tailcone access door.

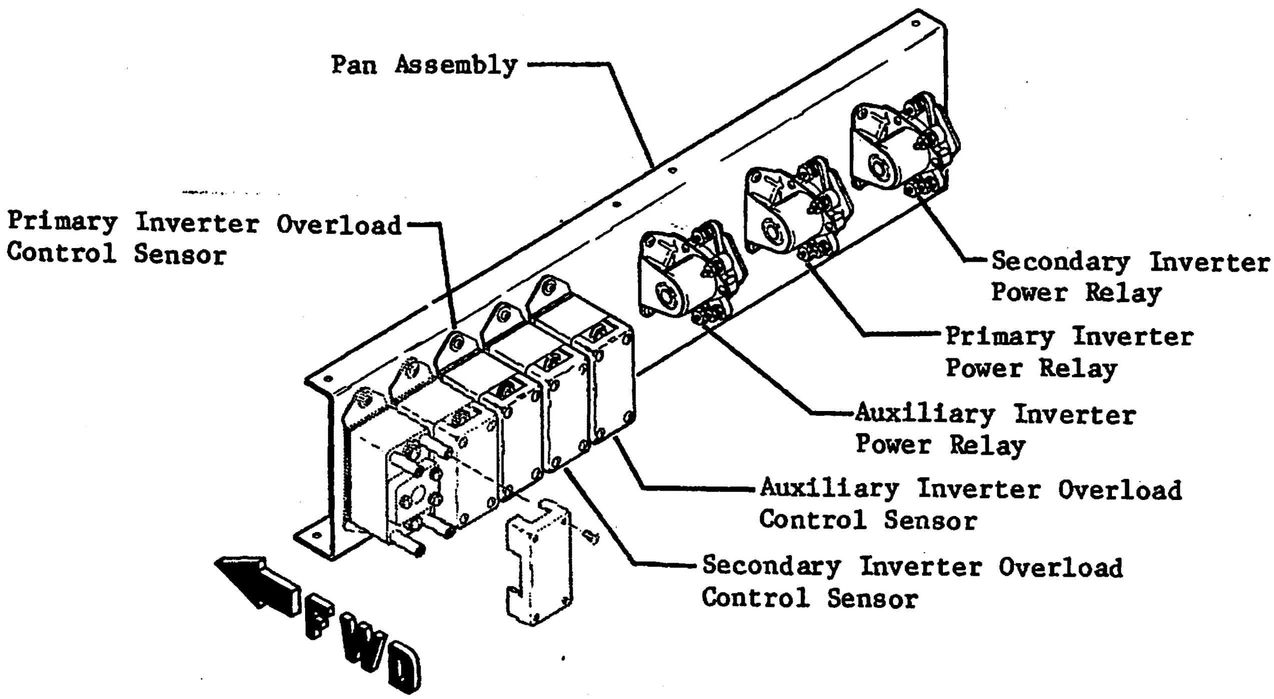


NOTE: Wiring Diagram Symbol Codes are stamped along lower edge of panel.

AC Overload Control Sensor Installation
Figure 201

INVERTER POWER RELAY - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION
   A. **Remove Power Relays** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Disconnect and tag electrical wiring.
      (4) Remove attaching parts and relay from aircraft.
   B. **Install Power Relays** (See figure 201.)
      (1) Install relay and secure with attaching parts.
      (2) Remove tags and connect electrical wiring to relay.
      (3) Connect aircraft batteries.
      (4) Perform operational check of AC power system. (Refer to 24-20-00.)
      (5) Secure tailcone access door.



Inverter Power Relay Installation
Figure 201

INDICATOR - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   A. Refer to 24-31-07 for removal and installation procedures.

DC GENERATION - GENERAL

1. DESCRIPTION

A. The aircraft primary 28 vdc requirements are supplied by two independent DC generator systems. Secondary sources of 28 vdc power are supplied by a DC battery system, an external power source, and an optional emergency DC battery system.

B. For further information on the external power system, refer to 24-40-00.

C. The DC generator system consists of a 30 volt, 400 ampere generator installed on each engine. Additional components are a generator control box, two voltage regulators, two reverse current diodes, two DC ammeters, a DC voltmeter, two system switches, two generator reset switches, and two generator caution lights. The DC generator system switches and two generator reset switches are located on the instrument panel. Two amber generator caution lights are located on the glareshield. The reverse current diodes are located on the aft side of the engine beam in the tailcone.

D. On aircraft equipped with nickel-cadmium batteries, the DC battery system consists of two 22 ampere-hour or two 40 ampere-hour, 24-volt nickel-cadmium batteries, a battery temperature indicating system, a battery overheat warning system, and two system switches. The system also utilizes the generator control box to connect the batteries to the aircraft electrical system.

E. On aircraft equipped with lead-acid batteries, the DC battery system consists of two 45 ampere-hour, 24-volt, lead-acid batteries and two system switches. The system also utilizes the generator control box to connect the batteries to the aircraft electrical system.

F. The emergency battery system consists of single or dual batteries rated at 25 vdc and 115 vac (+7, -10%) at 400 Hz, system switches and indicator light(s).

DC GENERATION SYSTEM - DESCRIPTION AND OPERATION

1. DESCRIPTION
   A. The DC generator system consists of a 30 volt, 400 ampere starter-generator installed on each engine, a generator control box, two voltage regulators, two reverse current diodes, two DC ammeters, a DC voltmeter, two system switches, two generator reset switches and two generator caution lights.
      (1) A starter-generator, located on each engine, is used as a starter until the engine is accelerated to operational speed and then it acts as a generator supplying regulated 28 vdc power for the aircraft electrical systems. Starter-generator cooling is provided by flexible ducting from the nacelle nose cap air intake.
      (2) Two solid state voltage regulators are installed in the tailcone. The regulators maintain a constant output voltage under varying engine speeds and load conditions by automatically adjusting the generator field current. The voltage regulators are preset at the factory following a complete functional test. The design of the regulators precludes voltage drift with time.
      (3) The overload control sensors are located in the tailcone. The overload control sensor consists basically of a 70-ampere thermal circuit breaker mechanically connected to a set of switch contacts. The overload sensors are electrically connected between the battery charging bus and the main bus of each circuit breaker panel.
         (a) During normal operation, the left main bus power relay is energized through the closed contacts of the overload control sensor and the L MAIN BUS circuit breaker. Operation for the right main bus power relay is the same as the left main bus power relay. If an overload condition exists, the thermal breaker contained within the sensor will position the switch contacts to remove power from the power relay and ground the applicable bus circuit breaker. The overload sensor will reset when the thermal breaker has cooled and the switch contacts have been returned to their original position. However, the 28 vdc required to energize the power relay will not be present due to the open bus circuit breaker. After the malfunction has been corrected and the applicable bus circuit breaker reset, the power relay will energize. When the power relay is energized, generator output voltage is connected to the main power buses.
      (4) Two reverse current diodes are installed on the aft side of the engine beam (one for each generator system). The reverse current diodes are electrically connected between the starter-generator and the battery charging buses, isolating the generator systems from each other.
      (5) The ammeters are installed on the instrument panel. The ammeters indicate the amperage output of their respective generator. The range of the ammeters are 0 to 400 amperes. Refer to Chapter 31 for instrument markings.

(6) The voltmeter is installed on the instrument panel. The voltmeter indicates generator output voltage supplied to the distribution buses. The range of the voltmeter is 0 to 30 volts. Refer to Chapter 31 for instrument markings.

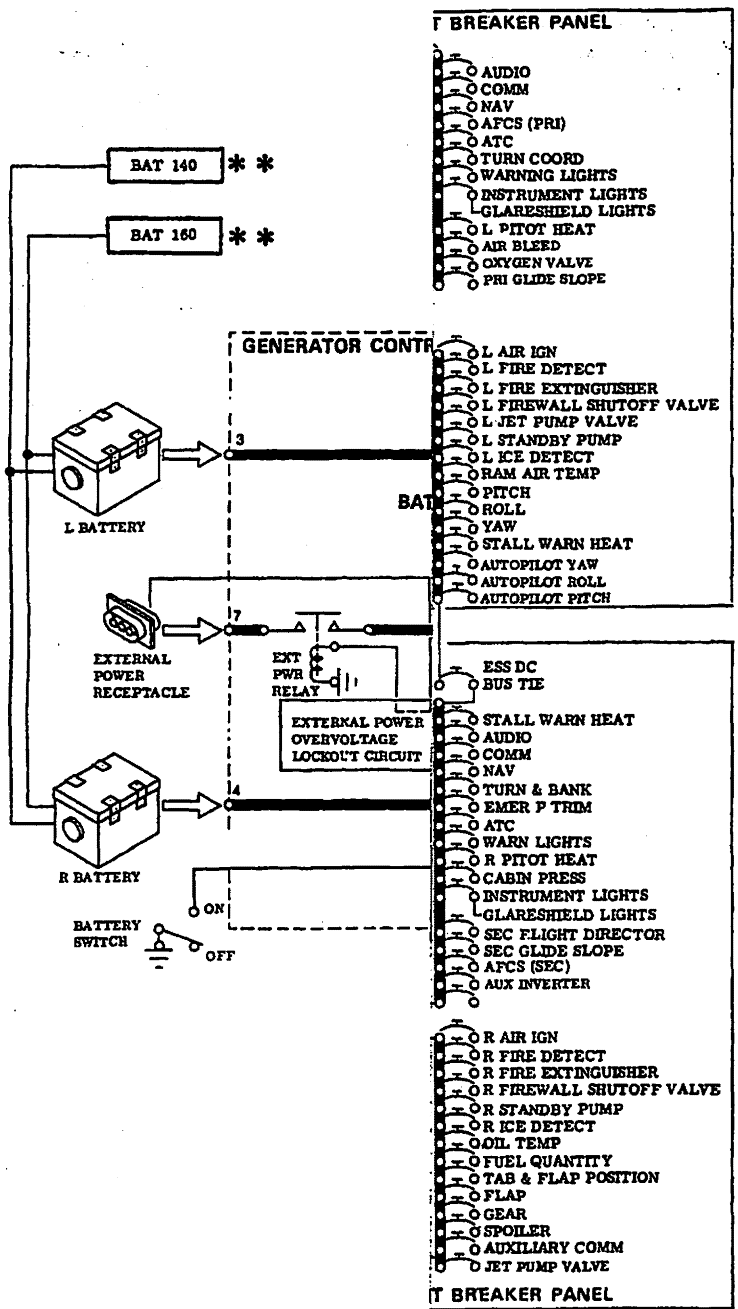
B. On French certified aircraft, each starter-generator is equipped with an overheat detector element which senses the cooling air as it exhausts from the starter-generator. The overheat detector element provides a visual indication to the crew by illuminating the GEN OV HT (red) warning light on the glareshield. The starter-generator GEN OV HT warning light circuit is checked by utilizing the GEN DUCT TEST on the pedestal switch panel. The generator overheat control units (E159 and E160) are installed on the RH electrical equipment tray adjacent to the fuel control box. The starter-generator should be allowed to cool to ambient after three consecutive start attempts.

2. **OPERATION** (See figure 1.)

A. When the Starter-Generator Switch is set to START, the applicable starter control relay (K1 or K2) will energize, completing the circuit from the fuel motive flow valve to the applicable starter relay (K13 or K14). When the motive flow valve closes, 28 vdc energizes the starter relay. With the starter relay energized, battery power is applied to the starter windings of the starter-generator. When engine speed exceeds idle rpm and the Starter-Generator Switch is set to GEN, the starter control relay is deenergized and the generation circuit relays (K3, K5, and K7 for left and K4, K6, and K8 for right) are energized. When the generation circuit relays are energized, three circuits are completed: (1) a circuit for 28 vdc to the voltage regulator (REG POWER), (2) a circuit for power to the refrigerant compressor motor relay, and (3) a circuit for the voltage regulator equalizer bus. Engine rpm acceleration raises generator voltage output to the voltage regulator setting. Operation of the opposite starter-generator is the same except when the Starter-Generator Switch is set to START, both starter relays are energized, allowing the output of the operating generator to assist in starting the opposite engine. When the opposite engine speed exceeds idle rpm and the Starter-Generator Switch is set to GEN, the opposite generation circuit relays are energized. With both generation circuits energized, both voltage regulator equalizer buses are connected. The voltage regulator equalizer buses sense any change in the applicable generator load or voltage drop across the reverse current diode. The voltage regulator will then automatically adjust its respective generator field until a balanced condition exists. If a generator should exceed an overvoltage condition (32.0 ±1 volts), a circuit in the voltage regulator completes a ground circuit to the generation circuit overvoltage relay (K9 and K10), energizing its applcable relay. With the overvoltage relay energized, ground is removed from the generator field. Setting the applicable Generator Reset Switch to RESET will reapply a ground to the generator field providing the overvoltage condition has been corrected.

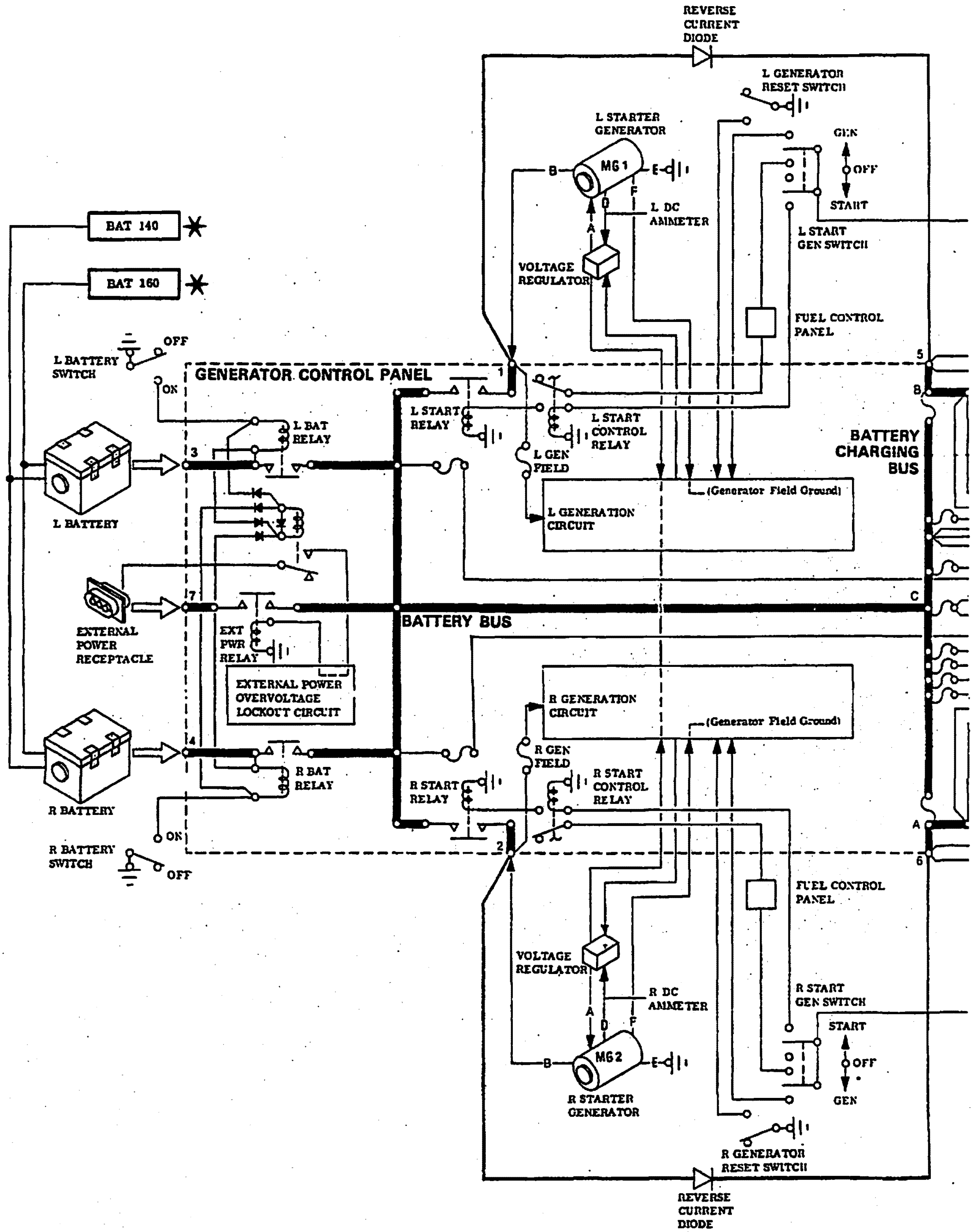
B. Generator Caution Light Operation (See figure 2.)

(1) The generator caution lights (LH) are lighted when a ground is applied to the lights through a set of contacts of deenergized relay (K7). When the Starter-Generator Switch is set to GEN, relay (K5) is energized and applies generator output voltage directly to relay (K7) and opens the generator ground circuit. Operation of the RH generator caution light is identical to the LH except relays (K6 and K8) are used.

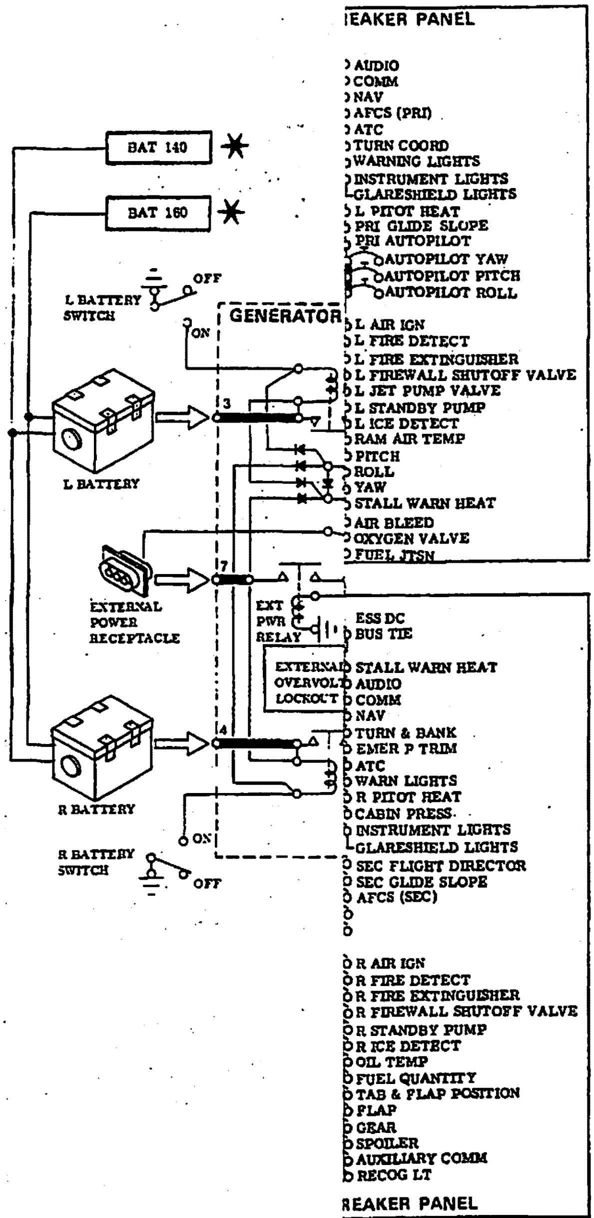


91 thru 25-102, separate circuit
tomatic cabin heat.

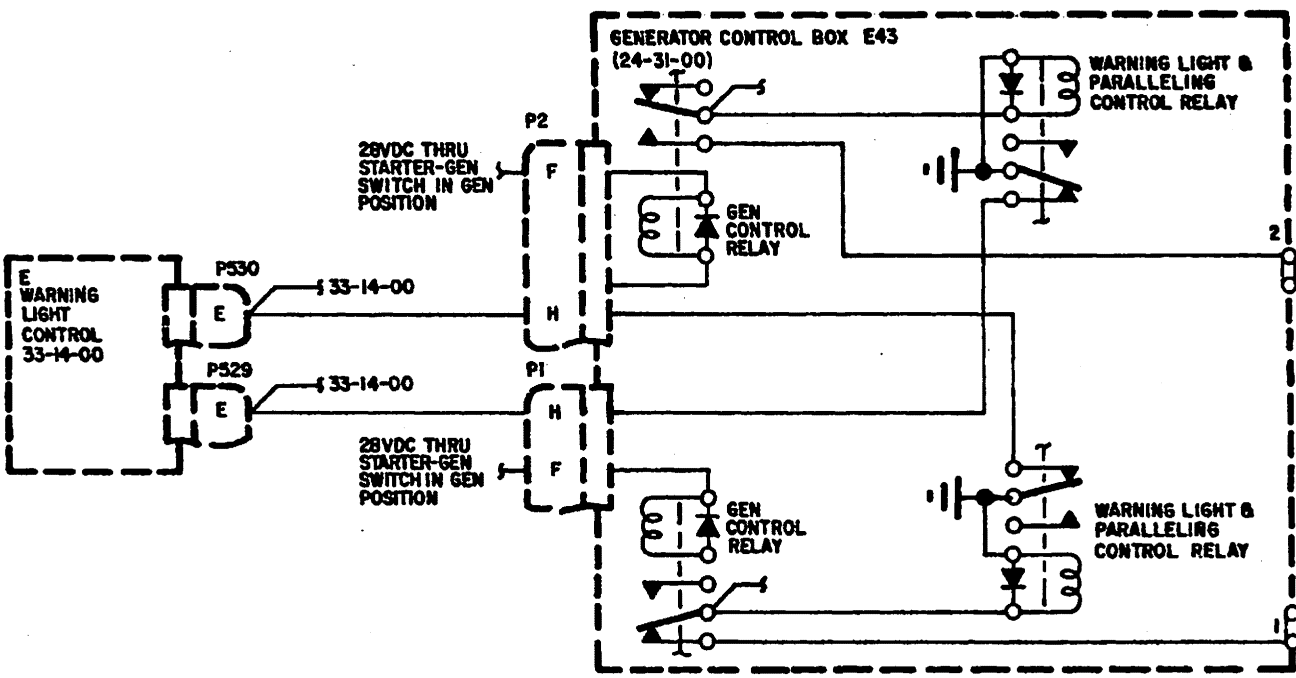
DC Generation
Figu
ective only on aircraft equipped with



DC Generation Electrical Control Schematic
Figure 1 (Sheet 2 of 3)



DC Generation Elective only on aircraft equipped with
Figure 1 (



Generator Caution Light Electrical Control Schematic
Figure 2

DC GENERATION SYSTEM - TROUBLE SHOOTING

1. **TROUBLE SHOOTING**
   A. Tools and equipment required for Trouble Shooting.

   NOTE: Equivalent substitutes may be used in lieu of the following items.

NAME	PART NUMBER	MANUFACTURER	USE
Voltmeter	Model 3430A	Hewlett Packard	Checking electrical circuits.
Voltmeter	Model 260	Simpson	Checking electrical circuits.

   B. Refer to figure 201 for Trouble Shooting procedure.
      (1) This figure is provided as an aid in isolating trouble in the starter-generator system.

BATTERY SWITCH(ES) SET TO ON AND STARTER GENERATOR SWITCH SET TO START, ENGINE WILL NOT START (MOTOR).

Note

IF ENGINE WILL MOTOR BUT WILL NOT START, REFER TO GENERAL ELECTRIC MANUAL SEI-186 FOR TROUBLE SHOOTING PROCEDURE.

THE FOLLOWING IS APPLICABLE FOR EITHER ENGINE.

MONITOR DC VOLTMETER, ASSURING THAT DC VOLTMETER CIRCUIT BREAKER IS DEPRESSED. A 28 VDC READING INDICATES THAT THE BATTERY RELAY IS FUNCTIONING PROPERLY AND 28 VDC IS PRESENT AT THE RH AND LH DC LOAD BUSES.

CHECK FOR LOOSE OR BROKEN WIRES AT GENERATOR CONTROL BOX AND STARTER GENERATOR CONNECTIONS.

CHECK THAT ALL CIRCUIT BREAKERS ARE PROPERLY POSITIONED. IF

NOT OK, RESET CIRCUIT BREAKERS. TIGHTEN OR REPLACE WIRING AS REQUIRED.

OK, SET L STARTER GENERATOR SWITCH OFF AND PULL L JET PUMP CIRCUIT BREAKER. REMOVE PLUG FROM L MOTIVE FLOW VALVE. RESET L JET PUMP CIRCUIT BREAKER AND SET L STARTER GENERATOR SWITCH TO START. CHECK 28 VDC BETWEEN PIN B AND AIRCRAFT GROUND. IF

NOT OK, VERIFY CIRCUIT FROM L MOTIVE FLOW VALVE TO LEFT JET PUMP CIRCUIT BREAKER. IF

OK, VERIFY CIRCUIT FROM L STARTER CONTROL RELAY TO L STARTER GENERATOR SWITCH. REPLACE OR REPAIR CIRCUIT AS REQUIRED.

NOT OK, REPLACE OR REPAIR CIRCUIT AS REQUIRED.

OK (28 VDC PRESENT), RECONNECT PLUG AND CHECK FOR 28 VDC BETWEEN TERMINAL A2 OF L START CONTROL RELAY AND AIRCRAFT GROUND. IF

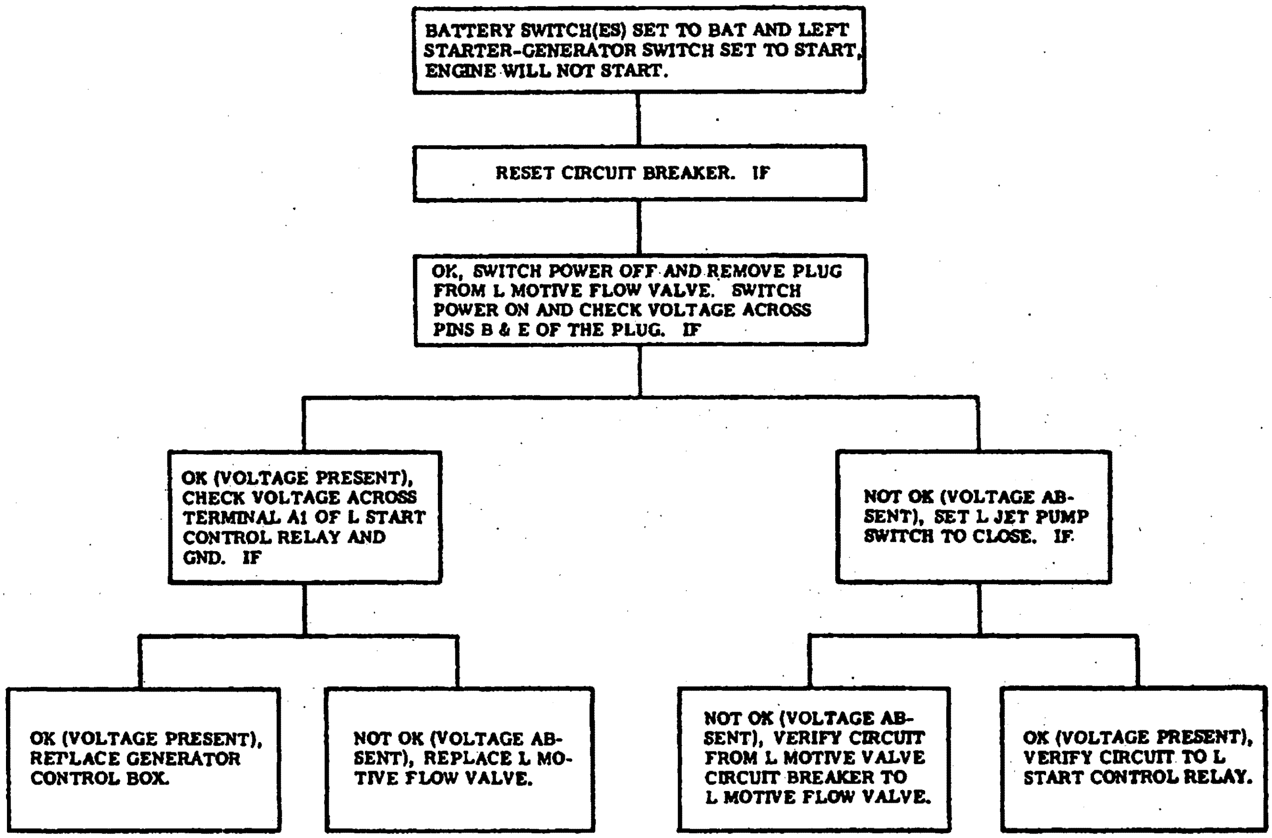
NOT OK (28 VDC ABSENT), REPLACE MOTIVE FLOW VALVE.

OK (28 VDC PRESENT), CHECK VOLTAGE BETWEEN STARTER GENERATOR BUS AND GROUND. IF

NOT OK (28 VDC ABSENT), REPLACE GENERATOR CONTROL BOX.

OK (28 VDC PRESENT), REPLACE STARTER GENERATOR.

Starter-Generator Trouble Shooting
Figure 101 (Sheet 1 of 2)



Starter-Generator Trouble Shooting
Figure 101 (Sheet 2 of 2)

DC GENERATION SYSTEM - MAINTENANCE PRACTICES

1. **INSPECTION/CHECK**

   A. It is recommended that the power distribution system be operationally checked for proper operation after maintenance of any type is performed in the tailcone area of the aircraft and in accordance with the current inspection intervals specified in Chapter 5. On aircraft equipped with dual battery switches, perform this operational check using the LH Battery Switch; then repeat check using RH Battery Switch. This operational check will also check the integrity of the battery relays. Perform operational check as follows:

   (1) **Right and Left Essential Buses and Bus Tie**

   (a) Set Battery Switch(es) to ON and pull ESS DC BUS TIE circuit breaker. This will isolate the L ESS bus and R ESS bus.

   (b) Set Left Standby Fuel Pump Switch to ON and wait until L FUEL PRESS warning light extinguishes. Pull L ESS BUS circuit breaker; the L FUEL PRESS warning light should illuminate. Set Left Standby Fuel Pump switch to OFF. Reset L ESS BUS circuit breaker.

   (c) Repeat steps (1)(b) using the right standby fuel pump, R FUEL PRESS warning light and R ESS BUS circuit breaker.

   (d) Reset ESS DC BUS TIE circuit breaker.

   (e) Set Left Standby Fuel Pump Switch to ON and wait until L FUEL PRESS warning light extinguishes. Pull L ESS BUS circuit breaker; the L FUEL PRESS warning light shall not illuminate. Set Left Standby Fuel Pump Switch to OFF. Reset L ESS BUS circuit breaker.

   (f) Repeat step (1)(e) using right standby fuel pump. R FUEL PRESS warning light and R ESS BUS circuit breaker.

   (2) **Right and Left Main Buses**

   (a) Pull MAIN BUS TIE circuit breaker. This will isolate the L MAIN BUS and R MAIN BUS.

   (b) Set Navigation Light Switch to ON and assure that lights are illuminated. Pull L MAIN BUS circuit breaker; navigation lights should extinguish. Set Navigation Light Switch to OFF and depress L MAIN BUS circuit breaker.

   (c) Set Beacon Light Switch to ON and assure that lights are illuminated. Pull R MAIN BUS circuit breaker; beacon lights shall extinguish. Set Beacon Light Switch to OFF and depress R MAIN BUS CIRCUIT BREAKER.

   (d) Depress MAIN BUS TIE circuit breaker.

   (3) **Right and Left Main Power Buses**

   (a) If above check (step 2) is satisfactory, right and left main power buses are functioning correctly.

   (4) **Right and Left Battery Buses**

   (a) Set Entry Light Switch ON. If lights are illuminated, left battery bus is functioning correctly.

   (b) Set R Stall Warning Switch to ON and move right stall vane. The right angle of attack indicator should move with changes in vane position. This indicates the right battery bus is functioning properly.

B. **Battery Switching Circuit Operational Check** (Aircraft equipped with Dual Battery Switches.)

(1) Set LH Battery and RH Battery Switches to OFF. Disconnect external power if connected.
(2) Disconnect LH battery quick disconnect. Do not disconnect RH battery quick disconnect.
(3) Set LH Battery Switch to ON; the warning lights on the glareshield shall not illuminate.
(4) Set LH Battery Switch to OFF and RH Battery Switch to ON. The warning lights on the glareshield shall illuminate.
(5) Set RH Battery Switch to OFF.
(6) Disconnect RH battery quick disconnect and connect LH battery quick disconnect.
(7) Set RH Battery Switch to ON; the warning lights on the glareshield shall not illuminate.
(8) Set RH Battery Switch to OFF and LH Battery Switch to ON. The warning lights on the glareshield shall illuminate.
(9) Set LH Battery Switch to OFF and connect RH battery quick disconnect.
(10) If the above conditions are not met, recheck wiring. Refer to applicable wiring diagram in Chapter 24-31-00 of the Wiring Manual.

C. **Generator Load and Paralleling Check**

(1) Start both engines. (Refer to Airplane Flight Manual for starting procedures.)
(2) Set Cabin Air Switch to OFF.
(3) Set engines at 70% rpm.
(4) Check ammeters for 50 to 100 amperes. If necessary, turn on equipment to obtain proper load.

NOTE: Both ammeters shall read approximately the same.

(5) Pull Main DC Bus Tie breaker and R Starter-Generator Switch to OFF.
(6) Amperage load of left generator shall be approximately double.

NOTE: This indicates the L Generator Current Limiter (FL1, 275A) is not blown.

(7) Set R Starter-Generator Switch to GEN. Ammeters shall return to the laod determined in step (4) and be approximately equal.
(8) Set L Starter-Generator Switch to OFF.
(9) Amperage load of right generator shall be approximately double.

NOTE: This indicates the R generator current limiter (FL2, 275A) is not blown.

(10) Set L Starter-Generator Switch to GEN and shut down engines.
(11) Reset Main Bus Tie Breaker.

D. Voltage Regulator Operational Check

(1) Start both aircraft engines. (Refer to Airplane Flight Manual.)

(2) Set Air Bleed Switch to OFF.

(3) Set left Starter-Generator Switch to GEN.

(4) Set left engine at 70% rpm. Using a Simpson (type 260) or Hewlett Packard (Model 3430A) voltmeter, check voltage between terminal No. 1 of generator control box and aircraft ground. On aircraft equipped with nickel-cadmium batteries, voltage shall read 29.0 (±0.3) volts. On aircraft equipped with lead-acid batteries, voltage shall read 29.3 (±0.3) volts.

(5) Set left Starter-Generator Switch to OFF and right Starter-Generator Switch to GEN.

(6) Set left engine at idle rpm and right engine at 70% rpm. Connect voltmeter between terminal No. 2 and aircraft ground. On aircraft equipped with nickel-cadmium batteries, voltage shall read 29.0 (±0.3) volts. On aircraft equipped with lead-acid batteries voltage shall read 29.3 (±0.3) volts.

(7) Disconnect voltmeter and shut down engines.

STARTER-GENERATOR - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

NOTE: The following procedure is applicable to either generator.

A. **Remove Generator** (See figure 201.)
   (1) Remove engine lower nacelle.
   (2) Remove electrical power from aircraft.
   (3) Loosen clamp and remove starter-generator cooling duct from starter-generator.
   (4) Remove terminal cover bolt and terminal cover form starter-generator. Disconnect and tag electrical wiring.
   (5) Remove lockwire from clamp bolt and loosen bolt sufficiently to allow removal of starter-generator.
   (6) Remove starter-generator from engine.
   (7) If replacement generator is to be installed, loosen and remove attaching parts and existing Q.A.D. adapter assembly and use new adapter assembly furnished with generator.

B. **Install Generator** (See figure 201.)
   (1) Before installing new Q.A.D. adapter assembly on engine, check that the three (3) screws attaching adapter plate to the adapter ring do not protrude above the mounting surface. Inspect Q.A.D. clamp for possible damage.
   (2) Check that Q.A.D. assembly slides freely over the gear box studs and fits flat against gear box.
   (3) Assure that gear box housing, generator housing, adapter plate mounting surface, Q.A.D. clamp and both splines are clean and free of foreign matter.
   (4) Install adapter assembly on gear box housing and secure with attaching parts. Torque nuts 160 to 190 inch-pounds plus drag torque.

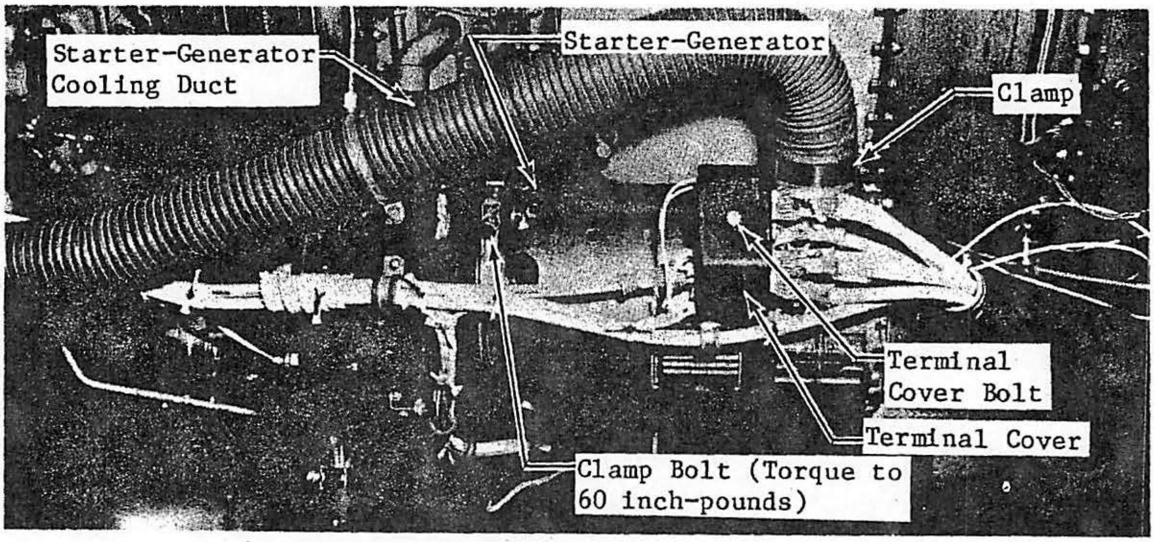
   CAUTION: ASSURE THAT BOTH SPLINES ARE CLEAN BEFORE APPLYING NEW GREASE. THIS WILL PREVENT CONTAMINATION OF NEW GREASE.

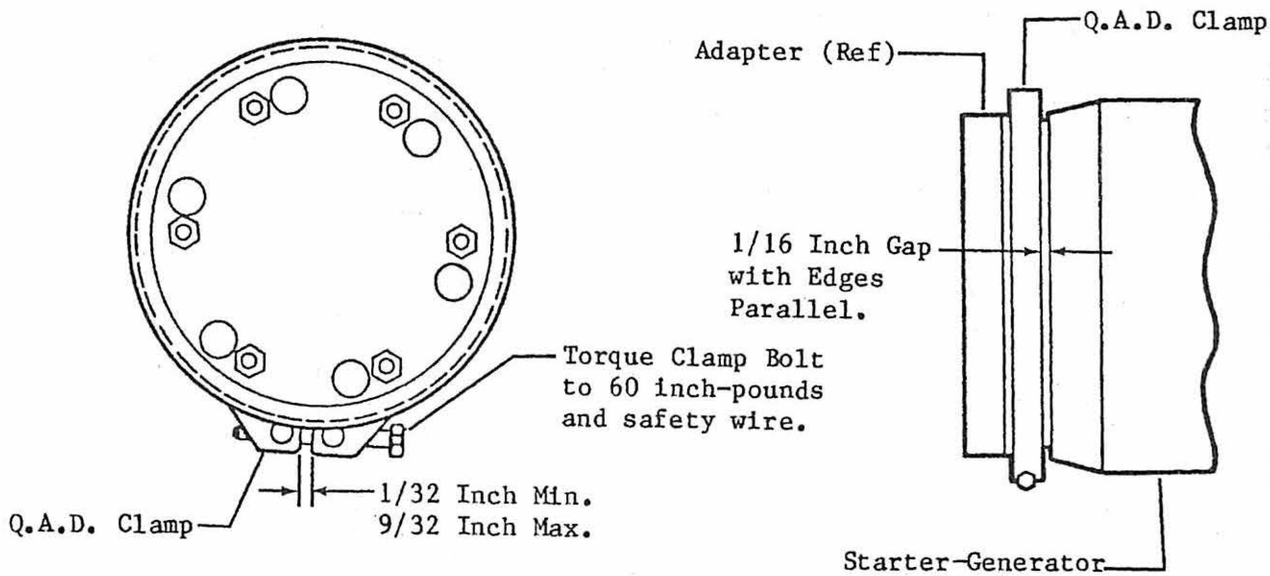
   (5) Apply Mobile Grease No. 29 or Braycote 664S (Bray Oil Co.) to male spline of starter-generator.
   (6) Apply anti-seize compound, TT-A-580 or JAN-A-669, to screw threads and V-groove of Q.A.D. clamp.
   (7) Torque clamp to 60 inch-pounds and safety wire. Check that space between ends of clamp is between 1/32 to 9/32 inch.
   (8) Connect wiring to starter-generator terminals. (Refer to applicable wiring diagram in the Wiring Manual.)
   (9) Install terminal cover and secure with attaching parts.
   (10) Install starter-generator cooling duct and secure with clamp.
   (11) Connect aircraft batteries and secure tailcone access door.
   (12) Perform operational check of starter-generator. (Refer to Airplane Flight Manual for engine starting procedures.
   (13) Install engine lower nacelle.

2. **INSPECTION/CHECK**

A. **Check Starter-Generator Brush Wear Check**

(1) Refer to Chapter 80 for starter-generator brush wear check.





Starter-Generator Installation
Figure 201

GENERATOR CONTROL PANEL - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

NOTE: Removal/Installation of generator control panel is described with the current limiter panel attached. Maintenance information on the current limiter panel is described in 24-50-02.

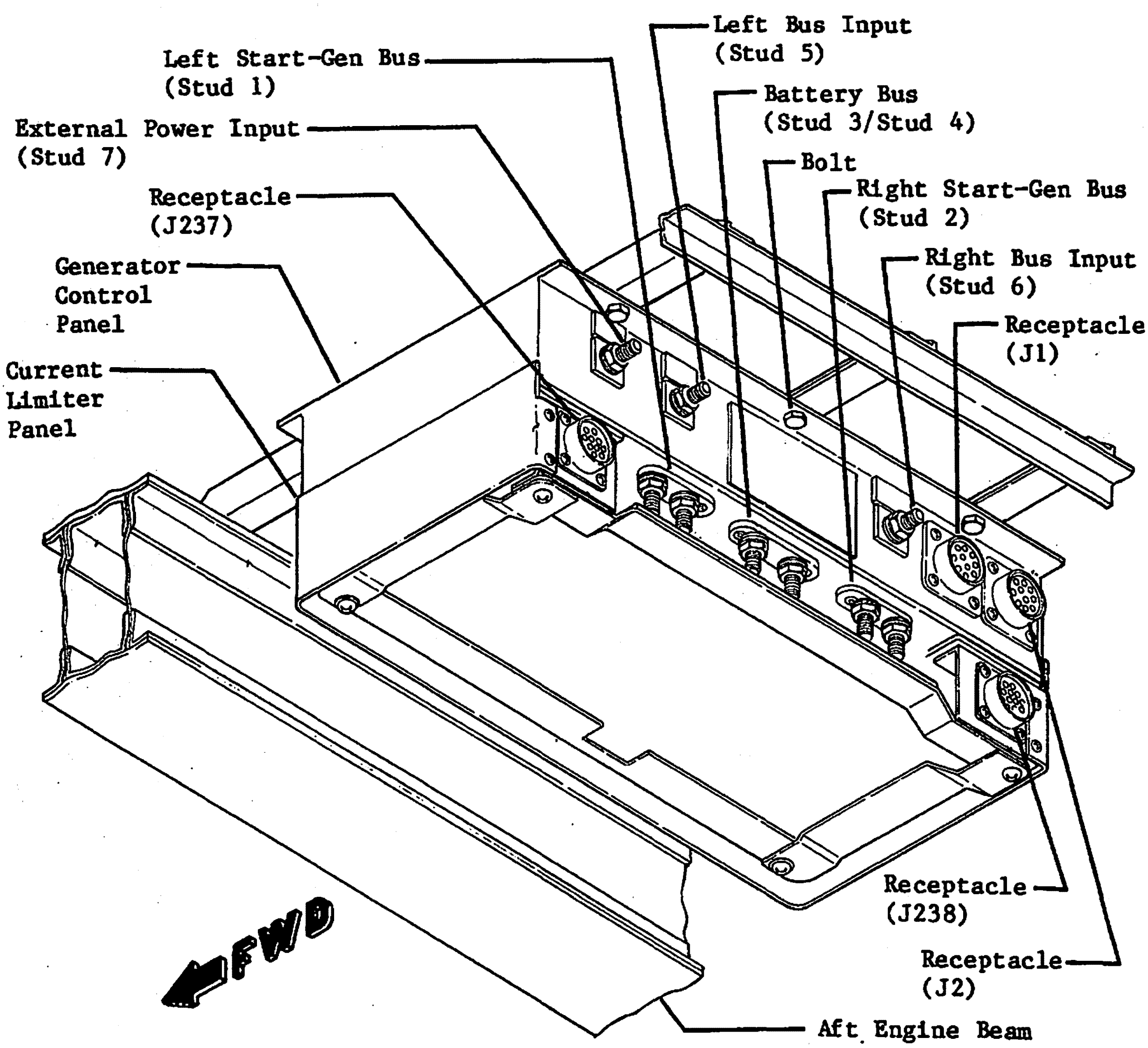
A. **Remove Generator Control Panel** (See figure 201.)
   (1) Disconnect aircraft batteries.
   (2) Disconnect electrical connectors from generator control panel and current limiter panel.
   (3) Remove current limiter panel cover assembly. (Refer to applicable procedure in 24-50-02.)
   (4) Disconnect wire terminals from current limiter panel. (Refer to applicable figure in 24-50-02.)

   NOTE: Install washer, lockwasher, and nut to the stud after wire terminals are disconnected.

   (a) Disconnect two number four gage wires from stud COM.
   (b) Disconnect left nacelle heat wire from stud securing current limiter FL33.
   (c) Disconnect right nacelle heat wire from stud securing current limiter FL34.
   (d) On Aircraft 25-076 thru 25-089, 25-091 thru 25-102, disconnect wires from fuse holder attached to current limiter panel guard assembly.
   (e) Aircraft modified per AMK81-7 or AMK81-9, Horizontal Stabilizer Trim and Autopilot Improvement; disconnect wire from battery charging bus that supplies electrical power to stabilizer actuator current limiter FL 70.
   (5) Disconnect wire terminals from generator control panel.

   NOTE: Install washer, lockwasher, and nut on stud after wire terminals are disconnected.

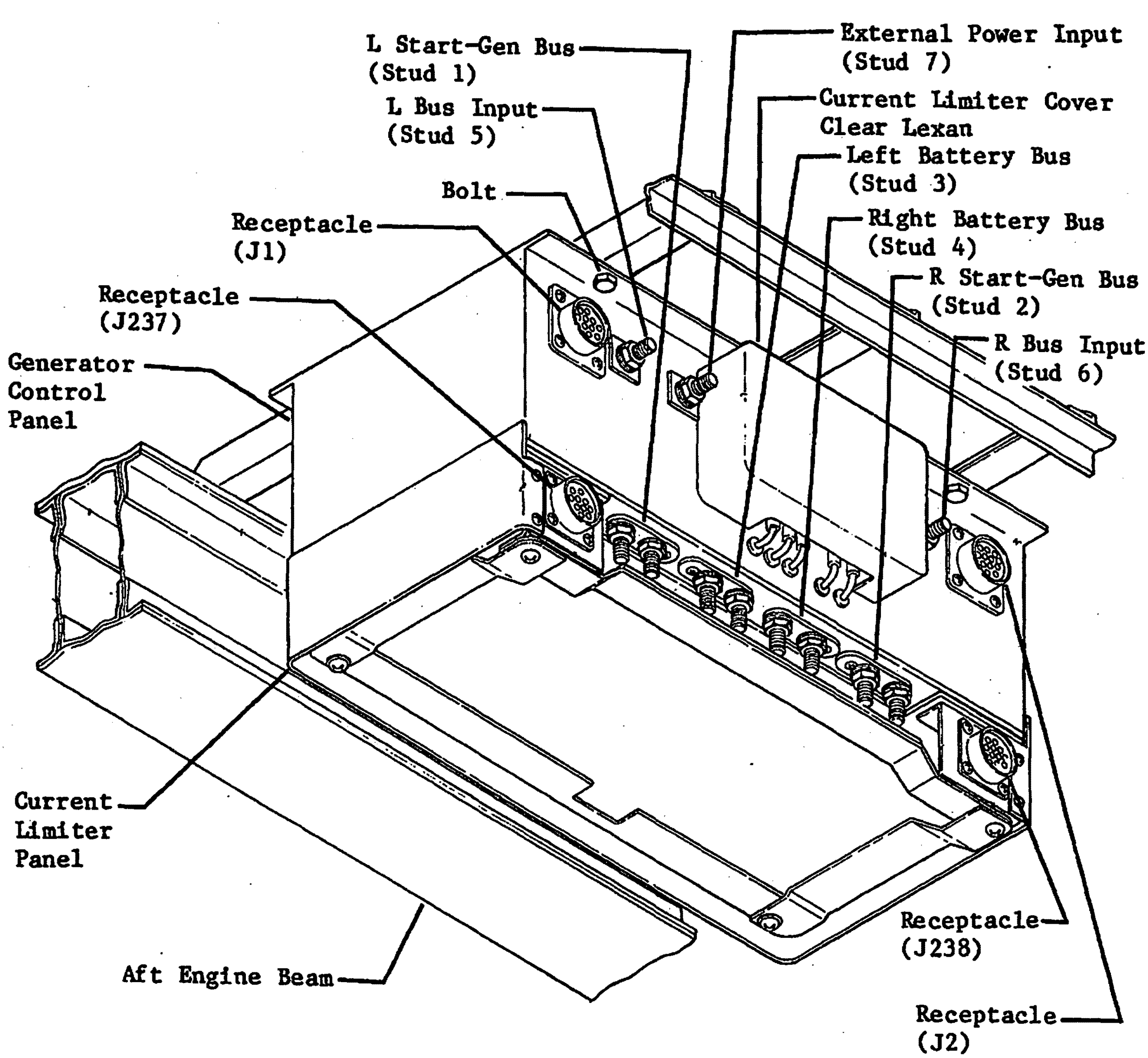
   (a) Stud 1 - Left starter-generator wires and wires to reverse current diode (left-hand).
   (b) Stud 2 - Right starter-generator wires and wires to reverse current diode (right-hand).
   (c) Stud 3 - Battery 1 wires.
   (d) Stud 4 - Battery 2 wires.
   (e) Stud 5 - Reverse current diode wires (left-hand).
   (f) Stud 6 - Reverse current diode wires (right-hand).
   (g) Stud 7 - External power receptacle wires.
   (h) On Aircraft 25-342 and Subsequent, remove current limiter clear Lexan cover and disconnect wire at current limit FL70.
   (6) Remove attaching bolts and generator control panel from aircraft.



NOTE: Refer to Current Limiter Panel (24-50-02) for additional wire connections. Torque nut on Stud 7 45-55 inch-pounds.

9-68C-3

Generator Control Panel Installation
Figure 201 (Sheet 1 of 2)



NOTE: Refer to Current Limiter Panel (24-50-02) for additional wire connections. Torque nut on Stud 7 45-55 inch-pounds.

9-68C-3

Generator Control Panel Installation
Figure 201 (Sheet 2 of 2)

B. **Install Generator Control Panel** (See figure 201.)

(1) Attach generator control panel to aircraft structure with bolts.

(2) Connect wire terminals to generator control panel.

NOTE:
- Remove nut, lockwasher, and washer from studs 1 through 7. The nuts are silver plated copper furnished with generator control panel.
- Refer to applicable Wiring Diagram for wire number code.

(a) Stud 1 - Left starter-generator wires and wires to reverse current diode (left-hand).

(b) Stud 2 - Right starter-generator wires and wires to reverse current diode (right-hand).

(c) Stud 3 - Battery 1 wires.

(d) Stud 4 - Battery 2 wires.

(e) Stud 5 - Reverse current diode wires (left-hand).

(f) Stud 6 - Reverse current diode wires (right-hand).

(g) Stud 7 - External power receptacle wires.

(h) On Aircraft 25-342 and Subsequent, remove current limiter clear Lexan cover and connect stabilizer actuator wire to current limiter FL70. Install current limiter clear Lexan cover.

(3) Install washer, lockwasher, and nut on studs 1 through 7. Torque nut 45 to 55 inch-pounds.

(4) Connect wire terminals to current limiter panel. (Refer to applicable figure in 24-50-02.)

NOTE:
- Remove nut, lockwasher, and washer from studs as required to connect wires.
- Refer to applicable figure in 24-50-02 and to applicable Wiring Diagram for wire number code.

(a) Connect two number four gage wires (wires to current limiters FL4 and FL5) to stud COM. Install washer, lockwasher, and silver plated copper nut on stud COM. Torque nut 30 to 42 inch-pounds.

(b) Connect left nacelle heat wire to stud securing current limiter FL33. Install washer, lockwasher, and nut on mounting stud. Torque nut 75 (±3) inch-pounds.

(c) Connect right nacelle heat wire to stud securing current limiter FL34. Install washer, lockwasher, and nut on mounting stud. Torque nut 75 (±3) inch-pounds.

(d) On Aircraft 25-076 thru 25-089, 25-091 thru 25-102, connect wires to fuse holder (current limiters FL37 and FL38) attached to current limiter panel guard assembly.

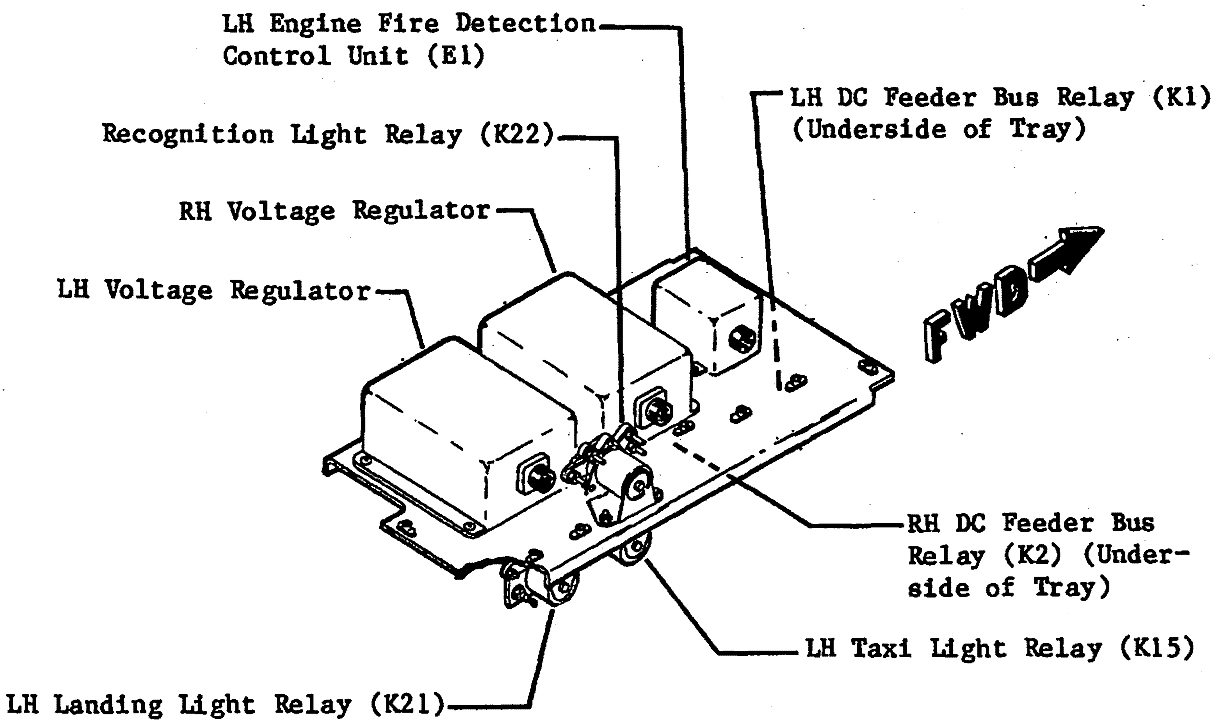
(5) Connect electrical connectors to generator control panel and to current limiter panel.

(6) Perform DC Feeder Bus Functional Test. (Refer to 24-50-02.)

(7) Perform DC Generation System Inspection/Check. (Refer to 24-31-00.)
(8) Perform Voltage Regulator Inspection/Check. (Refer to 24-31-03.)
(9) Install current limiter panel cover assembly. (Refer to 24-50-02.)

VOLTAGE REGULATOR - MAINTENANCE PRACTICES

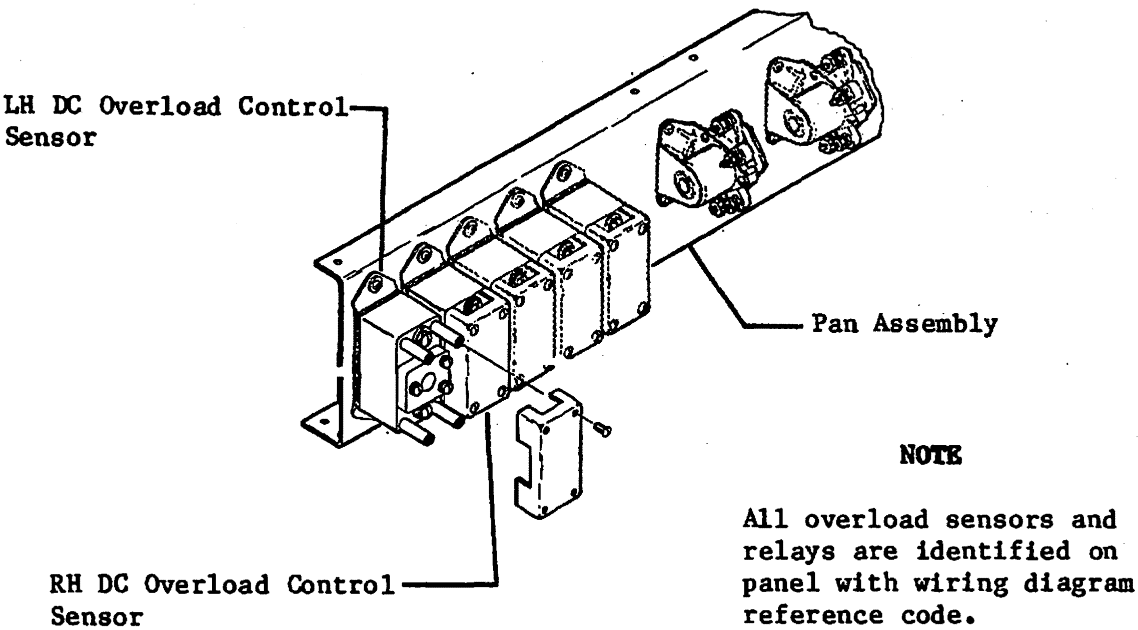
1. **REMOVAL/INSTALLATION**
   A. **Remove Voltage Regulator** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Disconnect electrical plug from regulator.
      (4) Remove attaching parts and regulator from aircraft.
   B. **Install Voltage Regulator** (See figure 201.)
      (1) Install regulator and secure with attaching parts.
      (2) Connect electrical plug.
      (3) Connect aircraft batteries.
      (4) Perform operational check of voltage regulators.
      (5) Secure tailcone access door.



Voltage Regulator Installation
Figure 201

DC OVERLOAD SENSORS - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   - A. **Remove Overload Sensors** (See figure 201.)
     - (1) Lower tailcone access door and disconnect aircraft batteries.
     - (2) Remove attaching parts and overload control sensor cover.
     - (3) Disconnect electrical wiring from overload control sensor. Tag electrical wiring.
     - (4) Remove attaching parts and overload control sensor from aircraft.
   - B. **Install Overload Sensor** (See figure 201.)
     - (1) Install overload control sensor and secure with attaching parts.
     - (2) Remove tags and connect electrical wiring to overload control sensor and install cover.
     - (3) Connect aircraft batteries.
     - (4) Perform functional check of DC generation. (Refer to 24-31-00.)
     - (5) Secure tailcone access door.



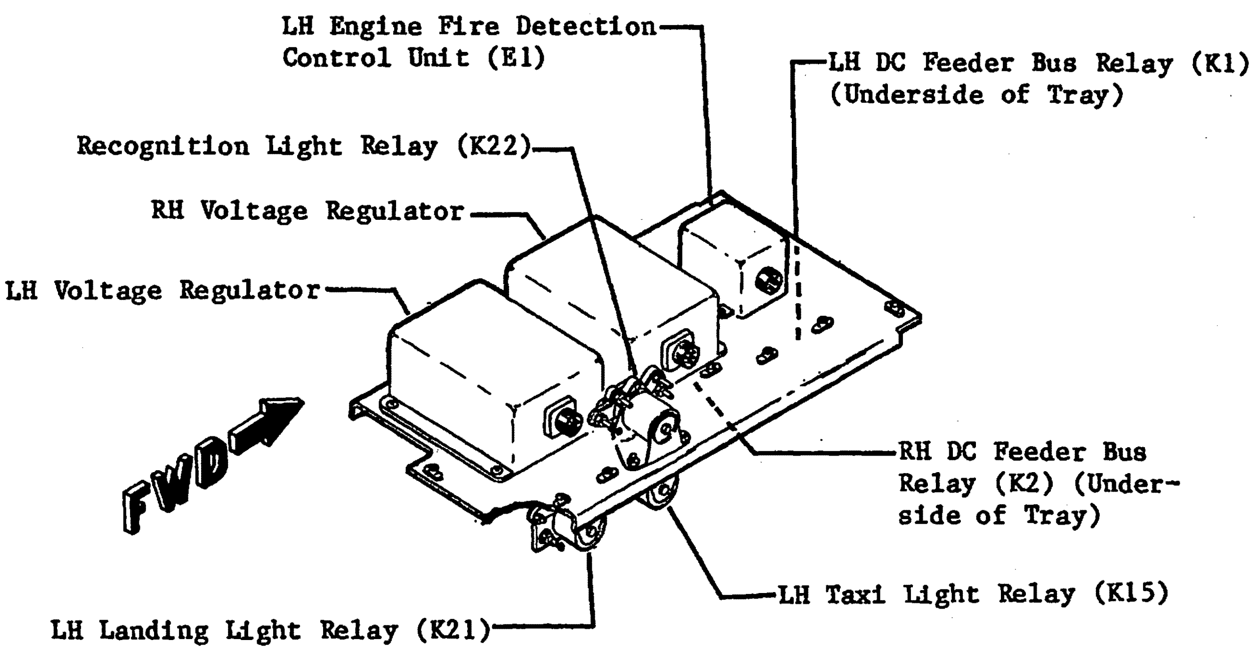
NOTE

All overload sensors and relays are identified on panel with wiring diagram reference code.

DC Overload Sensor Installation
Figure 201

FEEDER BUS RELAYS - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION
   A. **Remove Feeder Bus Relay** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Disconnect electrical wiring from relay. Tag wiring.
      (4) Remove attaching parts and relay from aircraft.
   B. **Install Feeder Bus Relay** (See figure 201.)
      (1) Install relay and secure with attaching parts.
      (2) Remove tags and connect electrical wiring.
      (3) Connect aircraft batteries.
      (4) Perform functional check of DC Generation System. (Refer to 24-31-00.)
      (5) Secure tailcone access door.



DC Feeder Bus Relay Installation
Figure 201

REVERSE CURRENT DIODE - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**

   A. **Remove Reverse Current Diode** (See figure 201.)

      (1) Lower tailcone access door and disconnect aircraft batteries.

      (2) Remove cap nut and diode cover from diode assembly.

      (3) Remove nut and washers and generator control box cables from cathode end of diode.

      (4) Remove starter-generator cable bolt and cables from base of diode assembly.

      (5) Remove attaching parts and diode assembly from aircraft. Remove inserts from diode base.

   B. **Install Reverse Current Diode** (See figure 201.)

      (1) Assure that base of diode assembly is free from burrs, dust, or other foreign particles.

      (2) Install inserts in base of diode assembly.

      (3) Apply a film of No. 340 Silicone Heat Sink Compound (manufactured by Dow Corning Corp., Electric Products Division, Hemlock, Michigan) to both sides of dielectric and position dielectric over inserts to base of diode assembly.

      (4) Install diode assembly and secure with diode assembly attachment bolt and washers. Torque diode assembly attachment bolt 70 to 80 inch-pounds.

      (5) Secure starter-generator cables to diode assembly with cable bolt and washers. Torque cable bolt 75 to 95 inch-pounds.

      (6) Connect generator control box cables and secure with nut and washer. Torque nut 18 to 22 inch-pounds.

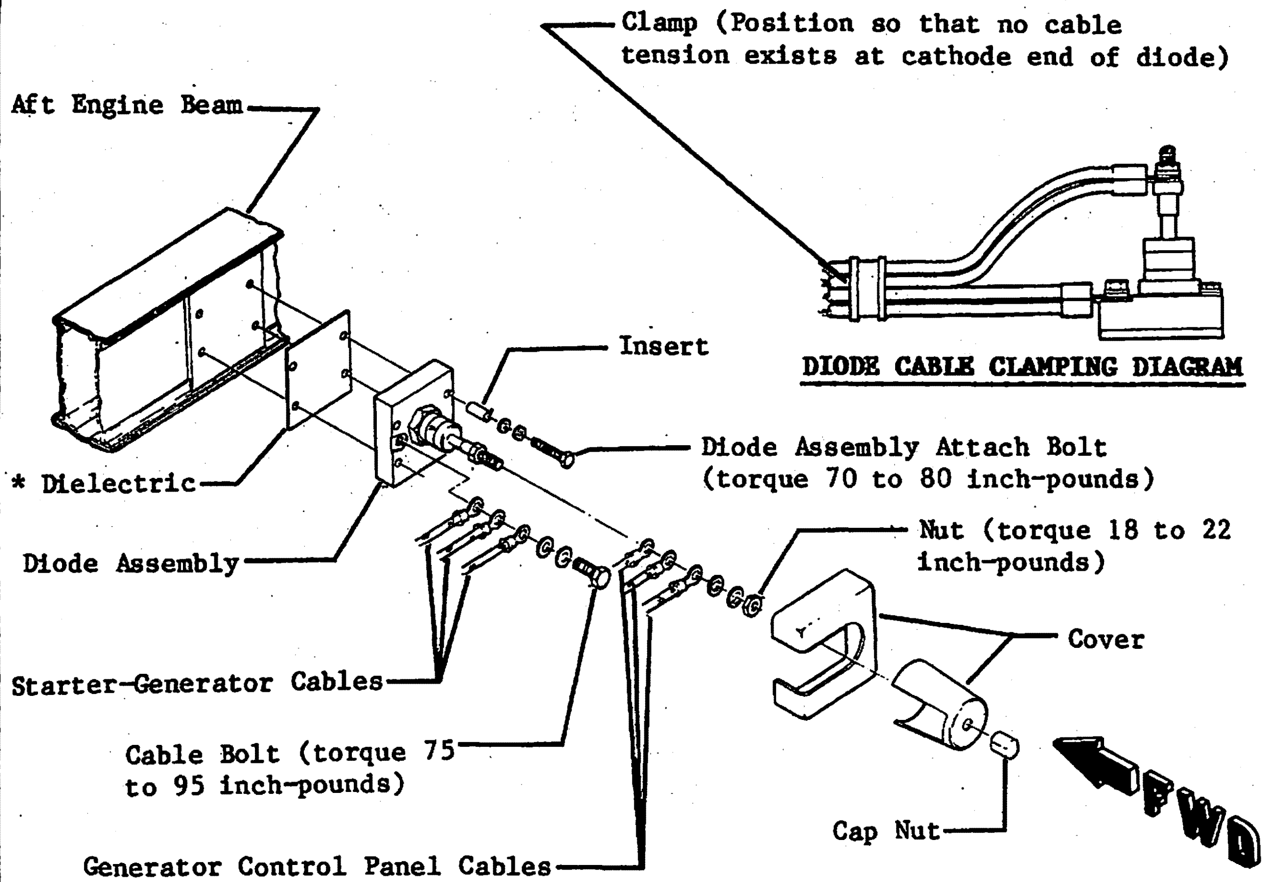
      (7) Position generator control panel cables and cable clamp as shown. Assure that no tension exists at cathode end of diode assembly when clamp is secured.

      (8) Install cover and apply Loctite screw lock compound (manufactured by Loctite Corp., Newington, Conn.) to the threads of the stud and secure with cap nut.

      (9) Connect aircraft batteries and perform operational check of DC power distribution system.

      (10) Secure tailcone access door.

* Apply a film of No. 340 Silicone Heat Sink Compound to both sides of dielectric.



Reverse Current Diode Installation
Figure 201

INDICATOR - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

   A. Remove Indicators

      NOTE: The voltmeter and ammeters are installed in the same instrument cluster on the instrument panel. Removal and installation procedures are identical, except for removal of applicable attaching screw of each indicator.

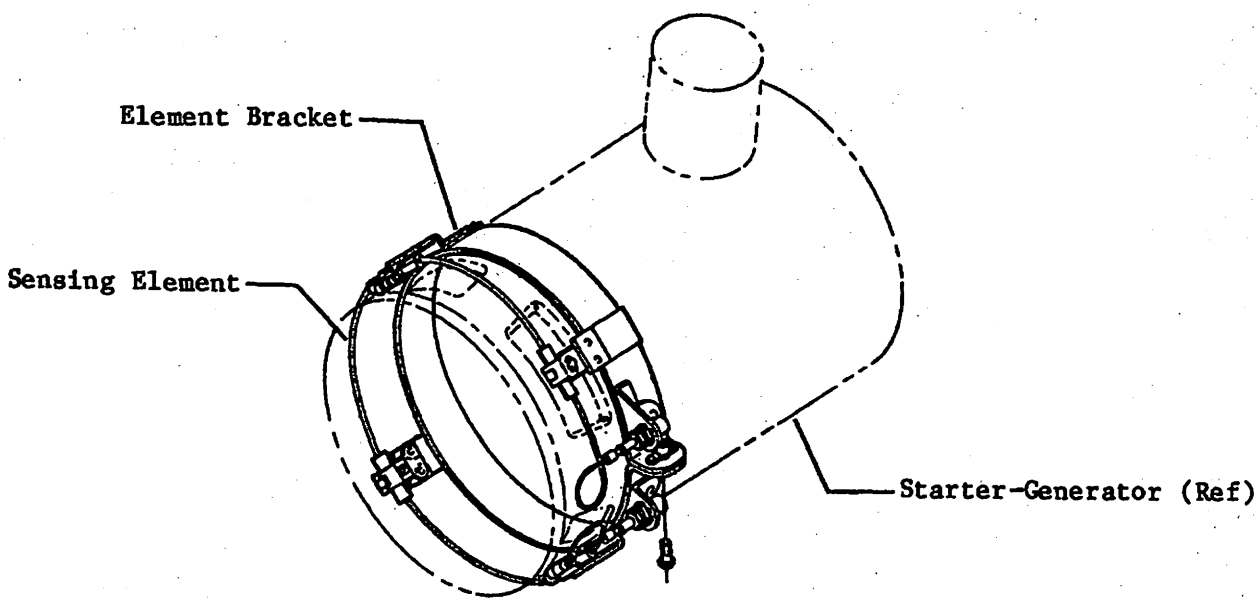
      (1) Loosen screw in upper RH corner of applicable instrument.
      (2) Remove instrument from cluster sufficiently to disconnect electrical plug.

   B. Install Indicator

      (1) Install indicator in cluster and secure with screw.

STARTER-GENERATOR OVERHEAT DETECTOR - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION
   A. Remove Detection Element (See figure 201.)
      (1) Remove engine lower nacelle.
      (2) Disconnect electrical plugs from detector elements.
      (3) Remove starter-generator from engine.
      (4) Release quick-release fasteners securing element and grommet.
      (5) Loosen and remove attaching parts securing element to element bracket.
   B. Install Detector Element (See figure 201.)
      (1) Install element and secure to element bracket with attaching parts.
      (2) Position element and grommet in quick-release fasteners. Assure that even pressure will be applied to element when fastener is secured.
      (3) Install starter-generator on engine.
      (4) Connect electrical plugs to elements.
      (5) Install engine lower nacelle.



Starter-Generator Overheat Detector System Installation
Figure 201

BATTERY SYSTEM - DESCRIPTION AND OPERATION

1. **DESCRIPTION (Nickel-Cadmium Batteries)**
   A. Battery power is supplied by two 22-ampere-hour 24-volt 19-cell nickel-cadmium batteries or by two optional 40 ampere-hour batteries. The batteries are installed in the tailcone.
   B. Batteries are normally shipped in a discharged state, complete with proper electrolyte level. It is only necessary to charge a battery to prepare it for service. No adjustment of liquid level will be necessary even though occasionally no liquid level is visible. The initial charge will cause the liquid to rise to proper level.

   CAUTION: SOME BATTERIES ARE SHIPPED WITH SHIPPING PLUGS IN THE FILL VENTS. IN THIS CASE, THE PROPER VENT VALVES ARE WITH THE BATTERY. REMOVE SHIPPING PLUGS AND INSTALL VENT VALVES AFTER UNPACKING. SHIPPING PLUGS MUST BE REMOVED PRIOR TO CHARGING OR DISCHARGING.

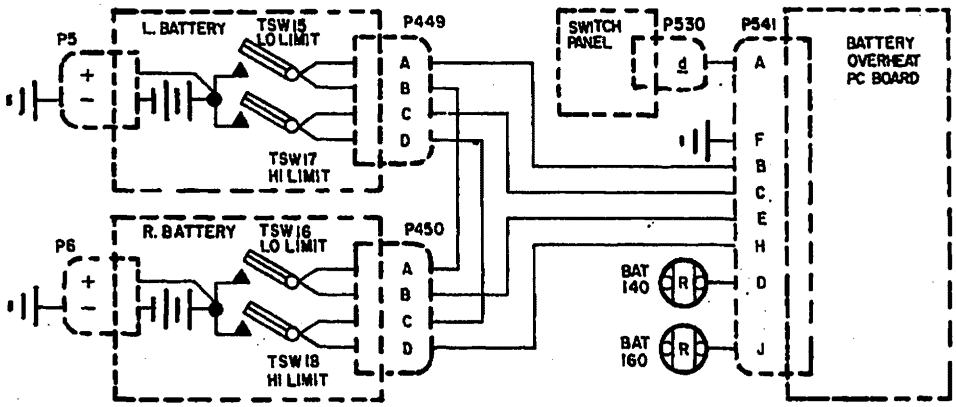
   C. The batteries can be stored indefinitely in the discharged state. Storage in any state of charge does not damage the cells, but the battery will slowly become discharged. The higher the existing temperature, the faster the self discharge. At normal temperature, the battery will deliver at least 85% of its rated capacity after 2 weeks of storage. It is recommended that a charged battery stored in excess of 2 months be charged prior to use. (Refer to Chapter 12 for servicing instructions.)
   D. Each battery consists of a stainless steel case containing 19 cells and a removable cover. The batteries are vented overboard through hoses connected to the battery case vent port and overboard vents. The overboard vents are scarfed to produce an airflow through the batteries to aid battery venting. The right drain is scarfed aft while the left drain is scarfed forward.
   E. The battery system also incorporates a battery overheat warning system. The system consists of two thermoswitches (lo-limit and hi-limit) installed in each battery and two warning lights. The warning lights are installed in pilot's subpanel or in the glareshield.
   F. On Aircraft 25-090, 25-103 and Subsequent, the optional temperature indicating system consists of a dual indicator mounted on the copilot's switch panel or in the center pedestal, a circuit breaker located on the copilot's circuit breaker panel and a temperature sensor located in each battery.

2. **DESCRIPTION (Lead-Acid Batteries)**
   A. Battery power is supplied by two 36-ampere-hour 24-volt lead-acid batteries. The batteries are installed in the tailcone.

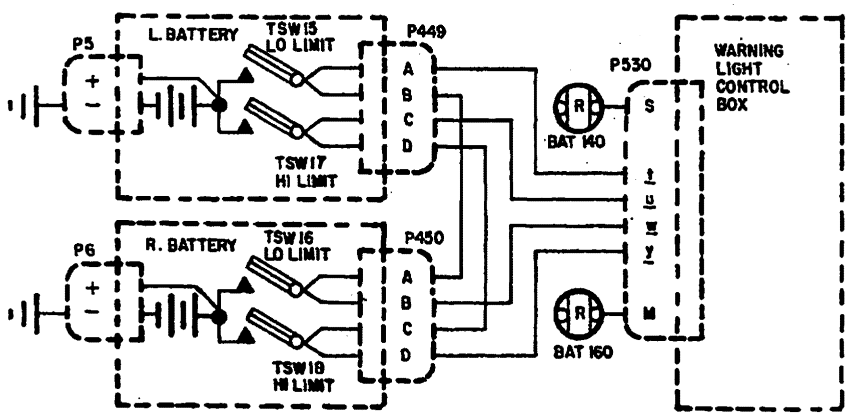
   CAUTION: BATTERIES MUST NOT BE ALLOWED TO STAND IN A DISCHARGED CONDITION OR WITHOUT THE PROPER AMOUNT OF ELECTROLYTE IN THEM OR THE PLATES WILL BECOME SULFATED.

B. Batteries shipped in a dry charged condition can be stored indefinitely if, during storage, batteries are kept clean in a dry place and at an ambient temperature not exceeding 26.7°C (80°F). The length of time required for charging when filled with electrolyte will vary with the length of storage and temperature during storage.
C. Batteries are received dry charged and will deliver a minimum of 50% of its rated capacity after the initial filling of electrolyte without further charging. Batteries should be given a booster charge prior to use.
D. State of charge or battery capacity is the physical as well as the chemical condition of the battery. Checking the specific gravity of the electrolyte is the best way to determine the state of charge of the battery. Specific gravity of a battery varies inversely with temperature and electrolyte level.

3. **OPERATION (Nickel-Cadmium Batteries)** (See figures 1 and 2.)
A. When the Battery Switch is set to ON, a ground circuit is completed to energize the battery relay. With the relay energized, battery power is applied to battery charging bus. On aircraft equipped with dual battery switches, the energized battery relay applies power to the battery bus.
B. On aircraft equipped with nickel-cadmium batteries, the battery overheat warning system alerts maintenance crews or the pilot of an impending battery overheat allowing corrective action to be taken. If battery temperature reaches 140°F, the lo-limit thermoswitch energizes the BAT 140 (red) warning light. If battery temperature reaches 160°F, the hi-limit thermoswitch energizes the BAT 160 (red) warning light. If at any time during flight or ground operation, including engine start, either overheat warning light illuminates, the batteries must be removed from the aircraft and the discharge-recharge reconditioning cycle must be performed. (Refer to Chapter 12.) On aircraft equipped with dual battery switches and the optional battery temperature indicator system, the overheated battery may be isolated from the system by setting its respective switch to OFF.
C. On aircraft equipped with nickel-cadmium batteries, the indicator face is divided into thirds by two white lines. The lower third indicates a battery tempreature of 100°F to 140°F, the center third indicates 140°F to 160°F and the upper third indicates 160°F to 200°F. If the temperature of either battery goes above 140°F (into center third on the indicator face) the BAT 140 light on the glareshield will illuminate. If battery temperature continues to rise and goes above 150°F (into upper third on indicator face) the BAT 160 light will also illuminate. By reading the indicator to determine which battery has malfunctioned, that battery may be isolated with its applicable Battery Switch.

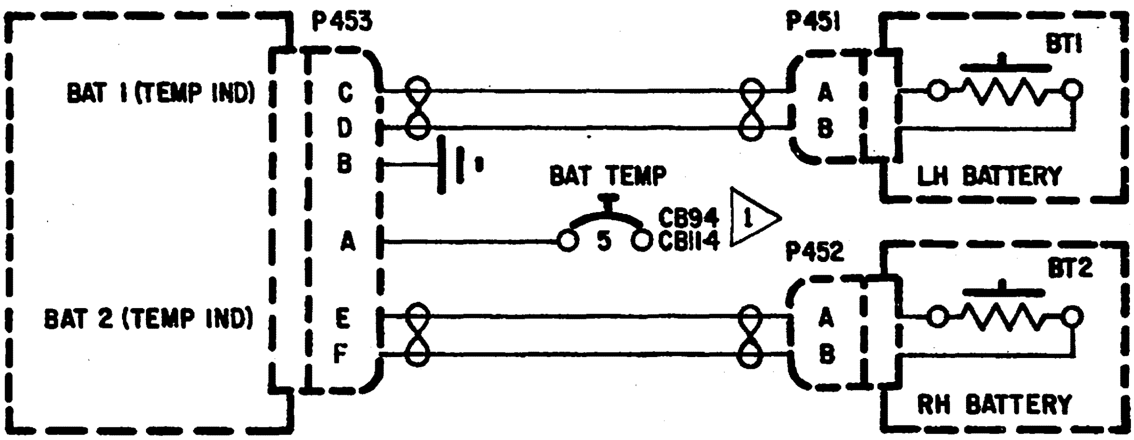


Aircraft 25-061, 25-070 thru 25-080



Aircraft 25-081 and Subsequent

Battery Overheat Warning System Schematic
Figure 1



1 Optional Circuit Breaker Number

Battery Temperature System Electrical Control Schematic
Figure 2

BATTERY SYSTEM - MAINTENANCE PRACTICES

1. **GENERAL**
   A. To extend the service life of the nickel-cadmium batteries, the batteries should be removed from the aircraft and a complete discharge and recharge cycle be performed in accordance with current inspection intervals. (Refer to Chapter 5.)
   B. To extend service life of the lead acid batteries, the batteries must be kept at or near full charge, proper electrolyte level maintained, battery kept clean and not overcharged.
   C. **Battery Maintenance Precautions (Nickel-Cadmium Batteries)**
      (1) The batteries are alkaline type and require absolute internal cleanliness. Avoid any contamination.

      CAUTION: DO NOT USE ANY TOOLS, JARS, OR INSTRUMENTS IN COMMON WITH ACID BATTERIES. ACID AND ALKALI DO NOT MIX.

      (2) Alkaline electrolyte has a strong corrosive effect on most metals. In case of spills, immediately neutralize area with either a solution of six ounces of boric acid to one gallon water or a solution of one part vinegar to three parts water. Do not flush with water prior to neutralizing as this will only enlarge the area of contamination. Do not neutralize battery.
      (3) Alkaline electrolyte is harmful to human skin. If a splash occurs, immediately rinse area with water then neutralize with either a solution of six ounces boric acid to one gallon water or a solution of one part vinegar to three parts water followed by washing with soap and water. Remove contaminated clothing to prevent slow burns to skin. If electrolyte get into eyes, wash with boric acid solution (six ounces to one gallon water) and consult physician. Be sure to advise the physician the the electrolyte was alkaline not acid.
      (4) Do not use a wire brush to clean cells. Use a bristle brush and water. Remove batteries from aircraft when using water.

D. **Battery Maintenance Precautions (Lead-Acid Batteries)**

CAUTION:
- ° HYDROGEN GAS GIVEN OFF DURING CHARGING AND SHORTLY AFTER CHARGING IS HIGHLY EXPLOSIVE. OPEN FLAMES AND SPARKS MUST BE AVOIDED IN THESE AREAS.
- ° TURN BATTERY CHARGER OFF BEFORE CONNECTING BATTERY TO CHARGER AND BEFORE MOVING BATTERY FROM CHARGER.
- ° BATTERY CHARGING AREA MUST BE WELL VENTILATED.
- ° SULPHURIC ACID IN THE ELECTROLYTE IS INJURIOUS TO SKIN. IF ACID IS SPILLED, NEUTRALIZE AT ONCE WITH A WATER SOLUTION OF SODIUM BICARBONATE (BAKING SODA).
- ° WHEN MIXING ELECTROLYTE, ALWAYS POUR ACID INTO WATER WHILE SLOWLY AND CONTINUALLY STIRRING. DO NOT POUR WATER INTO ACID. USE ONLY GLASS, HARD RUBBER OR OTHER SUITABLE CONTAINERS FOR MIXING ELECTROLYTE.
- ° DO NOT USE FRESHLY PREPARED ELECTROLYTE UNTIL IT HAS COOLED TO AT LEAST 90°F. CONSIDERABLE HEAT IS GENERATED, WHICH CAN DAMAGE THE BATTERY.

(1) DO NOT use a wire brush to clean battery. Use a stiff bristle brush. Wipe with cloth dampened with bicarbonate of soda solution (one part soda to 20 parts of water) to neutralize any spilled electrolyte solution.

2. **CLEANING/PAINTING (Nickel-Cadmium Batteries)**

A. **Clean Batteries**

(1) Remove batteries from aircraft.

CAUTION: IN ORDER TO AVOID SHORTING RESULTING IN POSSIBLE CELL DAMAGE, IT IS RECOMMENDED THAT THE BATTERIES BE DISCHARGED PRIOR TO DISASSEMBLY FOR CLEANING.

(2) Discharge batteries through a resistance high enough to permit a current flow of 20 or less amperes for 22 ampere-hour Marathon batteries (10 or less amperes for Gulton batteries), and 40 or less amperes for 40-ampere-hour Marathon batteries (17 or less amperes for Gulton batteries). When voltage is approximately 9.5 volts, placing a shorting clip (8-gage stranded wire, 6 inches long with insulated alligator clips) across each cell's terminals with the load applied. When 15 cells are shorted out, place a 1.0 ohm, 2 watt resistor across each of the remaining cells and allow the battery to remain shorted for 3 hours before removing the shorting straps and resistors.

CAUTION: UNDER NO CIRCUMSTANCES SHOULD A WIRE BRUSH BE USED FOR CLEANING. USE A BRISTLE BRUSH. WHEN CLEANING BATTERIES, USE EXTREME CARE TO PREVENT THIS MATERIAL FROM COMING INTO CONTACT WITH THE EYES. PROTECTIVE CLOTHING SUCH AS RUBBER GLOVES, AN APRON AND FACE SHIELD SHOULD BE WORN.

(3) After battery has been discharged, remove shorting clips and all intercell connecting links. During removal, mark all connecting hardware to ensure proper reinstallation. Remove any white deposits (potassium carbonate) from top of cells and case with a bristle brush.
(4) Loosen vent plugs, using a vent wrench.
(5) Remove cells from case. After removing cells, retighten the vents.
(6) After washing cells with tap water and drying, inspect for cracks and leaks. Store cells in a dry, clean area.
(7) Wash battery case with tap water and dry with compressed air.
(8) Wash intercell hardware with tap water and wipe dry. Remove any corrosive preventive with a solvent such as trichlorethane or acetone. Replace all parts that are damaged or have defective nickel plating. Replace burnt, cracked, bent or pitted terminals on battery power receptacles. Repair or replace bent battery cases and covers, loose or damaged cover gaskets and cell holddown bars.
(9) Inspect rubber case liners for deterioration or edges pulled away from the case. Liners may be recemented using a good metal-to-rubber cement (EC 1300 manufactured by 3M Co., or equivalent). Liners may be replaced using neoprene rubber of a 40-60 durometer which is the same thickness as the old liners.
(10) Replace cells in case, making certain to insert with the polarity symbols in the right direction. Cells are connected plus to minus.

NOTE: To aid in installation of battery cells, install cells inward from the end of each row until the center cell in each row remains. Apply a light coating of petroleum jelly on the sides of the last cells and adjacent cells and slide in last cells. Silicone reacts with electrolyte and is not recommended.

CAUTION: DO NOT USE "HOMEMADE" HARDWARE. VENDOR-SUPPLIED PARTS ARE SPECIFICALLY DESIGNED TO PRODUCE ADEQUATE ELECTRICAL CONNECTIONS.

(11) Install intercell connectors and tighten attaching hardware finger tight. Apply a thin coat of Shell Darina No. 2 or equivalent on all exposed hardware including contact surfaces and screw threads.
(12) Torque intercell connecting hardware 34 to 38 inch-pounds for Gulton batteries, 50 to 65 inch-pounds for Marathon 22 ampere-hour batteries and 35 to 50 inch-pounds for Marathon 40-ampere-hour batteries.
(13) Charge battery. (Refer to Chapter 12.)

3. **BATTERY STORAGE (Nickel-Cadmium Batteries)**
   A. The batteries can be stored indefinitely in the discharged state. Storage in any state of charge does not damage the cells, but the battery will slowly become discharged. The higher the existing temperature, the faster the self discharge. At normal temperature, the battery will delivery at least 85% of its rated capacity after 2 weeks of storage. It is recommended that a charged battery stored in excess of 2 months be charged prior to use. (Refer to Chapter 12 for servicing instructions.)

4. **BATTERY STORAGE (Lead-Acid Batteries)**
   A. Batteries shipped in a dry charged condition may be stored indefinitely, providing the batteries are kept in a clean, dry place at temperatures not exceeding 80°F (26.7°C). The length of charging time for a stored battery will vary with the length of storage and temperature during storage.
   B. When storing batteries which have had electrolyte added, it is important that the battery be thoroughly cleaned, serviced, and charged prior to storage. During storage, a lead acid battery will lose some of its charge and must be thoroughly charged at regulator intervals. When returning a 'wet' battery to service, it must be charged until specific gravity does not rise over a three-hour period.

BATTERY - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**

   A. **Remove Batteries** (See figure 201.)

   CAUTION: REMOVE THE QUICK-DISCONNECTS FROM BOTH BATTERIES EVEN THOUGH ONLY ONE BATTERY IS TO BE REMOVED. BATTERY VOLTAGE WILL FEED BACK FROM THE CONNECTED BATTERY AND ARCING WILL OCCUR IF THE QUICK DISCONNECT IS INADVERTENTLY SHORTED TO THE AIRCRAFT STRUCTURE.

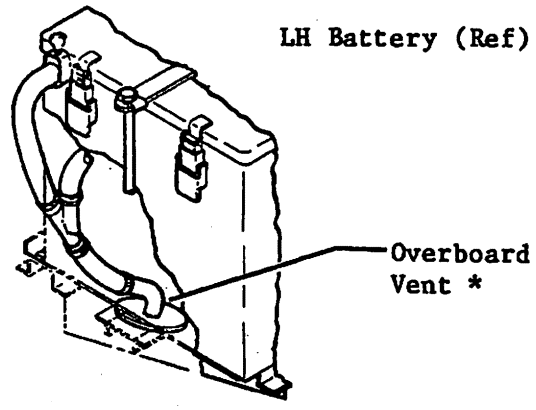
   (1) Lower tailcone access door.
   (2) Disconnect both battery quick-disconnects.
   (3) Disconnect battery overheat warning system plugs located directly below battery quick-disconnect.
   (4) Disconnect battery temperature indicator plug if installed.
   (5) Loosen clamps and remove vent hoses.
   (6) Loosen and remove retaining bar bolts.
   (7) Remove battery from aircraft.
   (8) Clean battery compartment if required.

   B. **Install Batteries** (See figure 201.)

   (1) Inspect battery to assure no damage has occurred during shipment. If new batteries are being put into service, refer to Chapter 12 for servicing procedures.
   (2) Check all electrical connections for tightness.

   CAUTION: POOR ELECTRICAL CONTACT COULD RESULT IN DAMAGE TO THE BATTERY.

   (3) Install battery.
   (4) Install retaining bar and bolts.
   (5) Connect vent hoses and secure with clamps.
   (6) Connect battery overheat warning system and battery temperature indicator plugs.
   (7) Connect battery quick-disconnects and safety wire per double twist method.
   (8) Secure tailcone access door.



TYPICAL VENT HOSE CONNECTION

Detail A

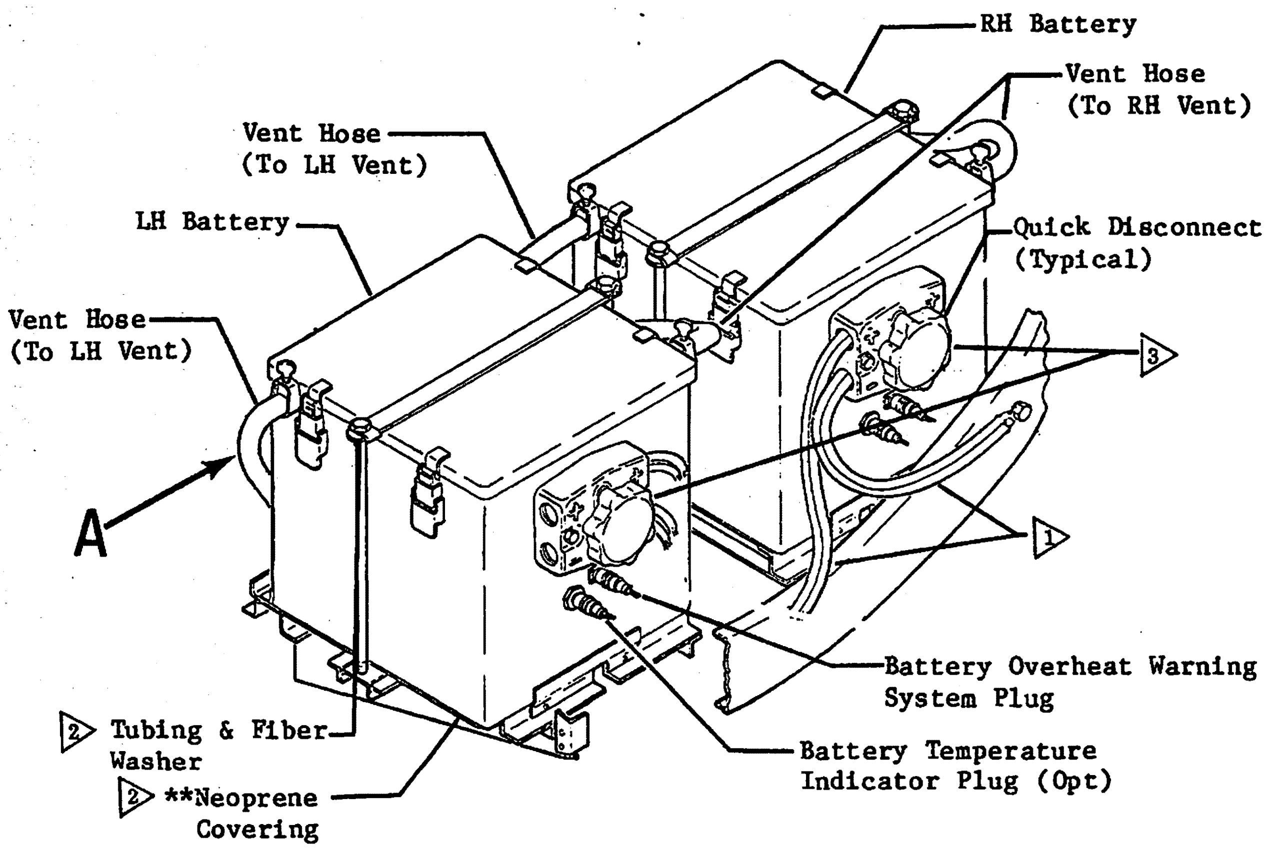
1 Minimum bend radius of Battery Wires (4 AWG) is 3 inches.

2 Effective on aircraft certified for Operation in France.

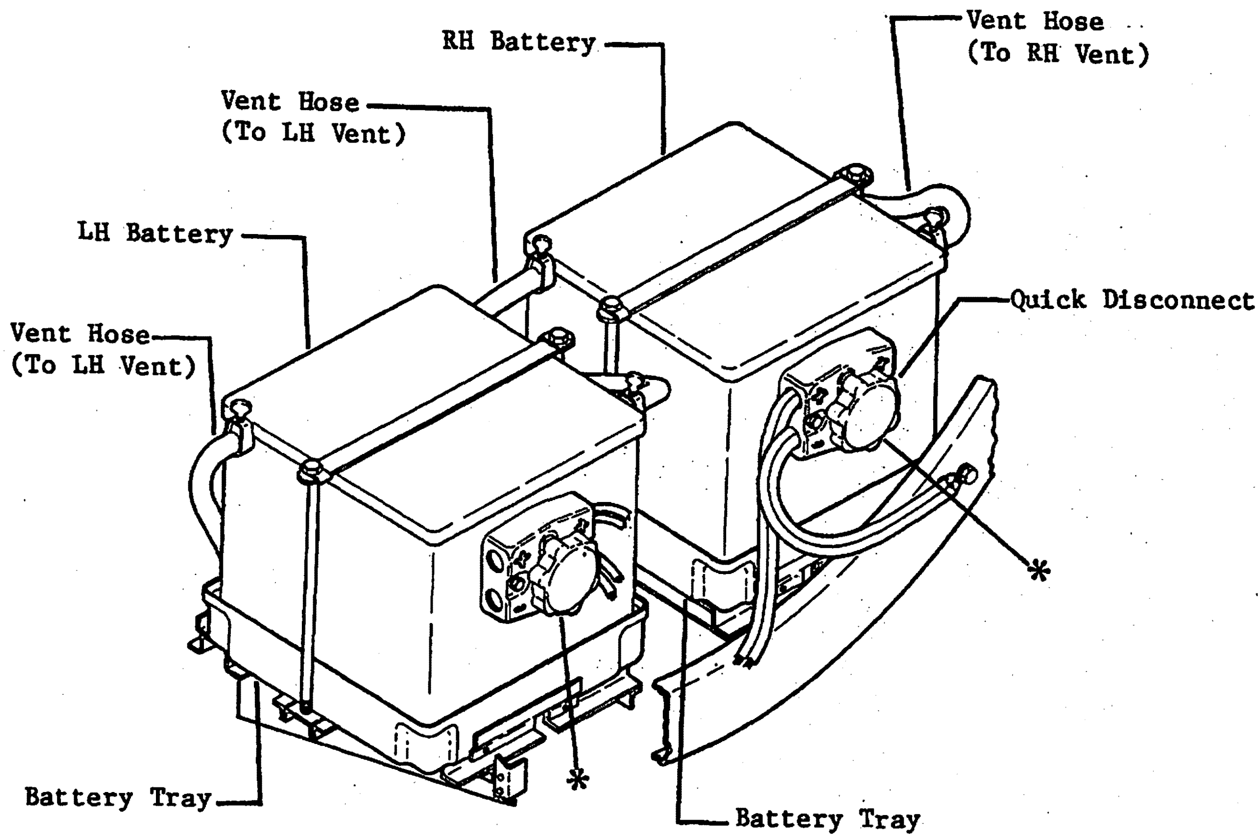
3 DO NOT use battery receptacles as hand holds when removing batteries. Battery damage could result.

** Cement Neoprene covering to Battery structure with EC-776 (mfg. by 3M Co.) or equivalent.

* The overboard vents have scarfed ends which provide positive battery ventilation.



Battery Installation
Figure 201 (Sheet 1 of 2)



* DO NOT use battery receptacles as hand holds when removing batteries. Battery damage could result.

Battery Installation
Figure 201 (Sheet 2 of 2)

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TEMPERATURE INDICATOR - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION
   A. Remove Indicator (See figure 201.)
      (1) Gain access to temperature indicator installation.
      (2) Disconnect electrical plug from temperature indicator.
      (3) Loosen screws securing indicator to mounting plates.
      (4) Loosen and remove mounting plate screws, mounting plate and indicator.
   B. Install Indicator (See figure 201.)
      (1) Install indicator and mounting plates and secure mounting plates with attaching parts.
      (2) Tighten screws on indicator, securing indicator to mounting plates.
      (3) Connect electrical plug to indicator.
      (4) Install any previously removed equipment.

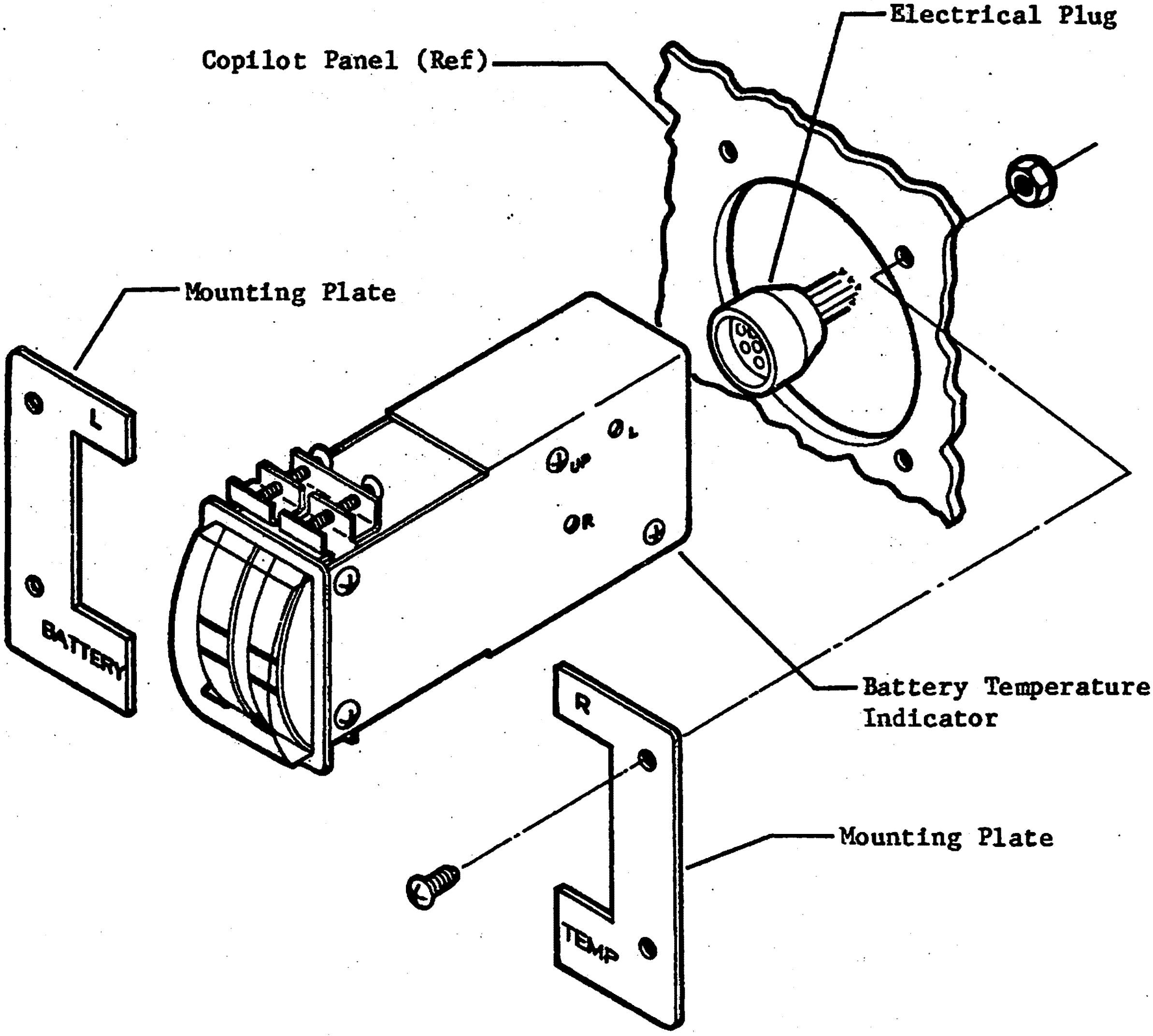
2. INSPECTION/CHECK
   A. Battery Temperature Indicator Operational Check

      NOTE: It is recommended that the battery temperature system be operationally checked for proper operation after maintenance of any type is performed in the tailcone area of the aircraft and in accordance with the time intervals specified in Chapter 5.

      (1) With Battery Switches off, disconnect both battery temperature indicator plugs (LH battery P451; RH battery P452) located left of centerline on the battery cases.
      (2) Connect an 1,800 ohm resistor between pins A and B in LH battery temperature indicator plug (P451).
      (3) Set Battery Switch ON. The LH battery temperature indicator should indicate an above normal reading.
      (4) Set Battery Switch OFF.
      (5) Disconnect resistor from LH battery temperature indicator plug and connect resistor between pins A and B in RH battery temperature indicator plug (P452).
      (6) Set Battery Switch ON. The RH battery temperature indicator should indicate an above normal reading.

          NOTE: If the above conditions are not met, rewire circuit and repeat steps (1) through (6).

      (7) Set Battery Switch to OFF and disconnect resistor from plug.
      (8) Reconnect LH and RH battery temperature indicator plugs.



Battery Temperature Indicator Installation
Figure 201

THERMISTOR - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**

A. **Remove Screw-In Type Thermistor** (See figure 201.)

CAUTION: EXERCISE EXTREME CAUTION NOT TO CAUSE ELECTRICAL SHORTING WHEN REMOVING CONNECTING LINK. SHORTING WILL DAMAGE THE BATTERY. DO NOT USE ANY TOOLS THAT HAVE BEEN USED ON LEAD ACID BATTERIES.

(1) Remove battery from aircraft. (Refer to 24-32-01.)
(2) Remove cover from battery.
(3) Remove cell connecting link with thermistor attached. (Refer to 24-32-01.)
(4) Remove epoxy covering thermistor and unscrew thermistor from link.
(5) Disconnect thermistor leads from electrical plug.

B. **Install Screw-In Type Thermistor** (See figure 201.)

(1) Clean cell connecting link with MEK or equivalent solvent.
(2) Screw new thermistor in link and tighten finger tight.
(3) Spread epoxy (EC-2216) over thermistor completely covering thermistor. Assure that thermistor wires are clean, not shorted together, or grounded to connector link.
(4) Allow epoxy to cure for 24 hours. Cure may be accelerated by heating at 150°F for two hours or 45 minutes at 200°F.
(5) Solder thermistor leads to electrical connector and apply RTV156 silicone rubber potting compound to receptacle.
(6) Install connecting link and secure with attaching parts. (Refer to 24-32-01.)
(7) Install battery cover.
(8) Install battery in aircraft. (Refer to 24-32-01.)

C. **Remove Epoxied-Type Thermistor** (See figure 201.)

CAUTION: EXERCISE EXTREME CAUTION NOT TO CAUSE ELECTRICAL SHORTING WHEN REMOVING CONNECTING LINK. SHORTING WILL DAMAGE THE BATTERY. DO NOT USE ANY TOOLS THAT HAVE BEEN USED ON LEAD ACID BATTERIES.

(1) Remove battery from aircraft. (Refer to 24-32-01.)
(2) Remove cover from battery.
(3 Remove all connecting link with thermistor attached. (Refer to 24-32-01.)
(4) Remove epoxy and thermistor from link.
(5) Disconnect thermistor leads from electrical plug.

B. **Install Epoxied-Type Thermistor** (See figure 201.)

(1) Clean cell connecting link with MEK or equivalent solvent.
(2) Spread a thin layer (.50 x .50 inch square) of EC-2216 epoxy on center of link.
(3) Place thermistor disc on connector link and apply a thin coat of epoxy on disc.

(4) Place thermistor in center of disc and cover completely with epoxy. Assure that thermistor wires are clean, not shorted together, or grounded to connector link.
(5) Allow epoxy to cure for 24 hours. Cure may be accelerated by heating at 150°F for two hours or 45 minutes at 200°F.
(6) Solder thermistor leads to electrical connector and apply RTV156 silicone rubber potting compound to receptacle.
(7) Install connecting link and secure with attaching parts.
(8) Install battery cover.
(9) Install battery in aircraft. (Refer to 24-32-01.)

2. INSPECTION/CHECK

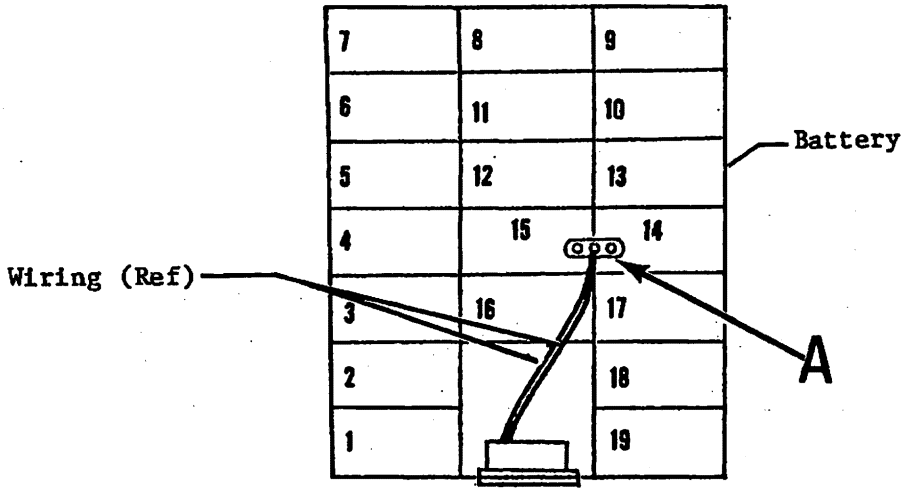
NOTE: Perform functional check of thermistor in accordance with the current inspection intervals specified in Chapter 5.

A. Perform Functional Check of Thermistor

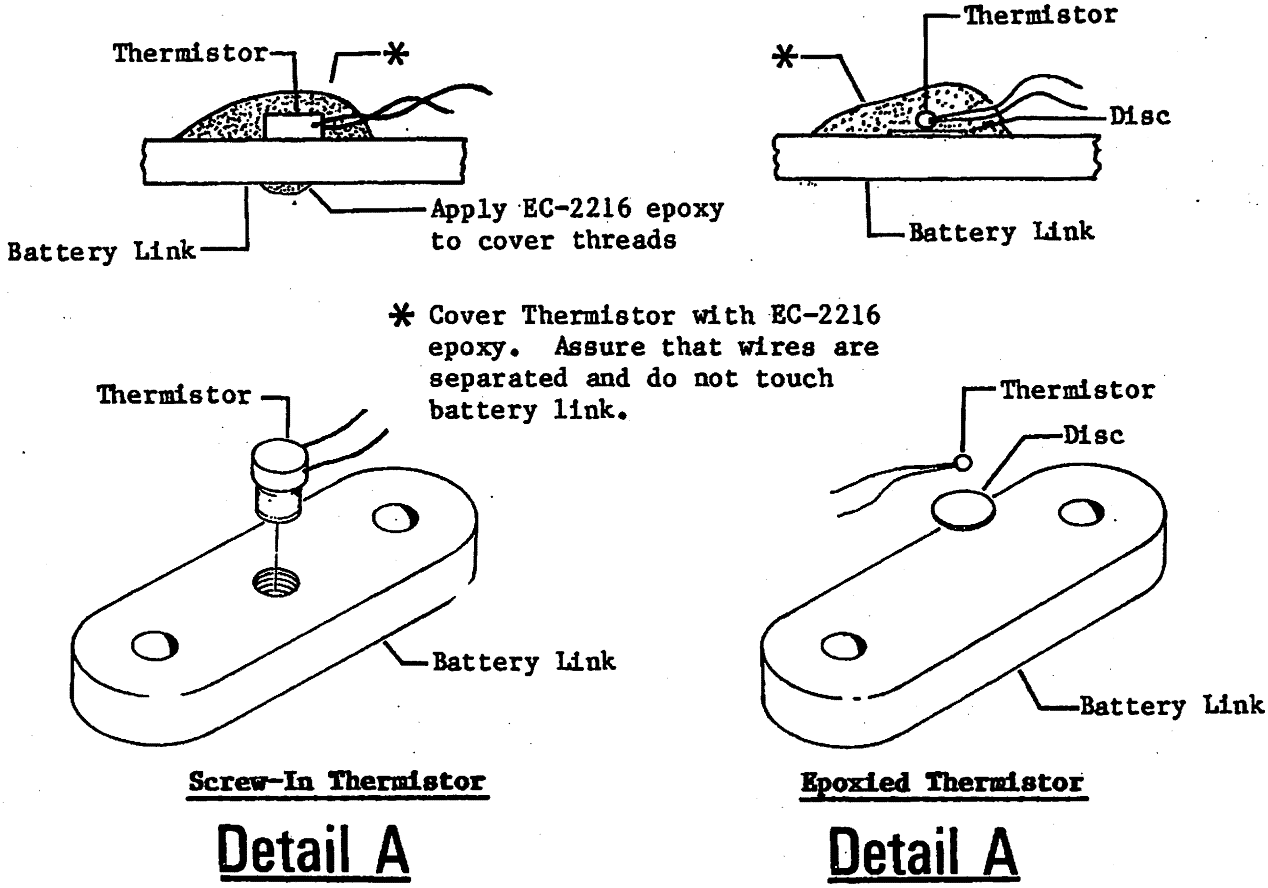
(1) Remove thermistor from battery. (Refer to Removal/Installation.)
(2) Place thermistor in an oil bath adjusted to 140°F (±1°) and connect wires to an ohmmeter.

NOTE: A minimum of four (4) inches of thermistor leads must be submerged in oil bath to achieve an accurate ohmmeter reading.

(3) Allow reading to stabilize. Check that ohmmeter reading is 295 (±11) ohms.
(4) Adjust oil bath temperature to 160°F (±1°) and allow temperature to stabilize. Check that ohmmeter reading is 213 (±10) ohms.
(5) Thermistors that fail to meet any one of these checks must be replaced.
(6) Install thermistor in battery. (Refer to Removal/Installation.)



BATTERY CELL DIAGRAM



Thermistor Installation
Figure 201

TEMPERATURE SWITCH - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

NOTE: Two temperature switches (hi-limit and lo-limit) are installed in each battery. The switches alert the crew, through the Battery 140 and 160 warning lights, if a battery overheat condition should exist.

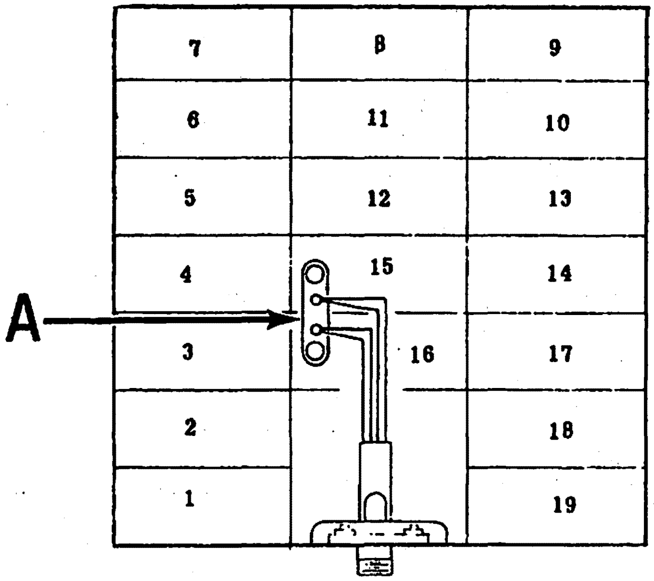
A. **Remove Battery Temperature Switch** (See figure 201.)

NOTE: The temperature switches are installed adjacent to each other on the same battery link. The hi-limit switch is identified by a red dot and the lo-limit by a blue dot.

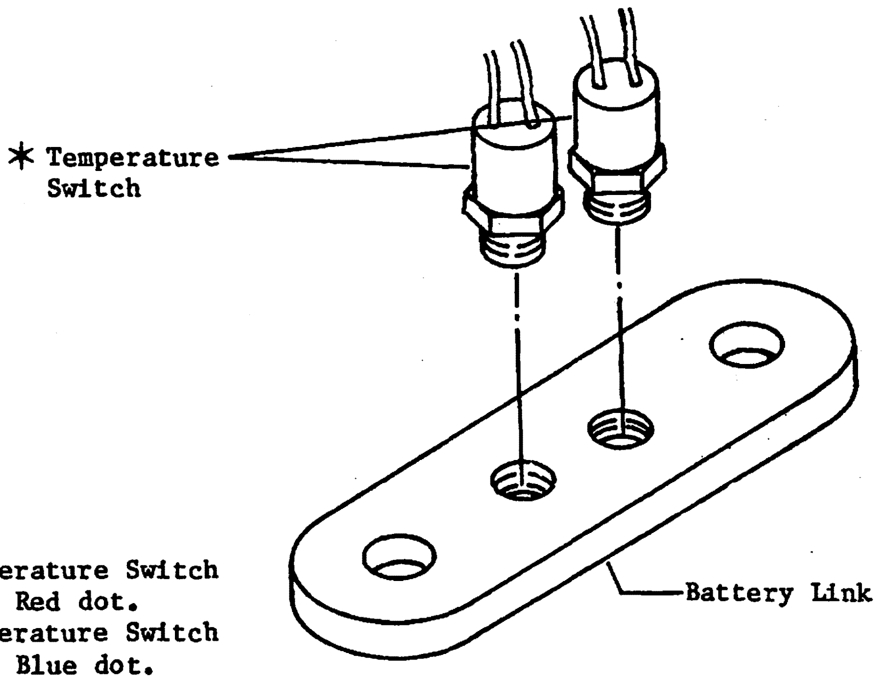
(1) Lower tailcone access door and disconnect aircraft batteries.
(2) Remove batteries from aircraft. (Refer to 24-32-01.)
(3) Cut wires approximately half way between temperature switch and defective temperature switch.
(4) Remove defective switch from the battery link.

B. **Install Battery Temperature Switch** (See figure 201.)

(1) Install new temperature switch in battery link.
(2) Splice new temperature switch to battery wiring using appropriate splice. (Refer to Chapter 20, Wiring Manual for splice callout.)
(3) Install battery link on battery.
(4) Install battery in aircraft. (Refer to 24-32-01.)
(5) Raise and secure tailcone access door.



BATTERY CELL IDENTIFICATION



* Hi-Limit Temperature Switch identified by Red dot.
Lo-Limit Temperature Switch identified by Blue dot.

Detail A

Battery Temperature Switch Installation
Figure 201

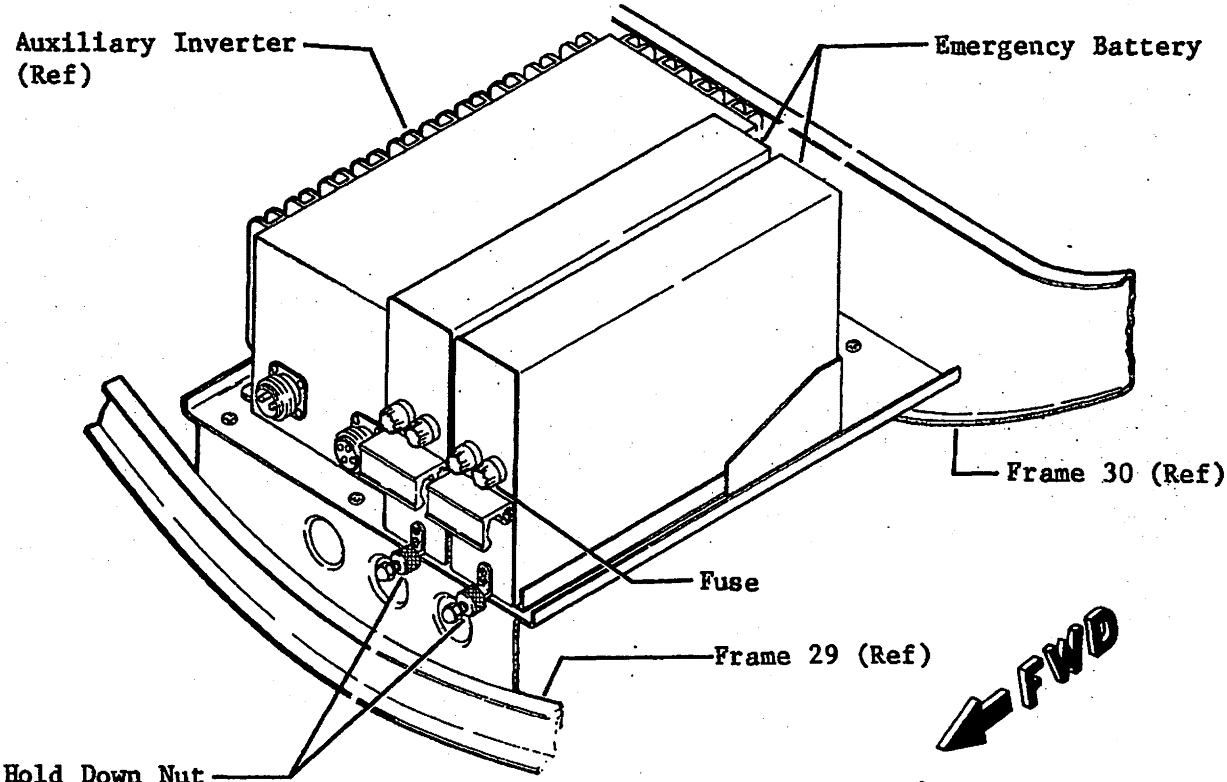
EMERGENCY BATTERY SYSTEM - DESCRIPTION AND OPERATION

1. DESCRIPTION
   A. The aircraft is equipped with either a single or dual emergency battery system. The battery(s) are installed in the nose compartment or the tailcone.

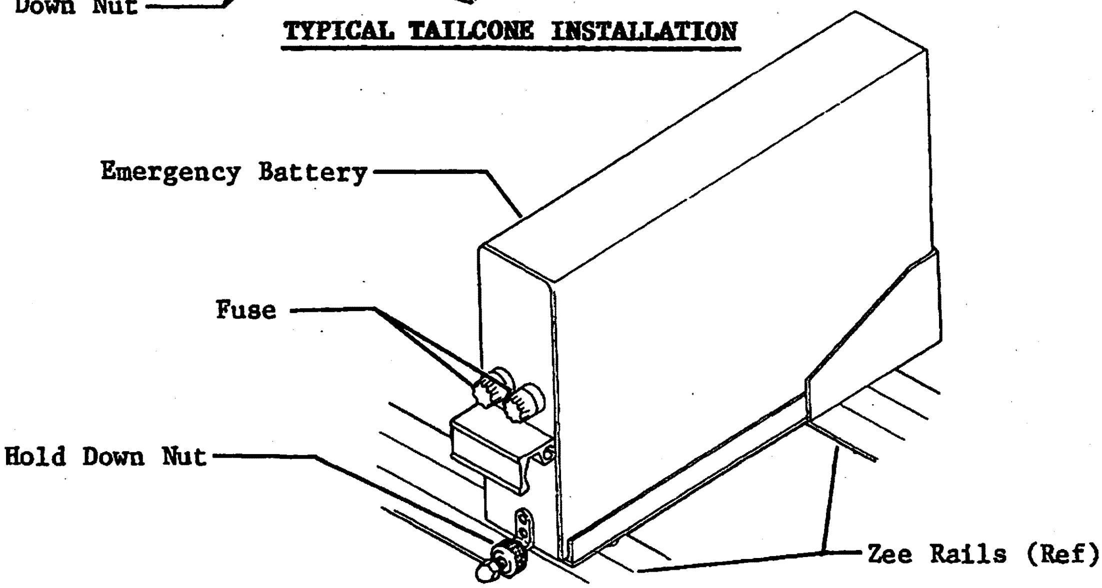
2. OPERATION
   A. On Aircraft equipped with single emergency battery, the system is controlled by ON-STBY-OFF Emergency Power Switch on the pilot's instrument panel. In the STBY position, the emergency battery is providing 115 vac, 4.6 vac and 28 vdc. Also, in the STBY position, the 28 vdc is provided to illuminate the emergency battery indicator (amber) light. However, the indicator light will not illuminate as long as normal power is available to the main buses. If normal power is lost, the indicator will illuminate. The Emergency Power Switch it then set to ON and supplies power to the applicable circuit breaker configuration. The battery is rated 25 vdc and 115 vac (+7, -10%) at 400 Hz. Emergency battery charge is maintained through normal power 28 vdc input.
   B. On Aircraft equipped with dual emergency batteries, the system is controlled by the (BAT 1) ON-STBY-OFF and (BAT 2) ON-OFF Emergency Power Switches on the pilot's instrument panel. During normal operation, the ON-OFF Switch is set to ON and the ON-STBY-OFF Switch is set to STBY. With the (BAT 2) switch set to ON, the emergency battery (BAT 2) is furnishing 115 vac, 4.6 vac for various avionics systems and 28 vdc power. The 28 vdc power is applied to the indicator (amber) light. However, the indicator light will not illuminate as long as normal power is avaiable to the main buses. The avionics equipment will be receiving power from their normal sources. With the (BAT 1) Switch set to STBY, the emergency battery (BAT 1) is furnishing 115 vac, 4.6 vac and 28 vdc power. The 115 vac and 4.6 vac is provided to various avionics systems and the 28 vdc is applied to the indicator (amber) light. However, the indicator (amber) light will not illuminate as long as normal power is available to the main buses. If normal power is lost, both indicator (amber) lights will illuminate. The (BAT 1) switch is then set from STBY to ON, supplying emergency power to the applicable circuit breaker configuration. Each battery is rated at 28 vdc and 115 vac (+7, -10%) at 400 Hz. Emergency battery charge is maintained through normal power 28 vdc input.

EMERGENCY BATTERY - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   A. **Remove Emergency Battery (Tailcone)** (See figure 201.)
      (1) Lower tailcone access door.
      (2) Disconnect aircraft batteries.
      (3) Loosen hold down nut and remove emergency battery from battery rack and aircraft.
   B. **Install Emergency Battery (Tailcone)** (See figure 201.)
      (1) Install emergency battery in battery rack and secure with hold down nut.
      (2) Connect aircraft batteries.
      (3) Secure tailcone access door.
   C. **Remove Emergency Battery (Nose Compartment)** (See figure 201.)
      (1) Lower tailcone access door and disconnect aircraft batteries.
      (2) Remove nose compartment access door.
      (3) Loosen hold down nut and remove emergency battery from battery rack and aircraft.
   D. **Install Emergency Battery (Nose Compartment)** (See figure 201.)
      (1) Install emergency battery in battery rack and secure with hold down nut.
      (2) Connect aircraft batteries and secure tailcone access door.
      (3) Install and secure nose compartment access doors.



TYPICAL TAILCONE INSTALLATION



TYPICAL NOSE COMPARTMENT INSTALLATION

Emergency Battery Installation
Figure 201

EXTERNAL POWER - DESCRIPTION AND OPERATION

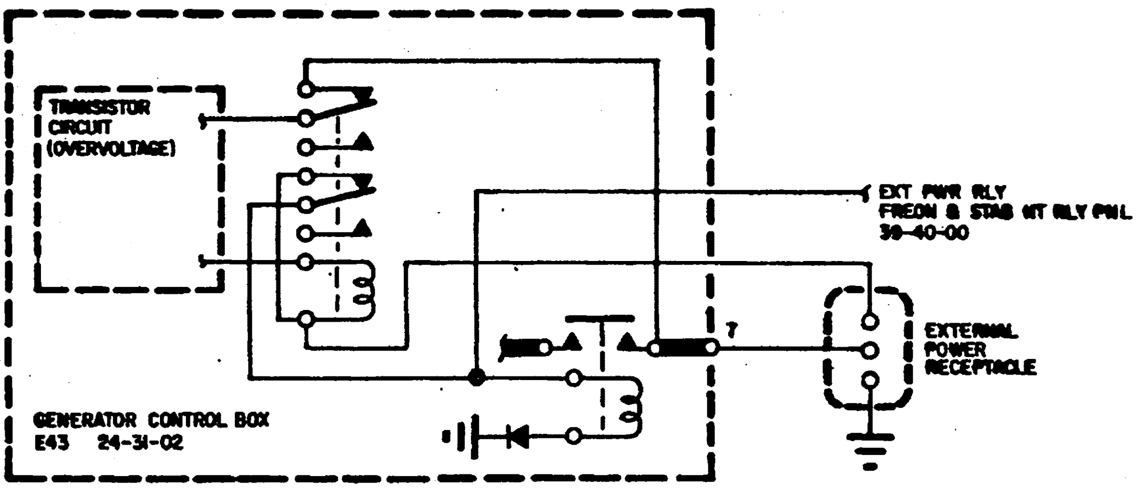
1. DESCRIPTION

A. External power is connected to the aircraft through a standard external power receptacle installed on the left side of the aft fuselage adjacent to the tailcone access door.

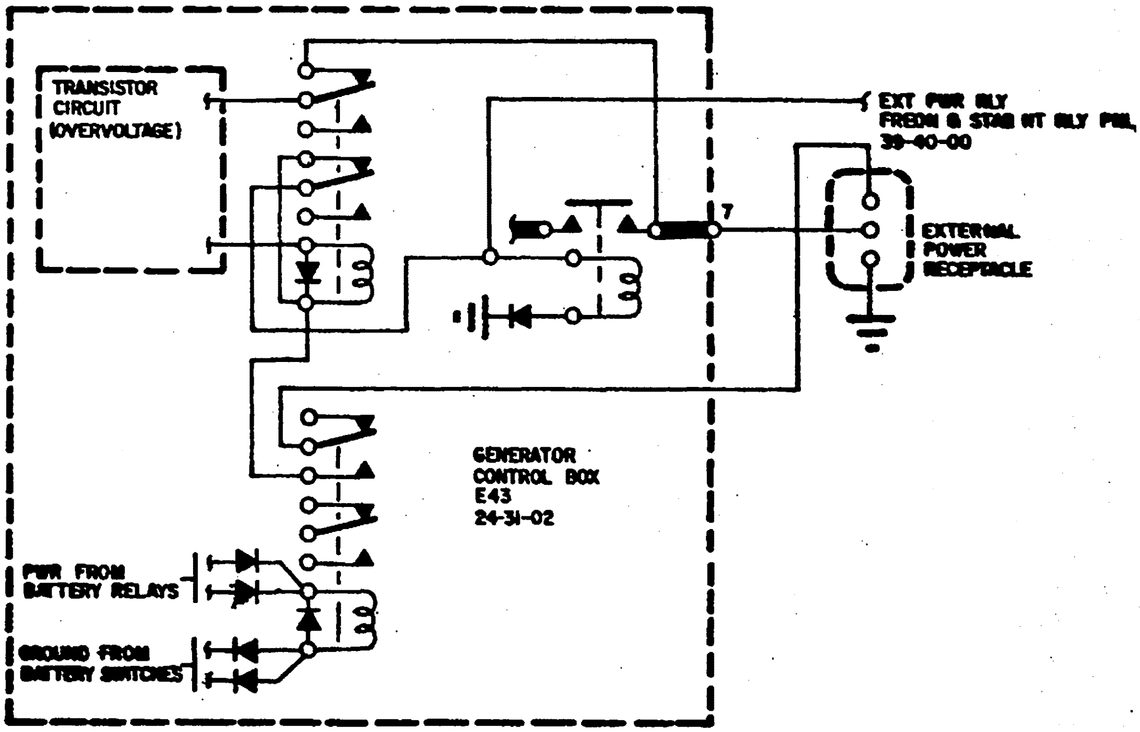
(1) All controls of the external power system are incorporated within the generator control box with the exception of the battery switches. Maintenance practices are limited to a visual check of wiring, replacement of the generator control box or the power receptacle.

2. OPERATION (See figure 1 and 2.)

A. With external power connected to the aircraft a power circuit is completed to the deenergized external power control relay. Setting either Battery Switch to ON, completes a ground circuit to energize the external power control relay. With the control relay energized, power is applied through the deenergized external power overvoltage cutout circuit to energize the external power relay. If external power input exceeds 32 (±1) volts, the zener diode breaks down and applies power to the gated diode. The gated diode will in turn break down and complete the ground circuit to the overvoltage cutout relay. The overvoltage cutout relay will energize and remove power from the external power relay, disconnecting external power from the aircraft.



External Power Electrical Control Schematic
Figure 1



External Power Electrical Control Schematic
Figure 2

EXTERNAL POWER RECEPTACLE - MAINTENANCE PRACTICES

1. REMOVAL/INSTALLATION

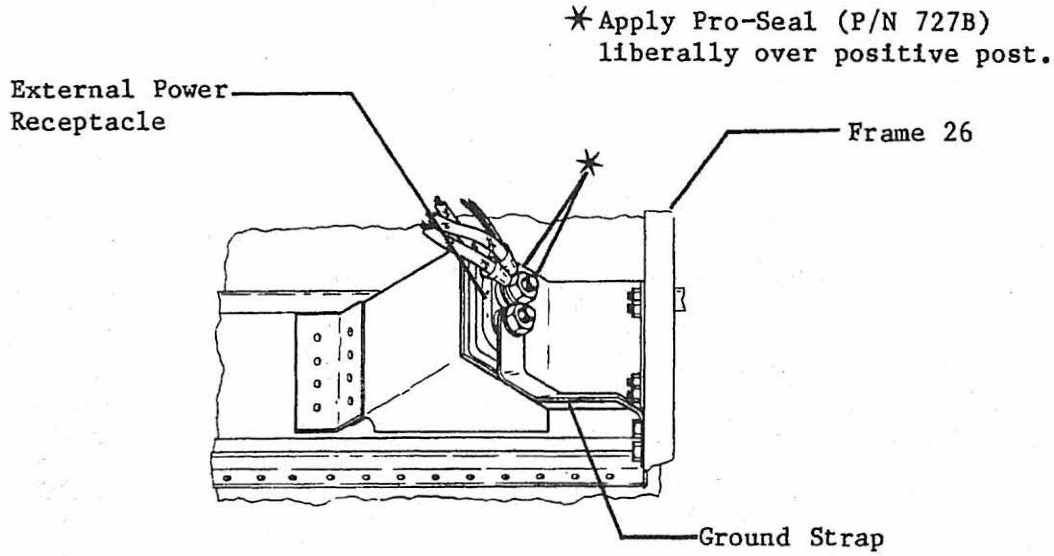
A. Remove External Power Receptacle (See figure 201.)

NOTE: Refer to 24-31-02 for removal and installation procedures of the generator control box.

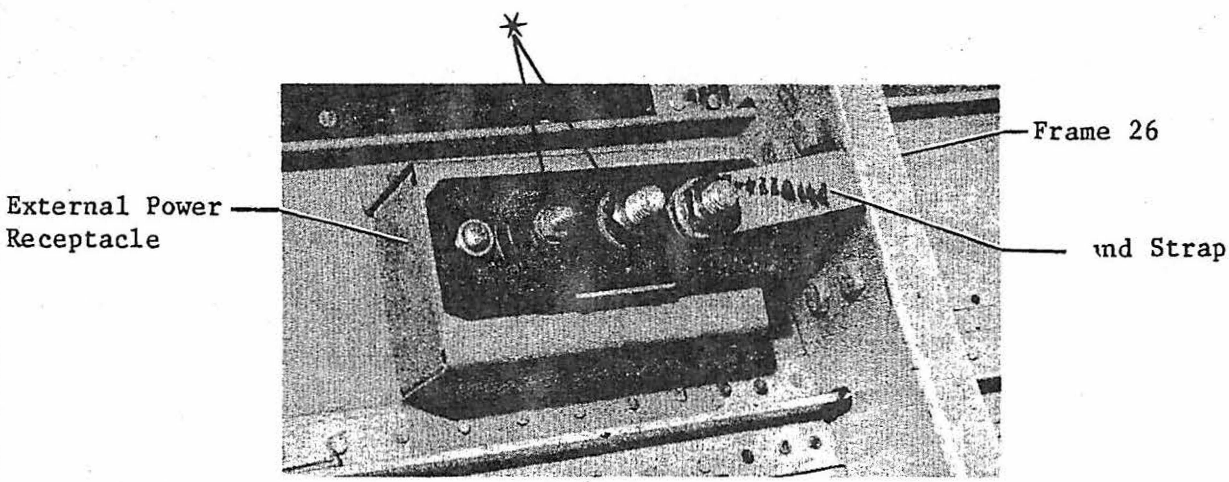
(1) Lower tailcone access door and disconnect aircraft batteries.
(2) Remove sealant from exterior surface of support structure opening and from positive (+) terminals on receptacle.
(3) Disconnect wiring and remove ground strap from terminals of receptacle. Tag wiring.
(4) Remove attaching parts and receptacle from aircraft.

B. Install External Power Receptacle (See figure 201.)

(1) Remove old sealant from exterior surface of support structure. Clean surface of support structure and outside of receptacle with methyl ethyl ketone.
(2) Install receptacle and secure with attaching parts.
(3) Seal around exterior surface of support structure and edge of receptacle using Pro Seal 890, Class B.
(4) Connect electrical wiring and ground strap to receptacle.
(5) Torque large terminals on receptacle 85 to 140 inch-pounds and small terminal 10 to 12 inch-pounds.
(6) Clean positive (+) terminals with O-A-51 acetone and allow to air dry for 15 minutes.
(7) Apply Pro Seal (727B) liberally over terminals as shown. Assure that all bare metal is covered.



Aircraft 25-070 thru 25-081



Aircraft 25-082 and Subsequent

VIEW LOOK OUTBOARD AT LH SIDE

External Power Receptacle Installation
Figure 201

ELECTRICAL LOAD DISTRIBUTION - DESCRIPTION AND OPERATION

1. DESCRIPTION

A. Electrical load distribution, from the power sources to the various using systems, is accomplished by wiring, current limiters and circuit breakers.

B. The circuit breaker panels are installed in the cockpit on the RH and LH sides. Push-pull type circuit breakers are installed to protect all electrical systems in the aircraft. All primary circuit breakers are located on the left panel and all secondary breakers on the right panel. The DC circuit breakers are thermal type and the AC are magnetic type.

(1) Each circuit breaker panel is divided into five major buses; ESS BUS, MAIN BUS, AC BUS, BAT BUS and MAIN PWR BUS.

C. The generator control panel is divided into four major buses: R GEN BUS, L GEN BUS, BAT BUS and BAT CHARGING BUS.

D. On Aircraft 25-318 and Subsequent, the maximum allowable bus loading shall not exceed the following limits. The values are the maximum allowable current with the "Bus-Tie" circuit breaker open.

	Each Ess Bus	Each Main Bus
Continuous	28 Amperes	49 Amperes
Intermittent (5-Min.)	31 Amperes	55 Amperes
Transient (5-Sec.)	51 Amperes	90 Amperes

E. If electrical equipment is added, an electrical load check must be performed. The load check may be performed with the additional equipment temporarily connected to the applicable bus.

F. AC power supplied by the inverters is applied through the paralleling control to the pilot's and copilot's circuit breaker panel AC load buses. The individual buses are protected by circuit breakers. Each bus (pilot's and copilot's) is connected by a bus tie circuit breaker.

G. The R and L Generator buses are connected by current limiters to the battery charging and battery bus. From these buses, power is distributed to various using systems and the major buses in each circuit breaker panel through current limiters. The ESS BUS and MAIN BUS are connected to its respective bus in the opposite panel.

H. The current limiter panel distributes electrical power from the batteries, generators, and external ground power to systems throughout the aircraft. Current limiters and current limiting fuses are protective devices utilized to protect against excessive current loads. The current limiter panel is installed on the generator control panel located in the tailcone just aft of the aft engine beam. The current limiter panel consists of a non-conductive plate assembly, bracket assemblies, bus bars, guard assembly, and cover assembly.

(1) **Plate Assembly**

(a) The non-conductive phenolic material has tee nuts cemented flush with the plate in countersunk holes. The tee nuts provide mounting for fuse holders, securing bus bars, and guard assembly.

(b) Enclosed link limiter type fuse (MS28937) is installed in the fuse holder. Ampere rating is marked on top surface of the limiter.

(2) Bracket assemblies contain electrical connectors. Electrical connectors provide quick electrical connect/disconnect during removal and installation.

(3) **Bus Bars**

(a) Silver plated copper bus bars are utilized in the current limiter panel. The bus bars are identified by nomenclature as: left generator bus, right generator bus, and battery charging bus.

(b) NAS bolts are silver soldered to the silver plated copper bars. These bolts provide the connecting points for current limiters, fuses, and other electrical circuits.

(4) **Guard Assembly**

(a) A guard assembly wraps around three sides of the current limiter panel as a protection for current limiter components. The guard assembly attaches to the plate assembly and provides attach points for the cover assembly.

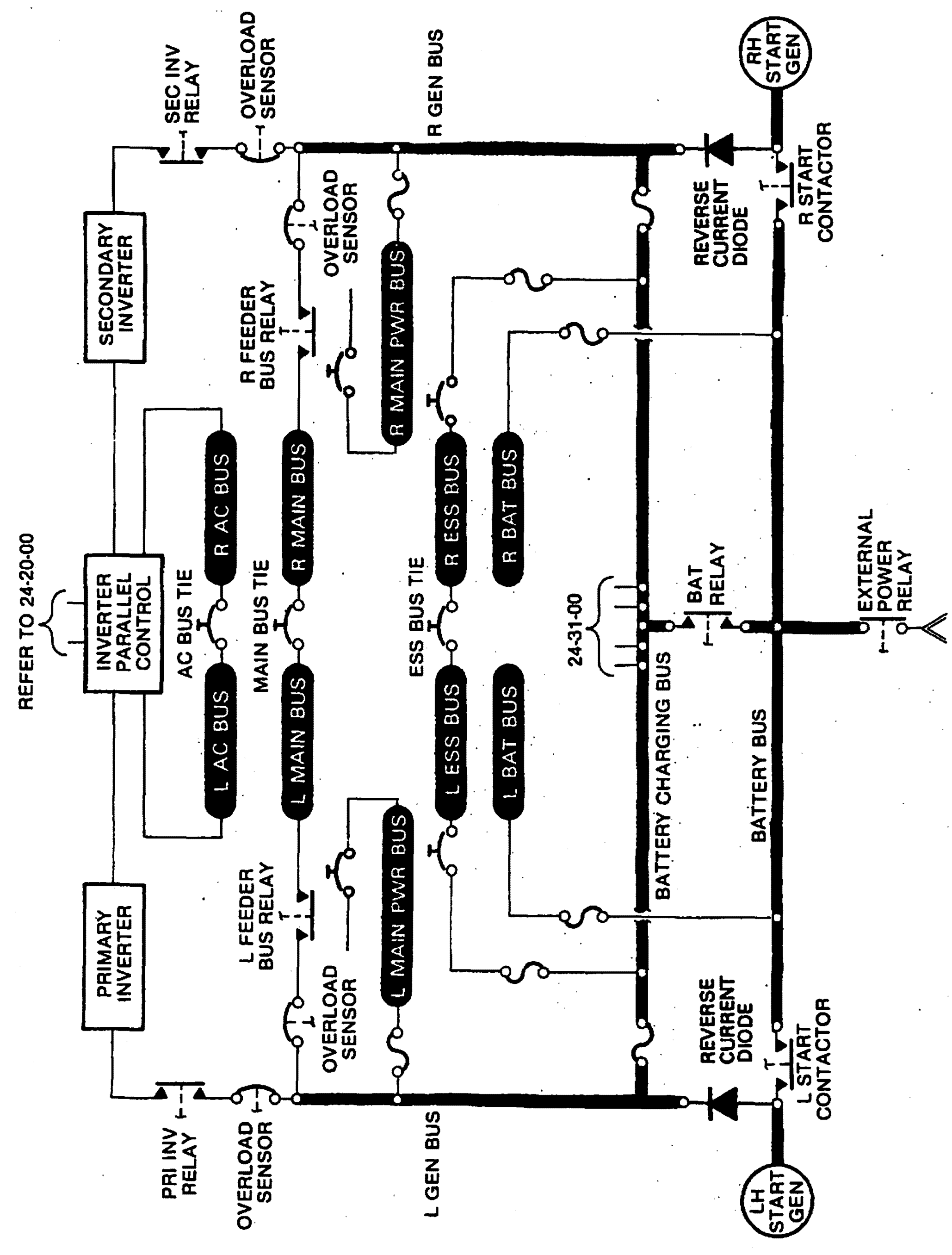
(5) **Cover Assembly**

(a) The cover assembly consists of a frame and plexiglass that attaches to the guard assembly with screws. Current limiters and fuses are installed in the current limiter panel so maintenance personnel can visually determine their condition (blown indicator) with cover assembly installed.

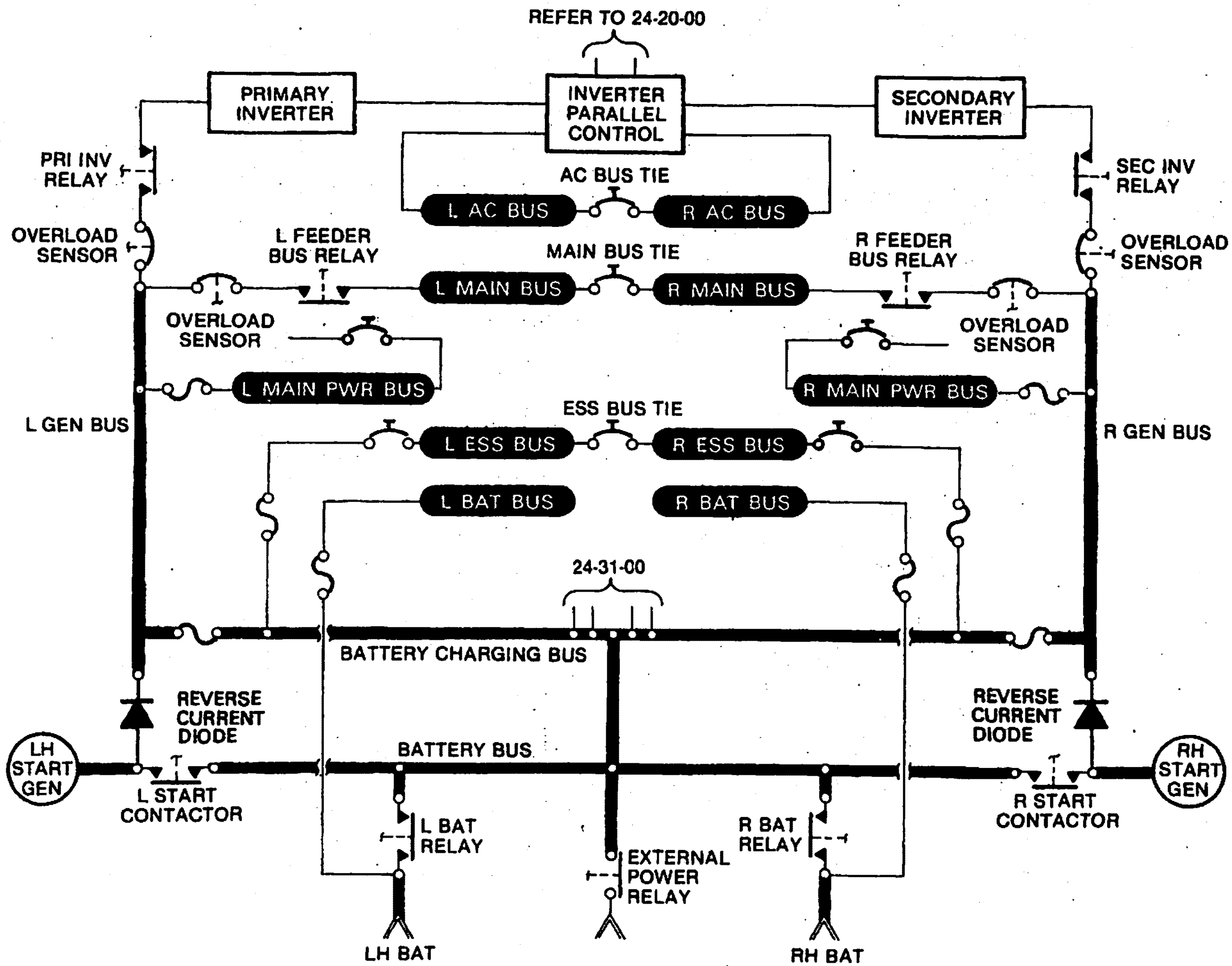
(b) The cover assembly has location, reference designator, and amperage of current limiters and fuses silkscreened on the plexiglass.

(6) **Spare Current Limiters and Fuses**

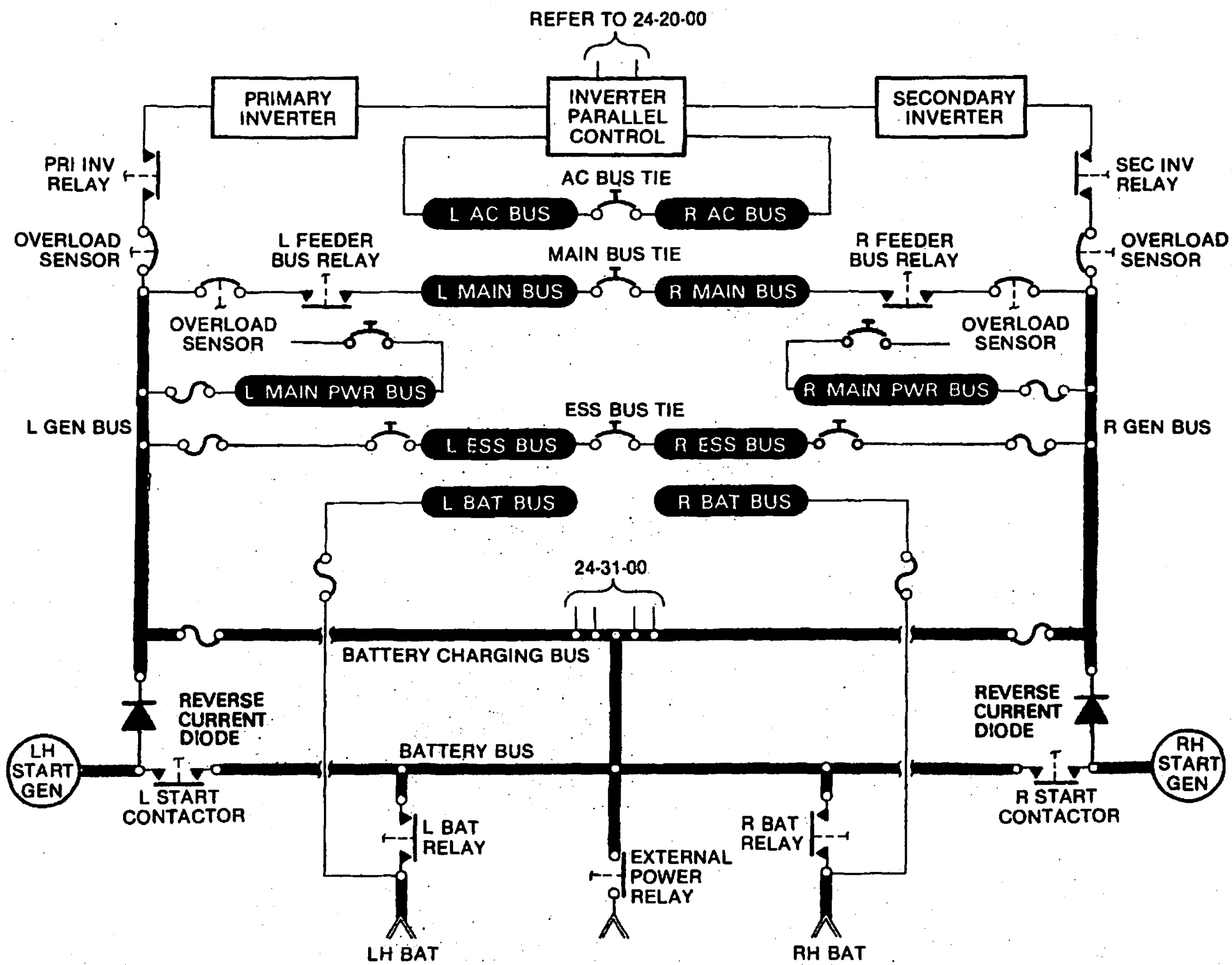
(a) A box assembly containing spare current limiters and fuses is installed on the lower side of the LH electrical equipment tray. To gain access to the spare current limiters and fuses, remove a screw at each end of the box assembly and remove box assembly from the electrical equipment tray.



Electrical Power Distribution System Schematic
Figure 1 (Sheet 1 of 3)



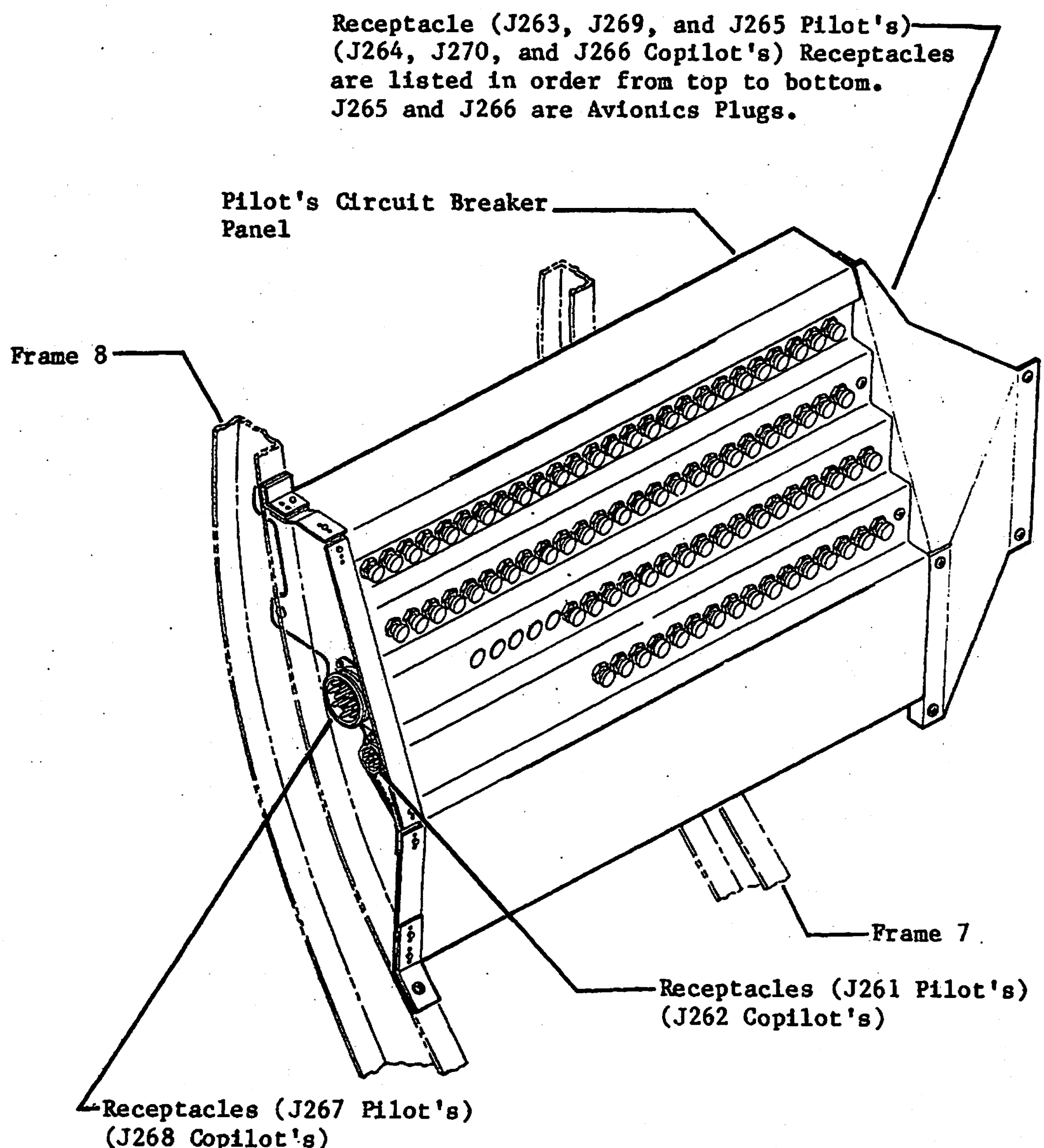
Electrical Power Distribution System Schematic
Figure 1 (Sheet 2 of 3)



Electrical Power Distribution System Schematic
Figure 1 (Sheet 3 of 3)

CIRCUIT BREAKER PANEL - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**
   A. **Remove Circuit Breaker Panel** (See figure 201.)
      (1) Disconnect aircraft batteries.
      (2) Remove screws securing circuit breaker panel to aircraft structure.
      (3) Tilt panel sufficiently to allow disconnection of electrical plugs.
      (4) Disconnect electrical plugs and remove circuit breaker panel from aircraft.
      (5) To allow removal of defective circuit breaker, the bus bar which is attached to the defective circuit breaker must be completely removed. Remove attaching parts and bus bar from circuit breakers.
      (6) Disconnect electrical wiring from remaining terminal of circuit breaker.
      (7) Remove jamnut, lockwasher, keyway, and circuit breaker from panel.
      (8) Place keyway on replacement circuit breaker and install circuit breaker in panel. Assure keyway is positioned in hole of panel doubler.
      (9) Secure circuit breaker to panel with lockwasher and jamnut.
      (10) Install previously removed bus bar on circuit breakers.
      (11) Connect electrical wiring to replacement circuit breaker.
   B. **Install Circuit Breaker Panel** (See figure 201.)
      (1) Install circuit breaker panel and connect electrical plugs.
      (2) Secure circuit breaker panel to aircraft structure.
      (3) Connect aircraft batteries.
      (4) Perform operational check of replacement circuit breaker system.

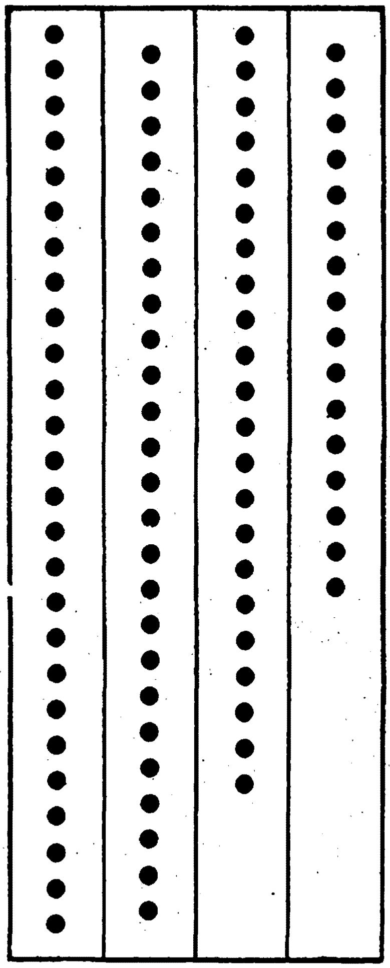


NOTE: All plug and receptacle wire bundles have yellow identification tags secured to them.

Circuit Breaker and Circuit Breaker Panel Installation
Figure 201 (Sheet 1 of 2)

FORWARD END OF PANEL

TOP
ROW



AFT END OF PANEL

Circuit Breaker Panel Numbering Diagram

NOTE:
- The circuit breakers and plug buttons in the pilot's panel are numbered 1, 3, 5, etc., starting with the top row and counting from the forward end to the aft end. The numbering continues with the forward end of row 2 and so on.
- The numbering of the copilot's circuit breaker panel is identical to the pilot's panel, except the circuit breakers are numbered 2, 4, 6, etc.

Circuit Breaker and Circuit Breaker Panel Installation
Figure 201 (Sheet 2 of 2)

CURRENT LIMITER PANEL - MAINTENANCE PRACTICES

1. **REMOVAL/INSTALLATION**

   A. **Remove Current Limiter Panel** (See figure 201 thru 204.)

   NOTE: Removal of the current limiter panel requires partial disassembly of the panel. Retain attaching hardware for installation.

   (1) Disconnect aircraft batteries.
   (2) Disconnect electrical connectors from current limiter panel.
   (3) Remove cover assembly from current limiter panel.
   (4) Remove panel guard assembly from current limiter panel.

   NOTE: On Aircraft 25-076 thru 25-089, 25-091 thru 25-102; remove screw and nut mounted fuse holder attached to the panel guard assembly or disconnect electrical wires from the fuse holder.

   (5) Remove nut, lockwasher, washer, and wire terminals (wire buses to current limiters FL4 cabin climate and FL5 stabilizer heat) from Stud COM (also attaches one end of FL3) on generator control panel.

   NOTE: Nut, lockwasher, and washer are furnished with generator control panel. The nut is silver plated copper and shall be reinstalled on Stud COM.

   (6) Remove nut, lockwasher, and washer are furnished with generator control panel. The nut is silver plated copper and shall be reinstalled on Stud 3 and Stud 4.
   (7) Remove nut, lockwasher, washer, and wire terminal (left nacelle heat and right nacelle heat) from studs securing current limiters FL33 and FL34.
   (8) Remove fuse buses and wire terminals from battery charging bus, left generator bus, and right generator bus. (Refer to removal, paragraph 1.C.)
   (9) Remove screws attaching current limiter panel plate assembly to generator control panel. Remove plate assembly.

   B. **Install Current Limiter Panel** (See figure 201 thru 204.)

   WARNING: IF CURRENT LIMITER PANEL IS BEING INSTALLED WITH GENERATOR CONTROL PANEL INSTALLED IN AIRCRAFT, VERIFY AIRCRAFT BATTERIES AND EXTERNAL POWER UNIT ARE DISCONNECTED DURING INSTALLATION OF CURRENT LIMITER PANEL.

   (1) Install fuse holders on current limiter panel plate assembly. (Refer to Fuse Holder installation, paragraph 1.J.)
   (2) Install bus bars (battery charging bus, left generator bus, and right generator bus) on generator control panel. (Refer to Bus Bar installation, paragraph 1.H.)

(3) Position current limiter panel plate assembly on generator control panel.

NOTE: Use care to prevent wires being trapped between current limiter panel plate assembly and generator control panel.

(4) Position panel guard assembly to current limiter plate assembly and generator control panel and attach with screws.

NOTE: On Aircraft 25-076 thru 25-089, 25-091 thru 25-102, attach fuse holder to panel guard assembly (screw and nut mounted). Connect electrical wires to fuse holder. (Refer to Wire Diagram Manual for wire hookup.)

(5) Attach FL15 fuse bus wire to Stud 3 and FL18 fuse bus wire to Stud 4 on generator control panel.

NOTE: Remove nut, lockwasher, and washer from Stud 3 and Stud 4. Attach the fuse bus wire terminals. (Refer to Generator Control Panel Installation 24-31-02 for nut torque value.)

(6) Install current limiter FL3. (Refer to Current Limiter FL3 installation, paragraph 1.F.)
(7) Install current limiters FL1 and FL2. (Refer to Current Limiter FL1 and FL2 installation, paragraph 1.D.)
(8) Install current limiter(s) FL33 and FL34. (Refer to Current Limiter Fl33 and FL34 installation, paragraph 1.K.)
(9) Attach solid copper buses, wire buses, and wire terminals to fuse holders and studs on battery charging bus, left generator bus, and right generator bus.

NOTE:
- Do not attach buses or wire terminals to studs attaching current limiter FL1, FL2, and FL3, except wires to current limiters FL4 and FL5.
- On left and right generator bus, stack buses and wire terminals on studs attaching current limiters FL33 and FL34.
- Use applicable current limiter panel installation figure and applicable Wiring Diagram.

(10) Install washer, lockwasher, and nut on studs (four steel studs) that attach fuse buses and wire terminals in the preceding step. Torque nuts to 75 (±3) inch-pounds.
(11) Perform functional check of DC Feeder Bus.

C. **Remove Current Limiter (FL1, FL2)** (See figure 201 thru 204.)
(1) Disconnect aircraft batteries.
(2) Remove cover assembly from current limiter panel.
(3) Remove nut, lockwasher, and washer from battery charging bus and left generator bus or right generator bus. Retain attaching hardware for installation.
(4) Lift current limiter FL1 or FL2 from bus bar studs.

D. **Install Current Limiter (FL1, FL2)** (See figure 201 thru 204.)

WARING: IF CURRENT LIMITER IS BEING INSTALLED WITH GENERATOR CONTROL PANEL INSTALLED IN AIRCRAFT, VERIFY AIRCRAFT BATTERIES AND EXTERNAL POWER UNIT ARE DISCONNECTED DURING INSTALLATION OF CURRENT LIMITER.

(1) Remove nut, lockwasher, and washer from battery charging bus and left generator bus or right generator bus.
(2) Position the current limiter so condition of the current limiter can be viewed when cover assembly is installed.
(3) Install FL1 and FL2 across battery charging bus and left generator bus or right generator bus.
(4) Install washer, lockwasher, and nut securing Fl1 and FL2. Torque nut 75 (±3) inch-pounds.

NOTE: If DC Feeder Bus Functional Check is required, do not torque nuts securing FL1 and FL2 until DC Feeder Bus Functional Check is completed.

(5) If current limiter is being installed with generator control panel installed in aircraft, install cover assembly on current limiter panel.
(6) Perform a current limiter check. (Refer to Inspection/Check in 24-31-01.)

E. **Remove Current Limiter (FL3)** (See figure 201 thru 204.)
(1) Disconnect aircraft batteries.
(2) Remove cover assembly from current limiter panel.
(3) Remove nut, lockwasher, and washer from Stud C and Stud COM on generator control panel.

NOTE: Nut, lockwasher, and washer are furnished with generator control panel. The nut is silver plated copper and shall be reinstalled on Stud C and Stud COM.

(4) Remove wire terminals (wire buses to current limiters FL4 cabin climate and FL5 stabilizer heat) from Stud COM.
(5) Lift current limiter FL3 from Stud C and Stud COM.

F. **Install Current Limiter (FL3)** (See figure 201 thru 204.)

WARING: IF CURRENT LIMITER IS BEING INSTALLED WITH GENERATOR CONTROL PANEL INSTALLED IN AIRCRAFT, VERIFY AIRCRAFT BATTERIES AND EXTERNAL POWER UNIT ARE DISCONNECTED DURING INSTALLATION OF CURRENT LIMITER.

(1) Remove nut, lockwasher, and washer from Stud C and Stud COM on generator control panel. Remove wire terminals (wires to current limiters FL4 and FL5) if installed on Stud COM.
(2) Position the current limiter so condition of the current limiter can be viewed when cover assembly is installed.
(3) Install washer(s) on Stud COM and install current limiter FL3 across Stud C and Stud COM.

NOTE: Washer(s) are installed on Stud COM to prevent damage to FL3 when nuts are torqued.

(4) Install current limiter (FL4 and FL5) wire terminals on Stud COM.
(5) Install washer, lockwasher, and nut (silver plated copper) on Stud C and Stud COM. Torque nuts 30 to 42 inch-pounds.
(6) If current limiter is being installed with generator control panel installed in aircraft, install cover assembly on current limiter panel.

G. **Remove Bus Bars (Battery Charging Bus, Left Generator Bus, Right Generator Bus)** (See figure 201 thru 204.)
(1) Disconnect aircraft batteries.
(2) Remove cover assembly from current limiter panel.
(3) Remove Battery Charging Bus
(a) Remove current limiters FL1 and FL2. (Refer to Current Limiter FL1 and FL2 removal, paragraph 1.C.)
(b) Remove current limiter FL3. (Refer to Current Limiter FL3 removal, paragraph 1.E.)
(c) Remove two screws, one at each end of battery charging bus, securing the battery charging bus to top panel on generator control panel.
(d) Remove battery charging bus and two spacers.
(4) Remove Left Generator Bus or Right Generator Bus
(a) Remove current limiter FL1 or FL2. (Refer to Current Limiter FL1 or FL2 removal, paragraph 1.C.)
(b) Remove nut, lockwasher, and washer from Stud A (right generator bus) or Stud B (left generator bus).

NOTE: Nut, lockwasher, and washer are furnished with the generator control panel. The nut is silver plated copper and shall be reinstalled on Stud A or Stud B.

(c) Remove nut, lockwasher, washer, fuse buses, and wire terminals from studs on left generator bus or right generator bus.

(d) Remove current limiter FL33 from left generator bus or current limiter FL34 from right generator bus. Remove nut, lockwasher, washer, and wire terminal from studs securing current limiters FL33 and FL34 to plate assembly.
(e) Remove one screw, near the center of the bus bar, securing generator bus bar to top panel on generator control panel.
(f) Remove left or right generator bus and spacer.

H. **Install Bus Bars (Battery Charging Bus, Left Generator Bus, Right Generator Bus)** (See figure 201 thru 204.)

WARNING: IF BUS BARS ARE TO BE INSTALLED WITH GENERATOR CONTROL PANEL INSTALLED IN AIRCRAFT, VERIFY AIRCRAFT BATTERIES AND EXTERNAL POWER UNIT ARE DISCONNECTED FROM THE AIRCRAFT DURING INSTALLATION OF BUS BARS.

(1) **Battery Charging Bus**
(a) Remove nut, lockwasher, and washer from Stud C.
(b) Position battery charging bus on Stud C.
(c) Position and align spacer, on each side of Stud C, between battery charging bus and top panel on generator control panel.
(d) Secure battery charging bus to top panel on generator control panel with screws.
(e) Install washer, lockwasher, and nut (silver plated copper) on Stud C. Do not tighten nut at this time.

(2) **Left Generator Bus**
(a) Remove nut, lockwasher, and washer from Stud B.
(b) Position left generator bus on Stud B.
(c) Position and align spacer between left generator bus and top panel on generator control panel.
(d) Secure left generator bus to top panel on generator control panel with a screw.
(e) Install washer, lockwasher, and nut (silver plated copper) on Stud B. Torque nut 30 to 42 inch-pounds.

(3) **Right Generator Bus**
(a) Remove nut, lockwasher, and washer from Stud A.
(b) Position right generator bus on Stud A.
(c) Position and align spacer between right generator bus and top panel on generator control panel.
(d) Secure right generator bus to top panel on generator control panel with a screw.
(e) Install washer, lockwasher, and nut (silver plated copper) on Stud A. Torque nut 30 to 42 inch-pounds.

(4) Install current limiter FL3. (Refer to Current Limiter FL3 installation, paragraph 1.F.)
(5) Install current limiter FL1 and FL2. (Refer to Current Limiter FL1 and FL2 installation, paragraph 1.D.)
(6) Install current limiters FL33 and FL34. (Refer to Current Limiter FL33 and FL34 installation, paragraph 1.K.)

(7) Attach solid copper buses and wire terminals to studs on battery charging bus, left generator bus, and right generator bus.

NOTE: Use applicable current limiter panel installation figure and applicable Wiring Diagram.

(8) Install washer, lockwasher, and nut on studs (four steel studs) that attach fuse buses and wire terminals in the preceding step. Torque nuts to 75 (±3) inch-pounds.

(9) Perform functional check of DC Feeder Bus.

I. **Remove Fuse Holder** (See figure 201 thru 204.)

(1) Disconnect aircraft batteries.
(2) Remove cover assembly from current limiter panel.
(3) Pull fuse from fuse holder (the fuse has knife blade type contacts).
(4) Disconnect electrical wires from fuse holder.
(5) Remove screws securing fuse holder to plate assembly.

J. **Install Fuse Holder** (See figure 201 thru 204.)

WARNING: IF FUSE HOLDER IS TO BE INSTALLED WITH GENERATOR CONTROL PANEL INSTALLED IN AIRCRAFT, VERIFY AIRCRAFT BATTERIES AND EXTERNAL POWER UNIT ARE DISCONNECTED DURING INSTALLATION OF FUSE HOLDER.

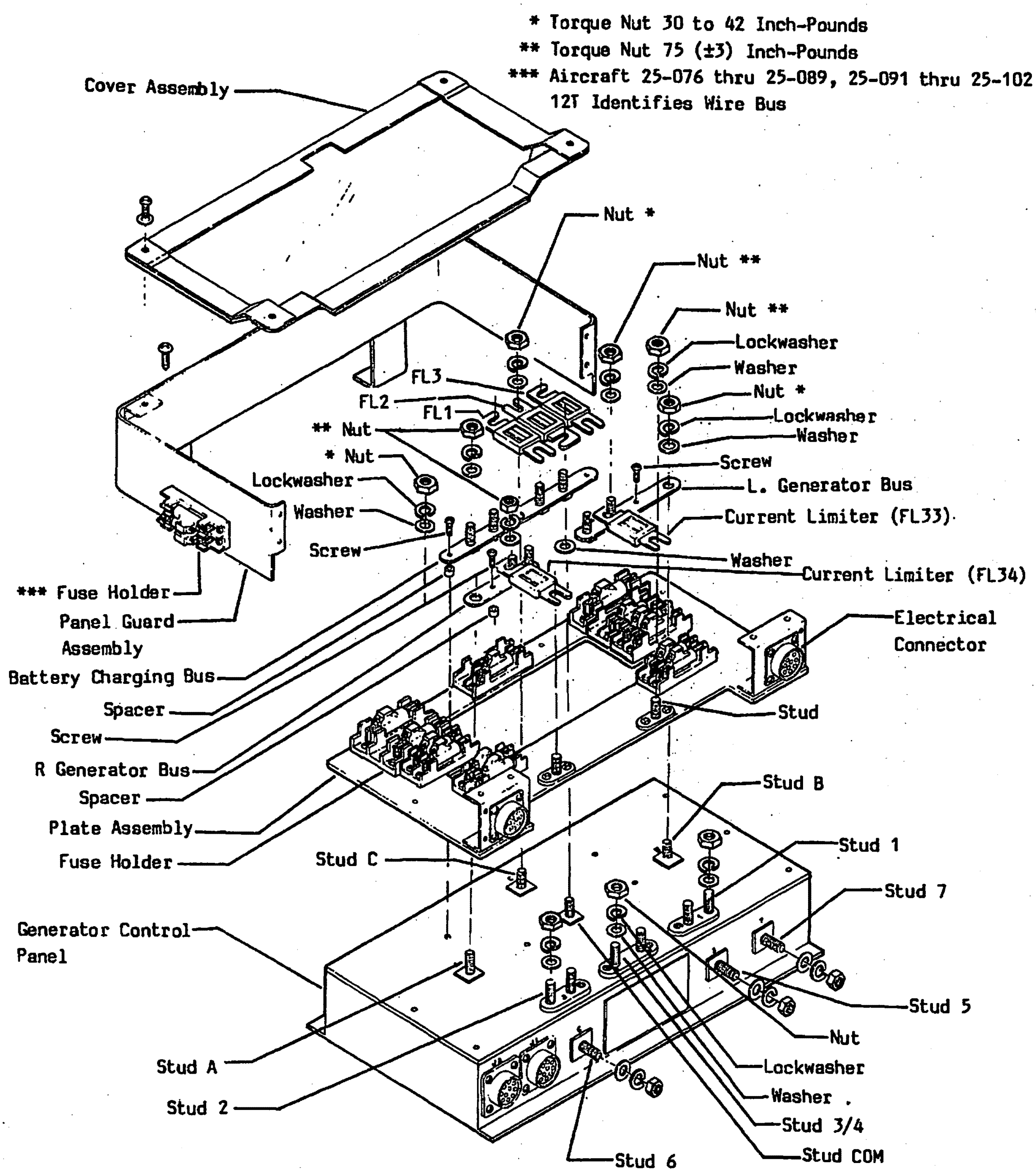
(1) Utilize cover assembly from current limiter panel to locate fuse holder.
(2) With fuse(s) removed from fuse holder, attach fuse holder to plate assembly with screws.
(3) Attach fuse bus and/or fuse wire to fuse holder. (Refer to Current Limiter Panel Installation figure and applicable Wiring Diagram.)
(4) Insert fuse in fuse holder so that blown indicator is visible when cover assembly is installed.
(5) If fuse holder is being installed with current limiter panel installed in aircraft, install cover assembly on current limiter panel.

K. **Install Current Limiters (FL33, FL34)**

(1) Remove nut, lockwasher, and washer from left or right generator bus and corresponding current limiter mounting stud.
(2) Install current limiter FL33 or FL34.

NOTE: If nacelle heat wire terminals are attached to mounting studs, lift wire terminals from the mounting studs and install current limiter FL33 and FL34. Reinstall nacelle heat wire terminals on mounting studs.

(3) Attach left nacelle heat wire on the mounting stud with current limiter FL33 and right nacelle heat wire on the mounting stud with current limiter FL34.
(4) Install washer, lockwasher, and nut removed in step 1.L.(1). Torque nuts on current limiter mounting studs to 75 (±3) inch-pounds. (Refer to Bus Bar installation, paragraph 1.H. for torquing nuts on left and right generator bus.)



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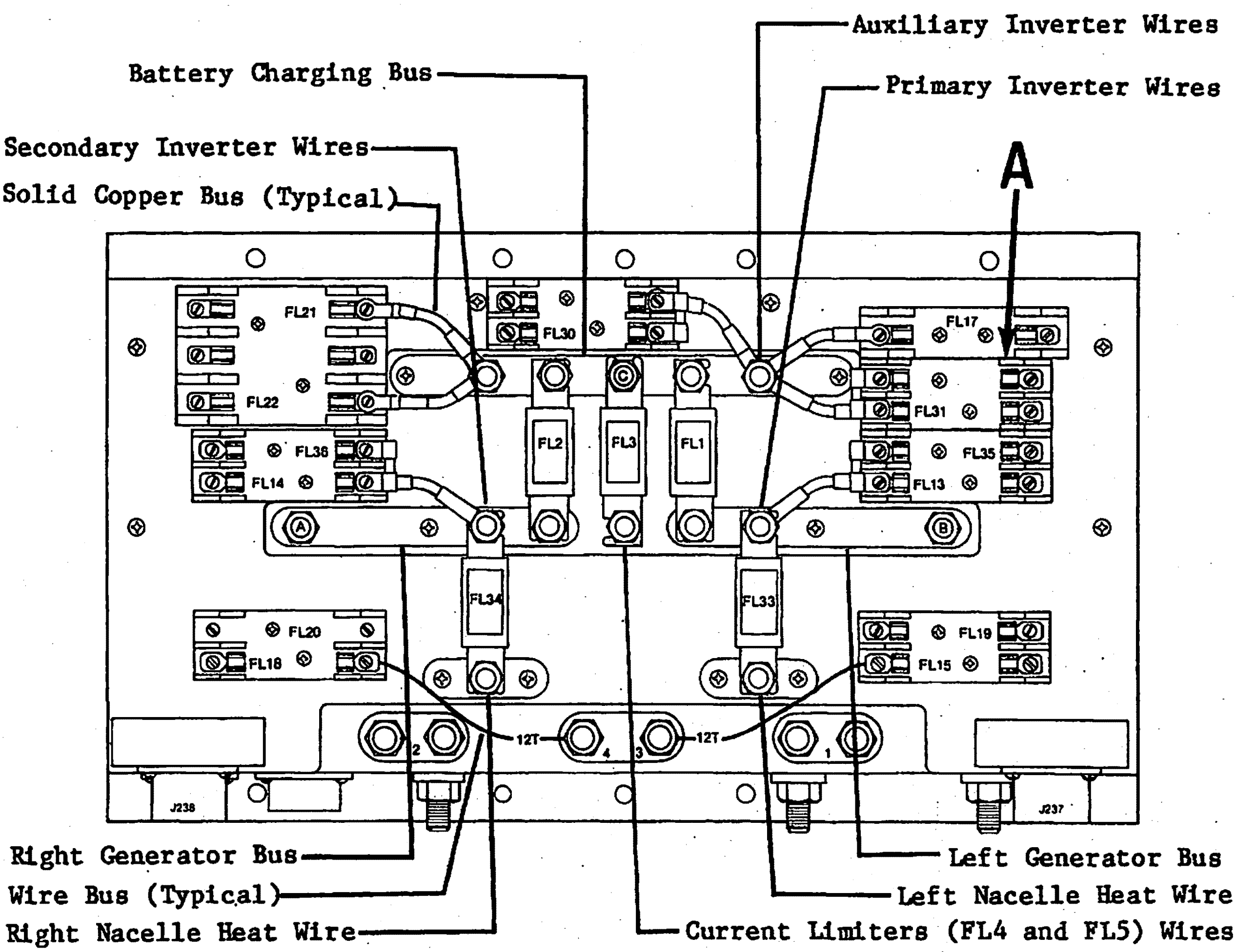
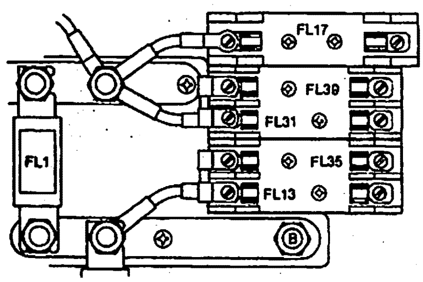
Current Limiter Panel Installation
Figure 201 (Sheet 1 of 3)





R.H. Recognition Light Option

Detail A

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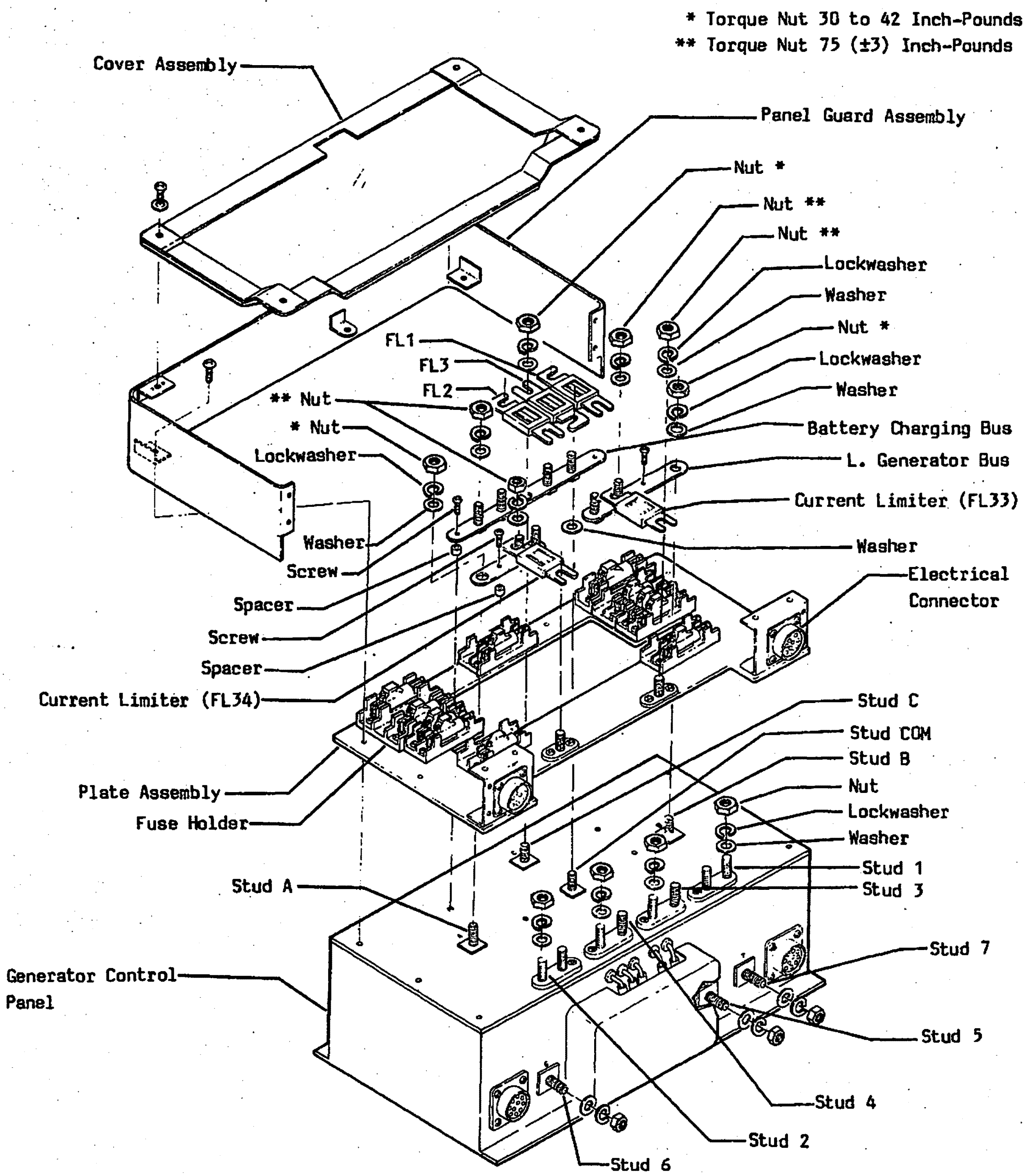
Current Limiter Panel Installation
Figure 201 (Sheet 2 of 3)

Current Limiter and Fuse Reference Designator	Amperage	Circuit	Power Source	Current Limiter Panel 2418053 -6	Current Limiter Panel 2418053 -8
FL1	275	L. Gen. Bus to Bat. Charging Bus		X	X
FL2	275	R. Gen. Bus to Bat. Charging Bus		X	X
FL3	275	Cabin Climate and Stabilizer Heat	Bat. Charging Bus	X	X
FL13	20	Left Landing Light	Left Generator Bus	X	X
FL14	20	Right Landing Light	Right Generator Bus	X	X
FL15	20	Left Battery Bus	Bat. No. 1 (Stud 3)	X	X
FL17	50	Left Essential Bus	Bat. Charging Bus	X	X
FL18	20	Right Battery Bus	Bat. No. 2 (Stud 4)	X	X
FL19	10	Left Generator Field	L. Start Gen. Bus (Stud 1)	X	X
FL20	10	Right Generator Field	R. Start Gen. Bus (Stud 2)	X	X
FL21	50	Electric Hyd. Pump	Bat. Charging Bus	X	X
F122	50	Right Essential Bus	Bat. Charging Bus	X	X
FL30	10	Fuel Flow	Bat. Charging Bus	X	X
FL31	5	Utility Light	Bat. Charging Bus	X	X
FL33	100	Left Nacelle Heat	Left Generator Bus	X	X
FL34	100	Right Nacelle Heat	Right Generator Bus	X	X
FL35	10	L. Main Power Bus	Left Generator Bus	X	X
FL36	10	R. Main Power Bus	Right Generator Bus	X	X
FL39	20	Recog. Light	Bat. Charging Bus		X
				-1	-11
				2418093 Cover Assembly	

-6 Current Limiter Panel - Aircraft 25-061, 25-070 thru 25-089 and 25-091 thru 25-102.

-8 Current Limiter Panel - Aircraft 25-061, 25-070 thru 25-089 and 25-091 thru 25-102 when Optional Recognition Light is installed.

Current Limiter Panel Installation
Figure 201 (Sheet 3 of 3)



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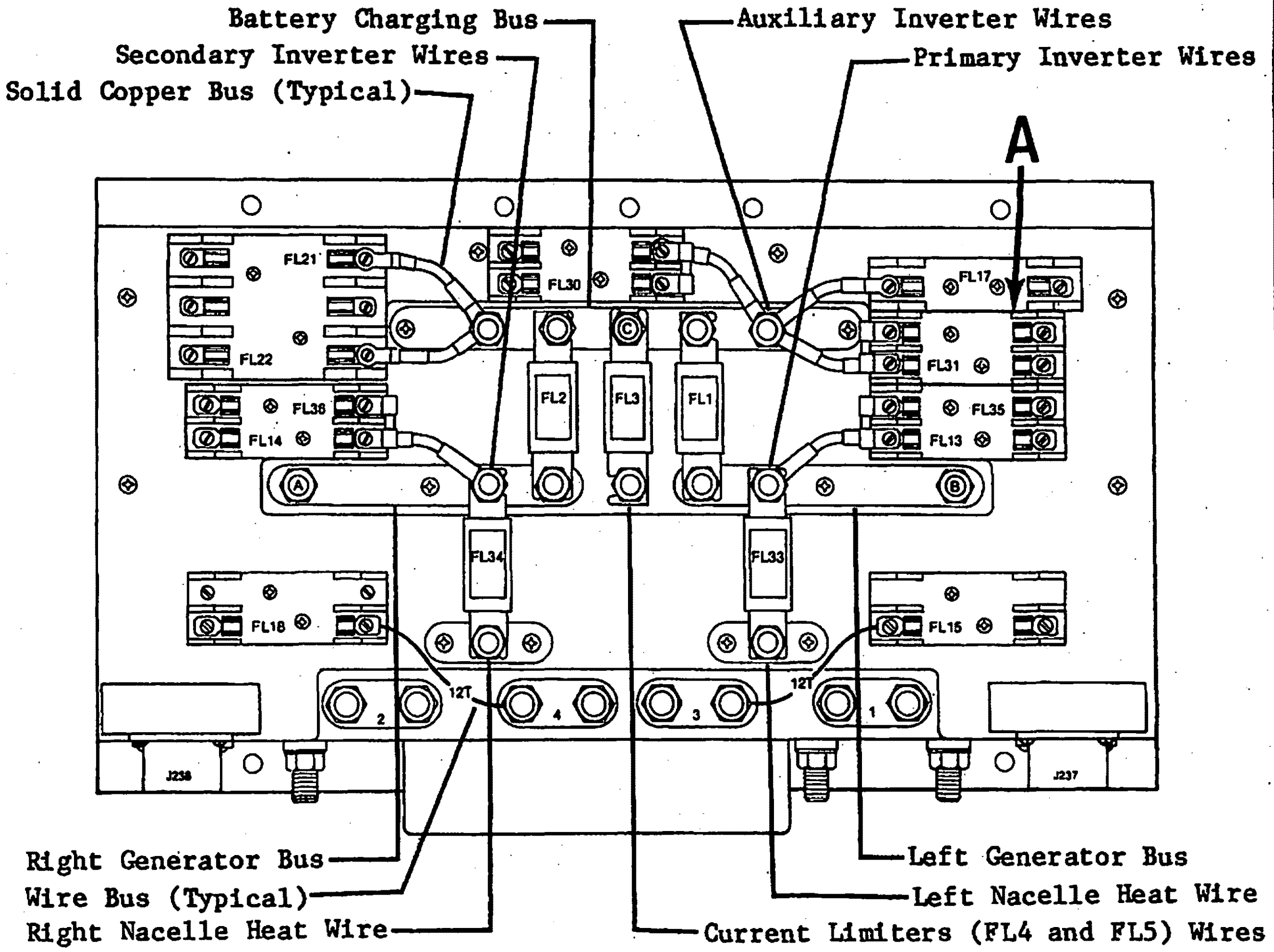
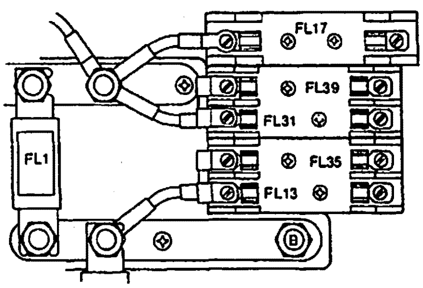
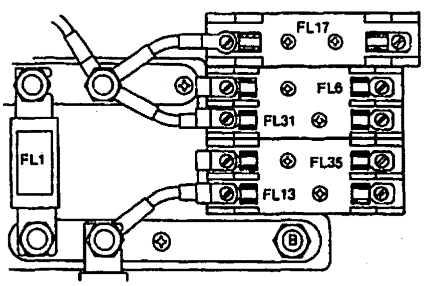
Current Limiter Panel Installation
Figure 202 (Sheet 1 of 3)





R.H. Recognition Light Option
(Aircraft 25-090, 25-103 thru 25-131)



R.H. Recognition Light Option
(Aircraft 25-132 thru 25-269 and 25-271)

Detail A

Current Limiter Panel Installation
Figure 202 (Sheet 2 of 3)

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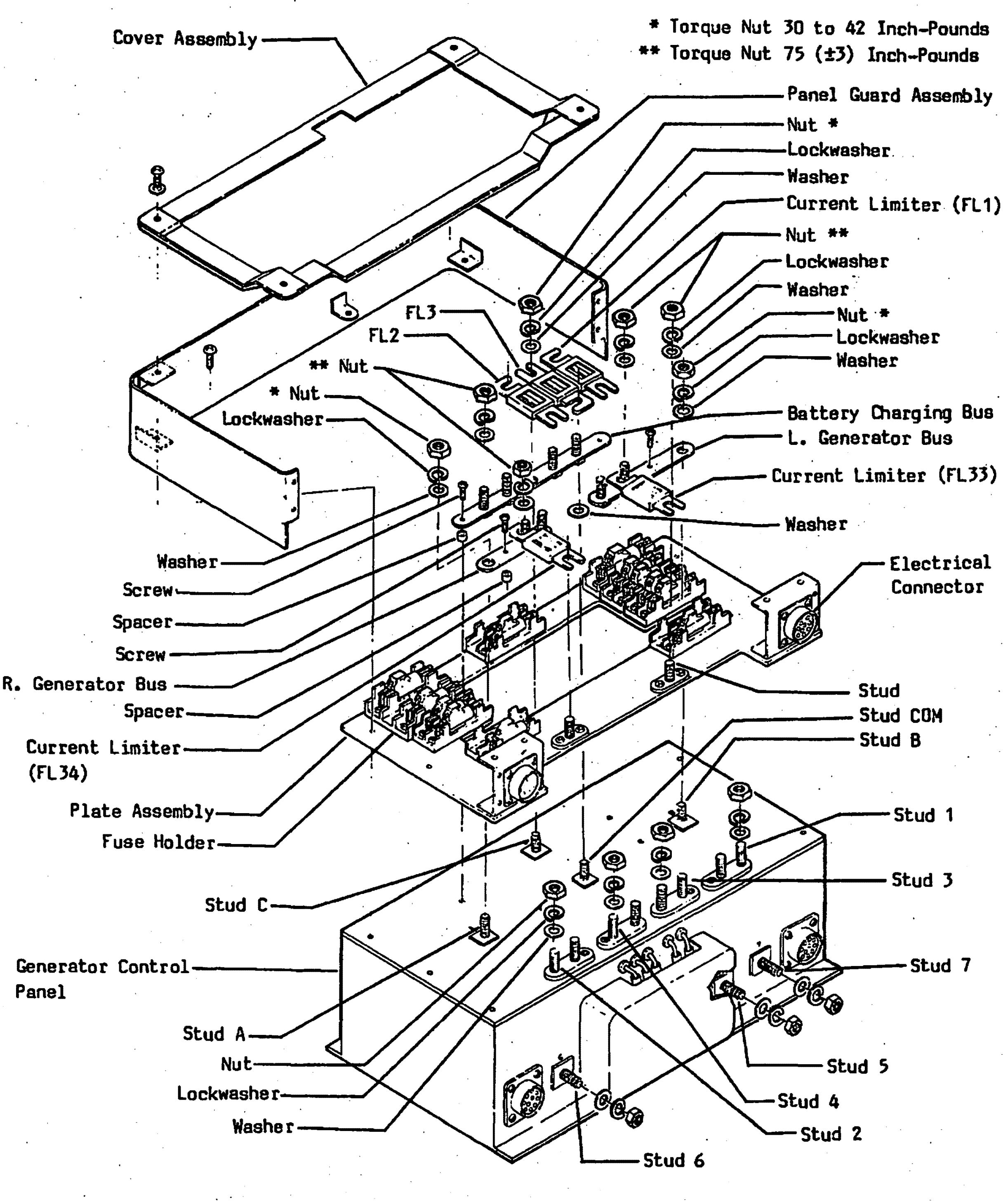
Current Limiter and Fuse Reference Designator	Amperage	Circuit	Power Source	Current Limiter Panel 2418053					
				-10	-12	-14	-17	-19	-20
FL1	275	L. Gen. Bus to Bat. Charging Bus		X	X	X	X	X	X
FL2	275	R. Gen. Bus to Bat. Charging Bus		X	X	X	X	X	X
FL3	275	Cabin Climate and Stabilizer Heat	Bat. Charging Bus	X	X	X	X	X	X
FL6	30	Recognition Light	Bat. Charging Bus			X		X	
FL13	20	Left Landing Light	Left Generator Bus	X	X	X	X	X	X
FL14	20	Right Landing Light	Right Generator Bus	X	X	X	X	X	X
FL15	20	Left Battery Bus	Bat. No. 1 (Stud 3)	X	X	X	X	X	X
FL17	50	Left Essential Bus	Bat. Charging Bus	X	X	X	X	X	X
FL18	20	Right Battery Bus	Bat. No. 2 (Stud 4)	X	X	X	X	X	X
FL21	50	Electric Hyd. Pump	Bat. Charging Bus	X	X	X	X	X	X
F122	50	Right Essential Bus	Bat. Charging Bus	X	X	X	X	X	X
FL30	10	Fuel Flow	Bat. Charging Bus	X	X	X	X	X	X
FL31	5	Utility Light	Bat. Charging Bus	X	X	X	X	X	X
FL33	100	Left Nacelle Heat	Left Generator Bus	X	X	X	X	X	X
FL34	100	Right Nacelle Heat	Right Generator Bus	X	X	X	X	X	X
FL35	10	L. Main Power Bus	Left Generator Bus	X	X	X	X	X	X
FL36	10	R. Main Power Bus	Right Generator Bus	X	X	X	X	X	X
FL39	20	Recog. Light	Bat. Charging Bus		X				
				-12	-13	-19	-12	-18	-12
				2418093 Cover Assembly					

-10 Current Limiter Panel - Aircraft 25-090, 25-103 thru 25-173
-12 Current Limiter Panel - Aircraft 25-090, 25-103 thru 25-131 when Optional Recognition Light is installed
-14 Current Limiter Panel - Aircraft 25-132 thru 25-210 when Optional Recognition Light is installed
-17 Current Limiter Panel - Aircraft 25-174 thru 25-210
-19 Current Limiter Panel - Aircraft 25-211 thru 25-269 and 25-271 when Optional Recognition Light is installed
-20 Current Limiter Panel - Aircraft 25-211 thru 25-269 and 25-271

Current Limiter Panel Installation
Figure 202 (Sheet 3 of 3)



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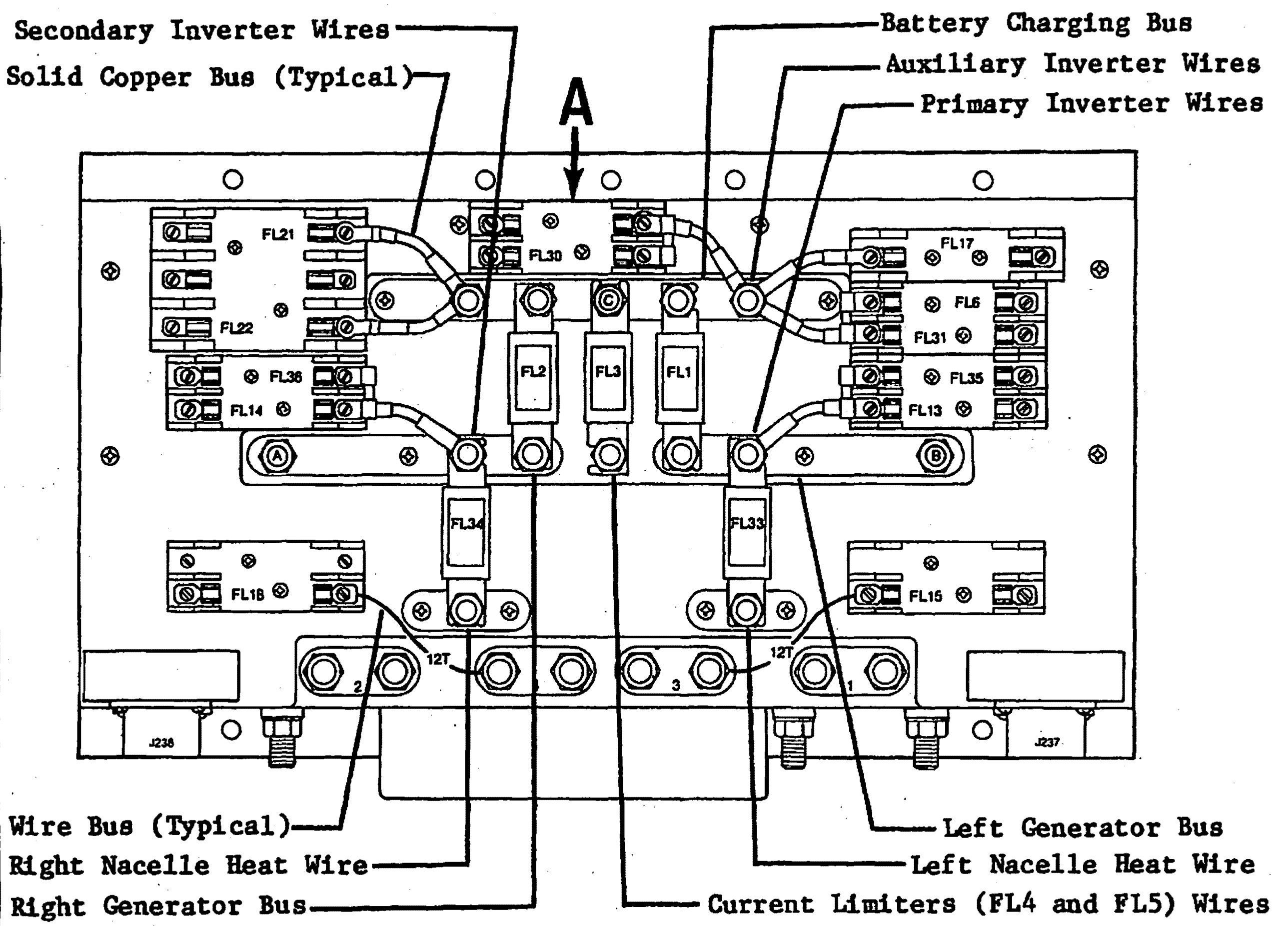
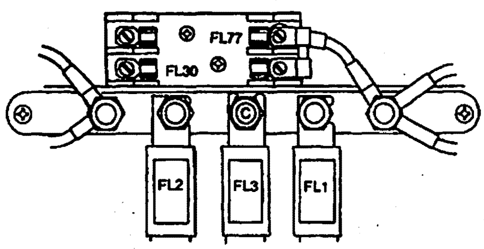
Current Limiter Panel Installation
Figure 203 (Sheet 1 of 3)





With L.H. Recognition Light Option

Detail A

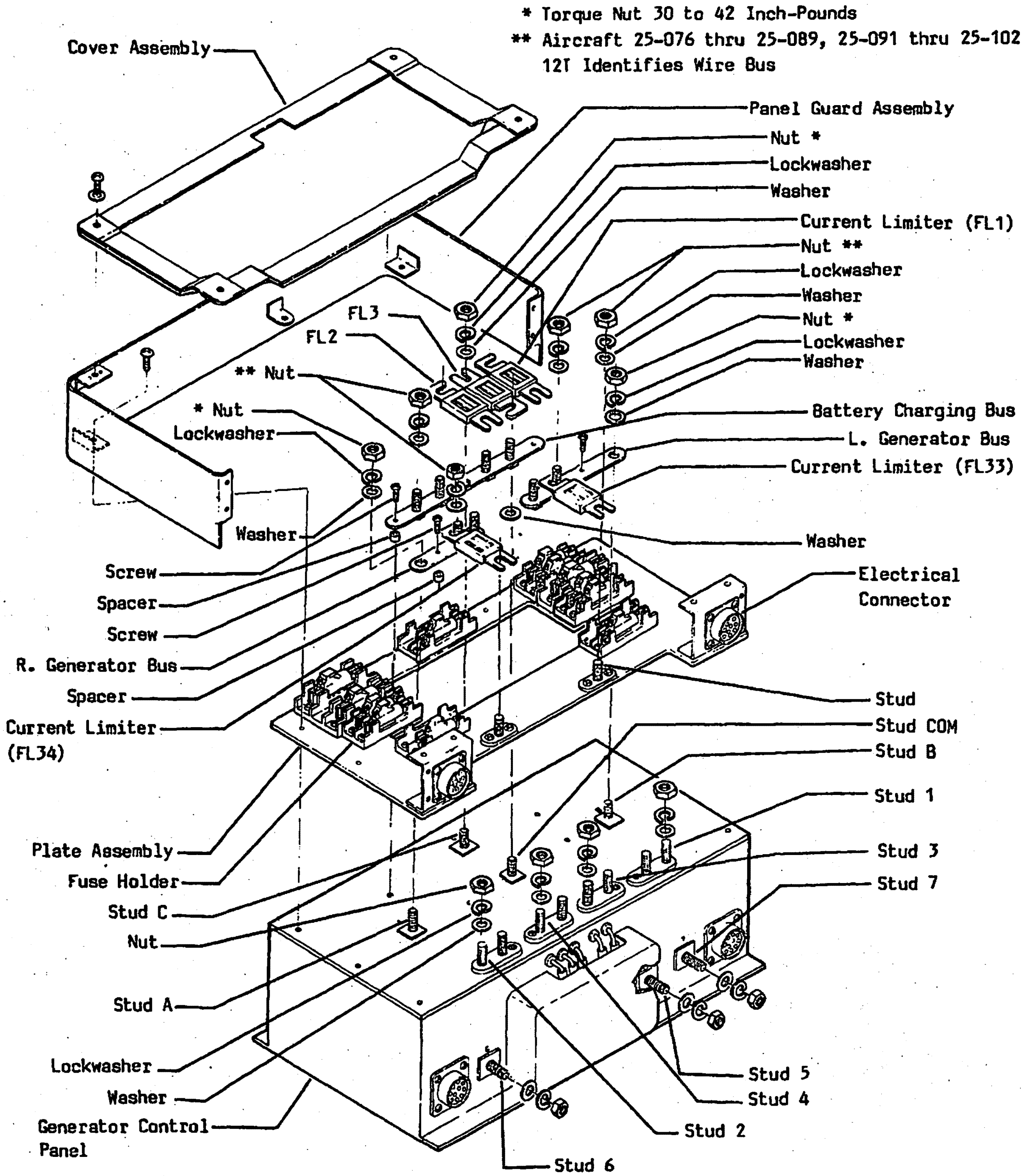
9-149C-10

Current Limiter Panel Installation
Figure 203 (Sheet 2 of 3)

Current Limiter and Fuse Reference Designator	Amperage	Circuit	Power Source	Current Limiter Panel 2418053 -21	-27	-29	-31
FL1	275	L. Gen. Bus to Bat. Charging Bus		X	X	X	X
FL2	275	R. Gen. Bus to Bat. Charging Bus		X	X	X	X
FL3	275	Cabin Climate and Stabilizer Heat	Bat. Charging Bus	X	X	X	X
FL6	30	Recognition Light	Bat. Charging Bus	X	X	X	X
FL13	20	Left Landing Light	Left Generator Bus	X	X	X	X
FL14	20	Right Landing Light	Right Generator Bus	X	X	X	X
FL15	20	Left Battery Bus	Bat. No. 1 (Stud 3)	X	X	X	X
FL17	50	Left Essential Bus	Bat. Charging Bus	X	X	X	X
FL18	20	Right Battery Bus	Bat. No. 2 (Stud 4)	X	X	X	X
FL21	50	Electric Hyd. Pump	Bat. Charging Bus	X	X	X	X
FL22	50	Right Essential Bus	Bat. Charging Bus	X	X	X	X
FL30	10	Fuel Flow	Bat. Charging Bus	X	X	X	X
FL31	5	Utility Light	Bat. Charging Bus	X	X	X	X
FL33	100	Left Nacelle Heat	Left Generator Bus	X	X	X	X
FL34	100	Right Nacelle Heat	Right Generator Bus	X	X	X	X
FL35	10	L. Main Power Bus	Left Generator Bus	X	X	X	X
FL36	10	R. Main Power Bus	Right Generator Bus	X	X	X	X
FL77	20	LH Recognition Light	Bat. Charging Bus				X
				-18	-18	-18	-87
				2418093 Cover Assembly			

-21 Current Limiter Panel - Aircraft 25-270, 25-272 thru 25-279
-27 Current Limiter Panel - Aircraft 25-280 thru 25-360
-29 Current Limiter Panel - Aircraft 25-361 thru 25-367
-31 Current Limiter Panel - Aircraft 25-361 thru 25-367 with Left-Hand Recognition Light

Current Limiter Panel Installation
Figure 203 (Sheet 3 of 3)



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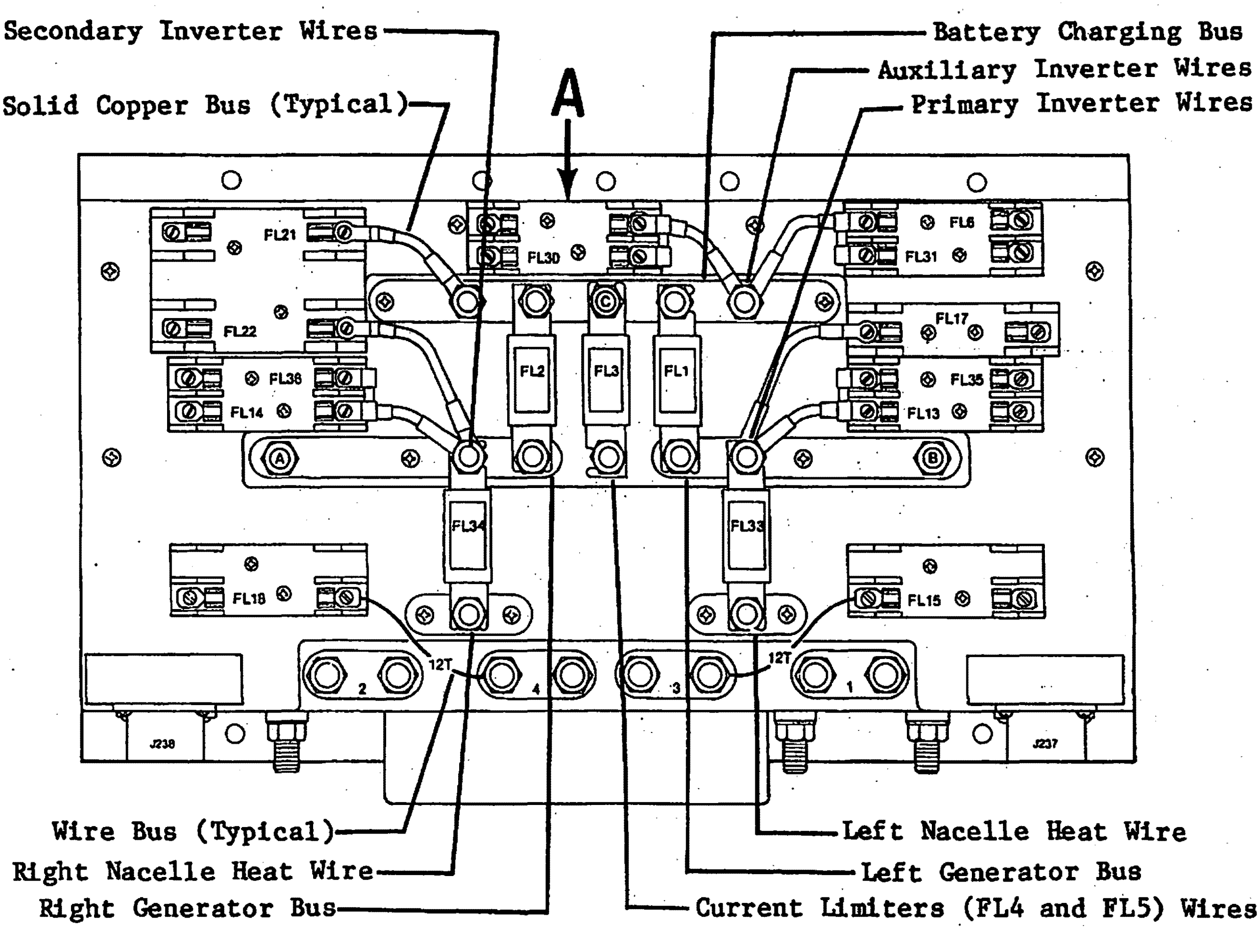
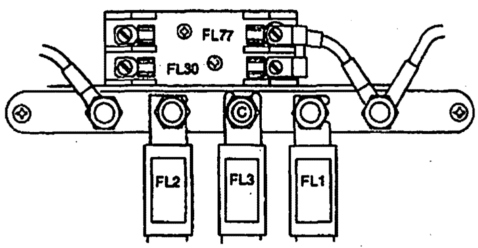
Current Limiter Panel Installation
Figure 204 (Sheet 1 of 3)





With L.H. Recognition Light Option

Detail A

9-149C-10

Current Limiter Panel Installation
Figure 204 (Sheet 2 of 3)

Current Limiter and Fuse Reference Designator	Amperage	Circuit	Power Source	Current Limiter Panel 2518325 -1	-2
FL1	275	L. Gen. Bus to Bat. Charging Bus		X	X
FL2	275	R. Gen. Bus to Bat. Charging Bus		X	X
FL3	275	Cabin Climate and Stabilizer Heat	Bat. Charging Bus	X	X
FL6	30	Recognition Light	Bat. Charging Bus	X	X
FL13	20	Left Landing Light	Left Generator Bus	X	X
FL14	20	Right Landing Light	Right Generator Bus	X	X
FL15	20	Left Battery Bus	Bat. No. 1 (Stud 3)	X	X
FL17	50	Left Essential Bus	Bat. Charging Bus	X	X
FL18	20	Right Battery Bus	Bat. No. 2 (Stud 4)	X	X
FL21	50	Electric Hyd. Pump	Bat. Charging Bus	X	X
FL22	50	Right Essential Bus	Bat. Charging Bus	X	X
FL30	10	Fuel Flow	Bat. Charging Bus	X	X
FL31	5	Utility Light	Bat. Charging Bus	X	X
FL33	100	Left Nacelle Heat	Left Generator Bus	X	X
FL34	100	Right Nacelle Heat	Right Generator Bus	X	X
FL35	10	L. Main Power Bus	Left Generator Bus	X	X
FL36	10	R. Main Power Bus	Right Generator Bus	X	X
FL77	20	LH Recognition Light	Bat. Charging Bus		X
				-90	-92
				2418093 Cover Assembly	

-1 Current Limiter Panel - Aircraft 25-368 and Subsequent
-2 Current Limiter Panel - Aircraft 25-368 and Subsequent with Left-Hand Recognition Light

Current Limiter Panel Installation
Figure 204 (Sheet 3 of 3)

2. INSPECTION/CHECK

NOTE: Perform this test following installation of generator control panel, current limiter panel, and when maintenance requires removal of battery charging bus, left generator bus, or right generator bus.

A. Operational Check of Left Main Bus

(1) Disconnect aircraft batteries.

WARNING: DO NOT REMOVE CURRENT LIMITER WHEN BATTERIES ARE CONNECTED.

(2) Remove current limiter FL2. (Refer to Current Limiter FL1, FL2 removal, paragraph 1.C.)
(3) Connect aircraft batteries.
(4) Pull (open) the MAIN BUS TIE circuit breaker.
(5) Set Battery Switch(es) on and check DC power on left main bus.
  (a) Verify proper operation of navigation lights.
  (b) While navigation lights are on, pull (open) L MAIN BUS circuit breaker. Navigation lights will extinguish verifying left main bus isolation.
  (c) Reset L MAIN BUS circuit breaker.
(6) Set Battery Switch(es) off.
(7) **Generator Feeder Wire Check**
  (a) Use a multimeter and check continuity.

NOTE: A diode is installed in the current and multimeter polarity is required.

Generator Test Point	Generator Control Panel Test Point	Result
L.H. Gen. Term B (+)	Stud 5 (-)	Continuity
L.H. Gen. Term B (-)	Stud 5 (+)	High Resistance
R.H. Gen. Term B (+)	Stud 6 (-)	Continuity
R.H. Gen. Term B (-)	Stud 6 (+)	High Resistance

(8) Disconnect aircraft batteries.

WARNING: DO NOT INSTALL CURRENT LIMITER WHEN BATTERIES ARE CONNECTED.

(9) Install current limiter FL2. (Refer to Current Limiter FL1, FL2 installation, paragraph 1.D.)

B. Operational Check of Right Main Bus

(1) Disconnect aircraft batteries.

WARNING: DO NOT REMOVE CURRENT LIMITER WHEN BATTERIES ARE CONNECTED.

(2) Remove current limiter FL1. (Refer to Current Limiter FL1, FL2 removal, paragraph 1.C.)

(3) Connect aircraft batteries.

(4) Pull (open) the MAIN BUS TIE circuit breaker.

(5) Set Battery Switch(es) on and check DC power on right main bus.

(a) Verify proper operation of anti-collision lights.

(b) While anti-collision lights are on, pull (open) R MAIN BUS circuit breaker. Anti-collision lights will extinguish verifying right main bus isolation.

(c) Reset R MAIN BUS circuit breaker.

(6) Pull (open) the L MAIN BUS circuit breaker.

(7) Reset MAIN BUS TIE circuit breaker. Verify voltage on the left main bus by checking that the navigational lights are operative.

(8) Set Battery Switch(es) off.

(9) Reset L MAIN BUS circuit breaker.

(10) Disconnect aircraft batteries.

WARNING: DO NOT INSTALL CURRENT LIMITER WHEN BATTERIES ARE CONNECTED.

(11) Install current limiter FL1. (Refer to Current Limiter FL1, FL2 installation, paragraph 1.D.)